



TIA STANDARD

Project 25

Inter-RF Subsystem Interface

Messages and Procedures for Voice

Services, Mobility Management, and RFSS

Capability Polling Services- Addendum 2-

Erratum 1 to Fix Errors and Omissions

**TIA-102.BACA-B-2
(Addendum to TIA-102.BACA-B)**

November 2016

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(From Project No. TIA-PN-102.BACA-B-2, formulated under the cognizance of the TIA TR-8 Mobile and Personal Private Radio TR-8.19 Subcommittee on Wireline System Interfaces).

Published by

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This document has been submitted to APCO/NASTD/FED by the Telecommunications Industry Association (TIA), as provided for in a Memorandum of Understanding (MOU) dated December, 1993. That MOU provides that APCO/NASTD/FED will devise a Common System Standard for digital public safety communications (the Standard), and that TIA shall provide technical assistance in the development of documentation for the Standard. This document has been developed by TR8.19 (Wireline Interfaces) with inputs from the APCO Project 25 Interface Committee (APIC), the APIC ISSI Task Group, and TIA Industry members. This document is being published to provide technical information on the emerging digital techniques for Land Mobile Radio Service.

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This document is an Erratum to TIA-102.BACA-B, November 2012, Project 25 Inter-RF Subsystem Interface (ISSI) Messages and Procedures for Voice and Mobility Management Services. The ISSI is the interface between Radio Frequency Subsystems (RFSSs) enabling relevant Project 25 user requirements to be satisfied (e.g., as described in the Inter-RF Subsystem Interface Overview (TSB-102.BACC-B)). Standardization efforts are planned to continue developing TIA standards defining compliance tests (e.g., measurement methods, performance recommendations, and interoperability tests) applicable to equipment implementing the TIA-102 ISSI protocols defined in this and other related TIA standards.

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DOCUMENT REVISION HISTORY

Version	Date	Description
Rev 0	September 2016	Approved for TIA publication.

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A. Addendum Introduction

This document is an Erratum to TIA-102.BACA-B, November 2012, "Project 25 Inter-RF Subsystem Interface Messages and Procedures for Voice and Mobility Management Services" [102BACA]. This addendum specifies the technical and editorial changes required to ensure correctness and clarity.

B. Addendum Scope

This addendum defines technical corrections and editorial clarifications to [102BACA].

This document will only include information relevant to the aforesaid scope. The reader is referred to [102BACA] for all other pertinent information. The next complete revision of [102BACA] will incorporate the changes contained in this addendum. Care must be taken when incorporating changes in this addendum to avoid conflicts with the changes made by other addendum that are incorporated at the same time. These include the addendum to add support for group emergency behaviors, and possibly others.

Changes to [102BACA] are described in this document starting at Section 0.1. *[Italic text between square brackets provides instructions on how to incorporate changes in a section.]* If a section contains subsections, and neither the section nor subsections thereof change, only the section will be identified in this document. Note that section numbers identified in this addendum correspond to the same section numbers in [102BACA], with new or modified text shown in orange to assist the reader.

C. Addendum References

This addendum references the following TIA documents.

C.1 Normative References

[102BACA] *Project 25 Inter-RF Subsystem Interface Messages and Procedures for Voice Services, Mobility Management, and RFSS Capability Polling Services*, TIA-102.BACA-B, TIA, November 2012

C.2 Informative References

[TIA_STYLE] *TIA Style Manual*, TIA, January 30, 1992

0.1 Minor Technical and Editorial Changes Throughout TIA-102.BACA-B

- “talk spurt” should be consistently spelled out as such (i.e., as 2 separate words, not as “talkspurt”, etc.) in the normal English text, as appropriate, throughout the document.
- “RTP resource” should consistently read “RTP resources”, as appropriate, throughout the document.
- “RF resource” should consistently read “RF resources”, as appropriate, throughout the document.
- **HEX-DIGIT** should consistently appear as such, not also appearing as **HEX DIGIT**, **hex-digit**, **HEXDIGIT**, etc. in the document.
- The term **setup** should consistently appear as such, not also appearing as **set up**, **set-up**, etc. in the document.
- The phrase **set up** should consistently appear as such, not also appearing as **set-up**, **setup**, etc. in the document.

NOTE: The actual changes will be implemented when **[102BACA]** is due for the next complete revision.

1. Introduction

[No change from existing text in this section of [102BACA], except for selected subsections thereof, as shown below.]

1.1 Scope

[No change from existing text in this section of [102BACA].]

1.2 Terminology

[No change from existing text in this section of [102BACA].]

1.3 Document Structure

[No change from existing text in this section of [102BACA].]

1.4 Definitions

[Replace existing text in this section of [102BACA] as shown below.]

[Modify the following definitions in this section of [102BACA] as shown below.]

FNE Home RFSS: The RFSS acting as the home of a General FNE address. See [102BACD].

FNE Serving RFSS: An RFSS that has expressed interest in receiving Supplementary Data addressed to a General FNE address. See [102BACD].

Full Rate: A voice coding method defined in [102BABA].

Half Rate: A voice coding method defined in [102BABA].

Listening (Serving) RFSS: An RFSS that has expressed interest in listening to (monitoring) supplementary data transactions of a particular unit or group. See [102BACD].

[Add the following definition to this section of [102BACA] as shown below.]

Transmitting SUID: SUID of the original audio source, which could be a mobile SU or a console SU, as appropriate.

1.5 Abbreviations

[Replace existing text in this section of [102BACA] as shown below.]

[Modify the following abbreviations in this section of [102BACA] as shown below.]

NASTD National Association of State Technology Directors

TSB Telecommunications Systems Bulletin

[Add the following abbreviations to this section of [102BACA] as shown below.]

CalledHome Called Home RFSS

CalledServ Called Serving RFSS

CallingHome Calling Home RFSS

CallingServ Calling Serving RFSS

CSSI Console Subsystem Interface (i.e., Ec)

FNEhome FNE Home RFSS

FNEserv FNE Serving RFSS

GroupHome Group Home RFSS

GroupServ Group Serving RFSS

Home Home RFSS

Listening	Listening (Serving) RFSS
Serving	Serving RFSS
UCS	Universal Character Set
UTF-8	Unicode Transformation Format, 8-bit encoding form

1.6 References

[No change from existing text in this section of [102BACA], except for subsections 1.6.1 and 1.6.2 below.]

1.6.1 Normative References

[Replace existing text in this section of [102BACA] as shown below.]

[Modify the following references as shown below.]

[102AABC] *Project 25 Trunking Control Channel Messages*, TIA-102-AABC-D, TIA, April 2015.

[102AABD] *Project 25 Trunking Procedures*, TIA-102.AABD-B, TIA, November 2014.

[102BABA] *Project 25 Vocoder Description*, ANSI/TIA-102.BABA-A, TIA, February 2014

~~**[102BABA-1]** *Project 25 Half Rate Vocoder Annex*, TIA-102.BABA-1 (Addendum to TIA-102.BABA), TIA, July 2009~~

[102BBAC] *Project 25 Phase 2 Two-Slot TDMA Media Access Control Layer Description*, TIA-102.BBAC, TIA, December 2010; and *Addendum 1* (February 2013)

[RFC0791] *Internet Protocol: DARPA Internet Program Protocol Specification*, RFC 0791, IETF, September 1981

[Add the following new references to this section of [102BACA].]

[RFC2474] *Definition of the Differentiated Services Field (DS Field) in the IPv4 and IPv6 Headers*, RFC 2474, IETF, December 1998

[RFC2597] *Assured Forwarding PHB Group*, RFC 2597, IETF, June 1999

[RFC2616] *Hypertext Transfer Protocol -- HTTP/1.1*, RFC 2616, IETF, June 1999

[RFC3246] *An Expedited Forwarding PHB (Per-Hop Behavior)*, RFC 3246, IETF, March 2002

[RFC3629] *UTF-8, a transformation format of ISO 10646*, RFC 3629, IETF, November 2003

[RFC5234] *Augmented BNF for Syntax Specifications: ABNF*, RFC 5234, IETF, January, 2008

1.6.2 Informative References

[Replace existing text in this section of [102BACA] as shown below.]

[Modify the following references as shown below.]

[102AACA] *Project 25 Digital Radio Over-The-Air Rekeying (OTAR) Messages and Procedures*, TIA-102.AACA-A, TIA, September 2014

[102BAEB] *Project 25 IP Data Bearer Service Specification*, TIA-102.BAEB-B, TIA, September 2014

~~*Editor's Note: The above document is a work in progress and cannot be referenced unless and until it is approved and published. Until such time*~~

as this Editor's Note is removed, the inclusion of the above document is for informational purposes only.

[T102] *Project 25 TIA-102 Documentation Suite Overview*, TSB-102-C, TIA, March 2016

[TIA_STYLE] *TIA Standards Style Guide For Engineering Committees*, TIA, July 2015

[Remove the following references from this section of [102BACA].]

~~**[102AABF]** *Project 25 Link Control Word Formats and Messages*, TIA-102.AABF-C, TIA, March 2011.~~

~~**[102BAAD]** *Project 25 Common Air Interface Description for Conventional Channels*, TIA-102.BAAD-A, TIA, February 2010~~

~~**[102BADA]** *Telephone Interconnect Requirements and Definitions (Voice Service)*, ANSI/TIA/EIA-102.BADA-2000, TIA, March 2000; and Addendum 1 (May 2006)~~

~~**[MFID]** *MFID Manufacturer's Identification Number Assignment Guide and Procedures*, Version 1.1, January 1999~~

~~**[P25 SoR]** *APCO Project 25 Statement of Requirements*, April 8, 2011~~

~~**[RFC0793]** *Transmission Control Protocol: DARPA Internet Program Protocol Specification*, RFC 0793, IETF, September 1981~~

~~**[RFC1034]** *Domain Names - Concepts and Facilities*, RFC 1034, IETF, November 1987~~

~~[RFC2543] SIP: Session Initiation Protocol, RFC 2543, IETF, March 1999~~

~~[RFC2736] Guidelines for Writers of RTP Payload Format Specifications, RFC 2736, IETF, December 1999~~

~~[RFC3629] UTF-8, a transformation format of ISO 10646, RFC 3629, IETF, November 2003~~

~~[RFC2474] Definition of the Differentiated Services Field (DS Field) in the IPv4 and IPv6 Headers, RFC 2474, IETF, December 1998~~

~~[RFC5234] Augmented BNF for Syntax Specifications: ABNF, RFC 5234, IETF, January, 2008~~

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2. Overview of Architecture and Protocol Suite

[No change from existing text in this section of [102BACA], except for subsections 2.1 and 2.2 below.]

2.1 Architecture Overview

[No change from existing text in this section of [102BACA], except for subsection 2.1.4 below.]

2.1.1 Call Models

[No change from existing text in this section of [102BACA].]

2.1.2 Subsystem Topologies

[No change from existing text in this section of [102BACA].]

2.1.3 Functional Model

[No change from existing text in this section of [102BACA].]

2.1.4 Service Access Point Model

[No change from existing text in this section of [102BACA], except for subsection 2.1.4.1 below.]

2.1.4.1 Mobility Management SAP

[No change from existing text in this section of [102BACA], except for subsection 2.1.4.1.4 below.]

2.1.4.1.1 MM Registration Transaction

[No change from existing text in this section of [102BACA].]

2.1.4.1.2 MM Serving Deregistration Transaction

[No change from existing text in this section of [102BACA].]

2.1.4.1.3 MM Home Deregistration Transaction

[No change from existing text in this section of [102BACA].]

2.1.4.1.4 MM Query Transaction

[Replace existing text in this section of [102BACA] with the text below.]

The Query Transaction is used by a Serving RFSS to request service profile information for a particular unit from a Home RFSS, and by a Home RFSS to verify that a Serving RFSS still has interest in a particular unit, group, or SN Context. It is implemented by the four primitives described below.

MM_Query.Request is used to ask another RFSS for the status of a unit or group. It includes the identity of the object in question, the identity of the RFSS being queried, and the type of query being requested which might be "Serving", "Normal Home", "Forced", or "Confirmed". It anticipates *MM_Query.Response* to complete the transaction.

MM_Query.Response provides the control function an answer to *MM_Query.Request*. Regardless of the type of query, the answer includes a status indication (Successful, Not Found, Forbidden, or Failed). The answer to a successful serving query includes the service profile information for the unit. The answer to a successful normal home, forced or confirmed query indicates whether or not the subject RFSS still has need for the object to be registered.

MM_Query.Indicate includes the identity of the unit or group being queried, the identity of the requesting RFSS, and the type of query being requested. It requires *MM_Query.IndicateResponse* in reply. A Serving RFSS that receives a "Forced" or "Confirmed" query from the Home RFSS of the subject object is expected to take particular steps (described in Section 5) to validate the registration of the object prior to responding to the query.

MM_Query.IndicateResponse provides the reply to *MM_Query.Indicate*. It includes a status (Successful, Not Found, Forbidden or Failed) that is expected to be relayed to the requesting RFSS, and, for serving queries only, might include service profile information to be communicated with the status.

2.1.4.2 Call Control SAP

[No change from existing text in this section of [102BACA].]

2.1.4.3 Transmission Control SAP

[No change from existing text in this section of [102BACA].]

2.1.4.4 RFSS Service Capability Polling SAP

[No change from existing text in this section of [102BACA].]

2.2 ISSI Internet Protocol Suite

[No change from existing text in this section of [102BACA], except for subsection 2.2.2 below.]

2.2.1 Lower Layers

[No change from existing text in this section of [102BACA].]

2.2.2 Network Layer

[No change from existing text in this section of [102BACA], except for subsection 2.2.2.1 below.]

2.2.2.1 Internet Protocol Version 4 (IPv4)

[No change from existing text in this section of [102BACA], except for subsections 2.2.2.1.1 and 2.2.2.1.2 below.]

2.2.2.1.1 IPv4 Header Format

[Replace existing text in this section of [102BACA] with the text below.]

For IPv4:

- The IP header length is typically 20 octets.
- The minimum MTU for each IP packet (IP header & body) SHALL be 1500 octets.

Figure 23 shows the header format of a typical IPv4 packet.

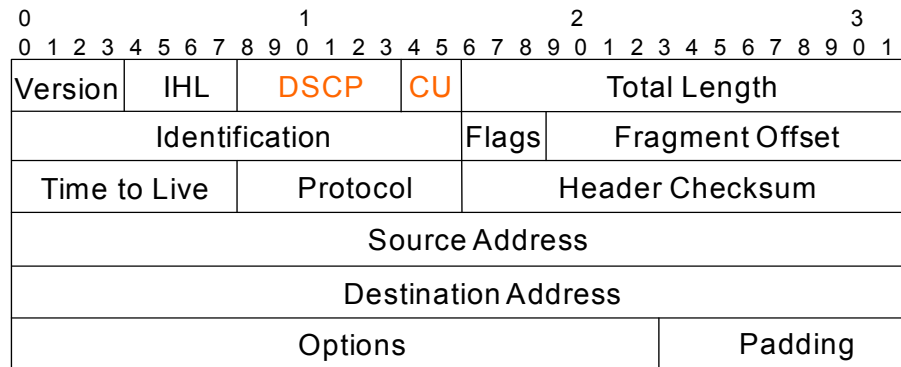


Figure 23 - Typical IPv4 Datagram Header Format

2.2.2.1.2 Header Field Definitions

[Replace existing text in this section of [102BACA] with the text below.]

Detailed generic definitions of the IPv4 header fields are contained in **[RFC0791]**.

For voice services over the ISSI, the IPv4 header fields SHALL be set as follows (all numbers are in decimal):

Version (4 bits):

4

IHL (4 bits):

(see **[RFC0791]**)

DS (8 bits):

The DS field comprises two subfields:

DSCP and CU **[RFC2474]²**

DSCP (6 bits):

² Editor's Note: **[RFC2474]** defines a replacement header field, called the DS (Differentiated Services) field, superseding the IPv4 TOS (Type of Service) octet. This new field includes a 6-bit DSCP (Differentiated Services Codepoint) which could be flexibly mapped to particular forwarding treatment, or per-hop behavior (PHB), at each Differentiated Services compliant network node along its path, for efficient routing of SIP messages and RTP packets to their destinations.

The DSCP value is owner/user configurable and **SHOULD** be set according to the actual QoS required for any given packet:

- a. For voice traffic (and associated in-band RTP control traffic), the DSCP value **SHOULD** be set to binary 101110, Expedited Forwarding (EF) **[RFC3246]**;
- b. For SIP control traffic, the DSCP value **SHOULD** be set to binary 100010, Assured Forwarding Class 4 Low Drop **[RFC2597]**.

CU (2 bits): (currently unused)

Total Length (16 bits):	(see [RFC0791])
Identification (16 bits):	0
Flags (3 bits):	0 or 1 (i.e., allow fragmentation)
Fragment Offset (13 bits):	0
Time to Live (8 bits):	(see [RFC0791])
Protocol (8 bits):	17 (i.e., UDP, see [RFC3232])
Header Checksum (16 bits):	(see [RFC0791])
Source Address (32 bits):	IPv4 address of the source RFSS
Destination Address (32 bits):	IPv4 address of the destination RFSS
Options (variable in length):	(see [RFC0791])
Padding (variable in length):	(used to ensure that the data begins on 32 bit word boundary. The padding is zero.)

NOTE - The assignment and coordination of IP addresses for the RFSSs are at the discretion of the user community, and are therefore beyond the scope of this document.

2.2.2.2 Internet Protocol Version 6 (IPv6)

*[No change from existing text in this section of **[102BACA]**.]*

2.2.3 Transport Layer

[No change from existing text in this section of [102BACA].]

2.2.4 Application Layer

[No change from existing text in this section of [102BACA].]

2.3 ISSI Data Transmission Order

[No change from existing text in this section of [102BACA].]

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3. SIP Messages and Parameters Definition for ISSI

[No change from existing text in this section of [102BACA], except for selected subsections thereof, as shown below.]

3.1 Preamble: Grammar

[Replace existing text in this section of [102BACA] with the text below.]

The grammar used to describe SIP parameters and headers is expressed in Augmented Backus- Naur Form (ABNF) as defined in **[RFC5234]** and used for defining SIP in **[RFC3261]** (Section 25), SDP in **[RFC4566]** (Section 9), and UTF-8 in **[RFC3629]** (Section 4).

Unless specified otherwise, values expressed as quoted strings are case-sensitive, in accordance with Section 7.3.1 of **[RFC3261]**.

Unless otherwise stated, tokens are case-insensitive, in accordance with sections 25.1 and 7.3.1 of **[RFC3261]**.

The following ABNF definitions are used:

```
integer          =  POS-DIGIT * (DIGIT)
                  ; integer without a leading 0

DIGIT            =  "0" / POS-DIGIT

POS-DIGIT       =  "1" / "2" / "3" / "4" / "5" / "6" / "7" / "8" / "9"

octet-string    =  1*unreserved
unreserved      =  alphanum / mark
alphanum        =  ALPHA / DIGIT
ALPHA           =  %x41-5A / %x61-7A ; A-Z / a-z
mark            =  "-" / "_" / "." / "!" / "~" / "*" / "'" / "(" / ")"
```

UTF8-octets = *(UTF8-char)
 UTF8-char = UTF8-1 / UTF8-2 / UTF8-3 / UTF8-4
 UTF8-1 = %x00-7F
 UTF8-2 = %xC2-DF UTF8-tail
 UTF8-3 = %xE0 %xA0-BF UTF8-tail / %xE1-EC 2(UTF8-tail) /
 %xED %x80-9F UTF8-tail / %xEE-EF 2(UTF8-tail)

 UTF8-4 = %xF0 %x90-BF 2(UTF8-tail) / %xF1-F3 3(UTF8-tail) /
 %xF4 %x80-8F 2(UTF8-tail)

 UTF8-tail = %x80-BF

In addition, the following definitions are used:

SP = %x20 ; the space character
 CRLF = %x0D %x0A ; internet standard newline
 DQUOTE = %x22 ; the double quote character
 boolean = "0" / "1"
 int = "0" / integer
 HEX-DIGIT = DIGIT / "A" / "B" / "C" / "D" / "E" / "F"

A number formed by concatenating "n" HEX-DIGIT is expressed as nHEX-DIGIT.

An RFSS SHALL send uppercase A-F for all HEX-DIGIT based parameters in the SIP URI, header fields, and message bodies of any originating SIP request (i.e., SIP request sent to the succeeding RFSS not as a result of receiving a similar SIP request from a preceding RFSS).

An RFSS SHALL accept uppercase A-F for HEX-DIGIT based parameters in the SIP URI, header fields, and message bodies of any SIP message received from another RFSS. If the received message is a SIP request from a preceding RFSS:

- a. When subsequently responding to the preceding RFSS, the RFSS SHALL send uppercase A-F for all HEX-DIGIT based parameters in the SIP URI, header fields, and message bodies of its SIP response.
- b. When subsequently sending a similar SIP request to any succeeding RFSS, the RFSS SHALL send uppercase A-F for all HEX-DIGIT based parameters in the SIP URI, header fields, and message bodies of its SIP request.

An RFSS SHALL also accept lowercase A-F for HEX-DIGIT based parameters in the SIP URI, header fields, and message bodies of any SIP message originally sourced by another RFSS of an older implementation. If the received message is a SIP request from a preceding RFSS:

- a. When subsequently responding to the preceding RFSS:
 - i. The RFSS SHALL send lowercase A-F for those HEX-DIGIT based parameters in the SIP URI and header fields of its SIP response that are copied per Table 13 of this document.
 - ii. The RFSS MAY send uppercase A-F or lowercase A-F for all other HEX-DIGIT based parameters in the SIP URI, header fields, and message bodies of its SIP response.
- b. When subsequently sending a similar SIP request to any succeeding RFSS, the RFSS MAY send uppercase A-F or lowercase A-F for all HEX-DIGIT based parameters in the SIP URI, header fields, and message bodies of its SIP request.

3.2 SIP Transport

[No change from existing text in this section of [102BACA].]

3.3 Request Line Construction

[No change from existing text in this section of [102BACA].]

3.4 SIP URI and related addressing logic

[No change from existing text in this section of [102BACA], except for Subsection 3.4.1 below.]

3.4.1 General Characteristics of SIP URIs

[Replace existing text in this section of [102BACA], and Subsection 3.4.1.3, as shown below.]

The SIP URIs of the ISSI use a limited form of SIP URIs as described in [RFC3261]. Specifically, they have the following form:

sip-uri-for-ISSI = "sip:" [radical-name "@"] domain-name [parameters]

where domain-name is described in Section 3.4.1.1, radical-name is described in Section 3.4.1.2, and parameters, if present, are as described in Section 3.4.1.3. The notation [...] indicates a field that is optional.

"sip:" and "@" are always present in the indicated positions in ISSI SIP URIs except for the Request URI of the SIP register message, where radical-name and @ are never present.

~~All fields in ISSI SIP URIs SHALL be case insensitive.~~

Handling of HEX-DIGIT based parameters is as specified in Section 3.1 of this document.

3.4.1.1 ISSI Domain Names

[No change from existing text in this section of [102BACA].]

3.4.1.2 ISSI Radical Names

[No change from existing text in this section of [102BACA].]

3.4.1.3 ISSI Parameters

[Replace existing text in this section of [102BACA] with the text below.]

The parameters are defined in the following way:

parameters	=	numplan-param / lr-param / branch-param
numplan-param	=	“;user=TIA-P25-SU” / “;user=TIA-P25-SG” / “;user=TIA-P25-SD-SU” / “;user=TIA-P25-SD-SG” / “;user=TIA-P25-SD-SGFNE” / “;user=TIA-P25-SDMON-SU” / “;user=TIA-P25-SDMON-SG” / “;user=TIA-P25-SDMON-SGFNE” / “;user=TIA-P25-PD-SNC” / “;user=TIA-P25-RFSS”
lr_param	=	“;lr”
branch-param	=	”;branch=z9hG4bK” branch-id
branch-id	=	token

where token is defined in Section 25.1 of **[RFC3261]**.

As stated in **[RFC3261]** Section 8.1.1, the branch-param parameter begins with the magic cookie “z9hG4bK”, followed by a globally unique branch-id parameter. The construction of the branch-id is manufacturer-dependent. However an example construction method is provided below for information only.

ISSI parameters have the following form, as shown in Table 9 below:

Table 9 - ISSI Parameters in the SIP URIs

ISSI Parameter	Syntax & Format description	Use
SU Numbering Plan	";user=TIA-P25-SU"	In any SIP URI where the su-id radical name is used
SG Numbering Plan	";user=TIA-P25-SG"	In any SIP URI where the sg-id radical name is used
SD Numbering Plans	";user=TIA-P25-SD-SU" ";user=TIA-P25-SD-SG" ";user=TIA-P25-SD-SGFNE"	For mobility management procedures only, to identify supplementary data interest in the given mobility object.
SD Monitoring Numbering Plans	";user=TIA-P25-SDMON-SU" ";user=TIA-P25-SDMON-SG" ";user=TIA-P25-SDMON-SGFNE"	For mobility management procedures only, to identify supplementary data monitoring interest in the given mobility object.
PD SNC Numbering Plan	";user=TIA-P25-PD-SNC"	For mobility management procedures only, to identify the SMDCP context of a given SU.
RFSS Service Capability Polling Numbering Plans	";user=TIA-P25-RFSS"	For RFSS service capability polling procedures only
lr parameter	"lr" Refer to Section 19.1.1 of the [RFC3261]	In any SIP URI of the Record-Route and Route header fields
Branch parameter	"branch=z9hG4bK<branch-id>" Where <branch-id> is the hexadecimal representation of the variable part of the branch ID.	In SIP URI of the Via header field Refer to Section 8.1.1.7 of [RFC3261]
Empty	"	In any other SIP URI

Following is an example method for calculating the branch-param (provided here for information only):

The branch-id is described as follows:

```

branch-id    =    loop-detector "." branch-number
loop-detector =    1*HEX-DIGIT
branch-number =    1*HEX-DIGIT

```

As recommended in the **[RFC3261]**, loop-detector is the hexadecimal representation of the 128-bit value, calculated according to MD5 Message-Digest algorithm defined in the **[RFC1321]** (The MD5 Message-Digest Algorithm).

The following "input" value is used for MD5:

```

input        =    To-field From-field Call-ID-field Request-URI CSeq-number

```

where:

- To-field, From-field, Call-ID-field are respectively the To, From, Call-ID header fields of the received request. For each header field xyz (located at line N in the request, with N>0), the xyz-field string used is located between the CRLF of line (N-1) and the CRLF of line N, CRLF excluded.
- Request-URI: The Request-URI of the received request
- CSeq-number is the sequence number from the CSeq header field.

The components of "input" are strung together - in the order given above, with no space or any other character between them - to form a sequence of UTF-8 characters and thus a sequence of bits.

branch-number is a counter starting at 1 for the first branch and incremented by one for the next branch.

Refer to Via header field in Section 3.5.17 for an example of a branch parameter ID.

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3.4.2 Request URI formatting Summary

[No change from existing text in this section of [102BACA].]

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3.5 SIP Header Encoding

[Replace existing text in this section of [102BACA], and selected subsections thereof, as shown below.]

General syntax for header fields is covered in Section 20 of **[RFC3261]** and in **[RFC1035]**.

When comparing header fields, field names are always case-insensitive, in accordance with Section 7.3.1 of **[RFC3261]**.

Unless otherwise stated in the definition of a particular header field, field values, parameter names, and parameter values are case-insensitive, in accordance with Section 7.3.1 of **[RFC3261]**.

This section provides additional requirements. Furthermore, the explicitly defined ABNF constructs for SIP header fields in this section are purposely more restrictive than those found in **[RFC3261]**.

Table 13 lists the header fields used in the SIP messages sent by the RFSSs over the ISSI. This table gives restriction rules on a similar table given in Section 20 of **[RFC3261]**.

The “where” Column describes the request and response types in which the header field can be used. Possible values in this column are:

Request: header field MAY only appear in requests
 Response: header field MAY only appear in responses
 2xx, 4xx, etc: A numerical value or range indicating response codes in which the header field can be used
 Copied: header field is copied from the request to the response

An empty entry in the “where” column indicates that the header field MAY be present in all requests and responses.

The “SIP Method” columns describe the requirement specific to the ISSI definition. Possible values are:

C-ISSI: The header field is conditional for ISSI and depends on the context of the procedure.
 C-RFC: Same as the “c” condition in Table 2 (Summary of header fields) in Section 20 of **[RFC3261]**.
 M-ISSI: The header field is mandatory for ISSI.
 M-RFC: Mandatory as defined in Table 2 (Summary of header fields) in the Section 20 of **[RFC3261]**.
 NO-ISSI: The header field is never sent out on ISSI.
 O-ISSI: The header field is optional for ISSI.

-: The void “-“ identifier is used to indicate the header field is not applicable for the SIP Method as defined in Section 20 of [RFC3261].

If an RFSS receives a SIP Request or SIP Response that contains a SIP header field specified as being “NO-ISSI” in this document, the RFSS SHOULD ignore this SIP header field. Upon receipt of the SIP Request, the RFSS SHOULD return a GEN_IGNORE_NON_ISSI_HEADER 2xx response (refer to Annex A of this document).

The RFSS SHOULD simply discard any additional, irrelevant SIP header fields received, and it SHOULD continue with the SIP transaction.

Table 13 SIP-message header fields used on the ISSI

SIP Methods							
SIP Header Field	Where	ACK	BYE	CANCEL	INVITE	REGISTER	OPTIONS
Accept	Request	-	NO-ISSI	-	M-ISSI	M-ISSI	M-ISSI
Accept	2xx	-	-	-	NO-ISSI	NO-ISSI	M-ISSI
Accept	415	-	-	-	M-ISSI	C-ISSI	M-ISSI
Allow	Request	-	M-ISSI	-	M-ISSI	M-ISSI	M-ISSI
Allow	2xx	-	M-ISSI	-	M-ISSI	M-ISSI	M-ISSI
Allow	Response	-	M-ISSI	-	M-ISSI	M-ISSI	-
Call-ID	Copied	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC
Contact	Request	NO-ISSI	-	-	M-RFC	C-ISSI(1)	NO-ISSI
Contact	1xx	-	-	-	NO-ISSI	-	-
Contact	2xx	-	-	-	M-RFC	C-ISSI(1)	NO-ISSI
Content-Disposition		C-ISSI(7)	C-ISSI(7)	-	C-ISSI(7)	C-ISSI(7)	C-ISSI(7)
Content-Length		M-ISSI	M-ISSI	M-ISSI	M-ISSI	M-ISSI	M-ISSI
Content-Type		C-ISSI(2)	C-ISSI(2)	-	C-ISSI(2)	C-ISSI(2)	C-ISSI(2)
Cseq	Copied	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC
Expires	Request, 2xx response	-	-	-	NO-ISSI	C-ISSI(1)	-
Expires	Non-2xx response	-	-	-	NO-ISSI	NO-ISSI	-
From	Copied	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC
Max-Forwards	Request	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC
MIME-Version		C-ISSI(3)	C-ISSI(3)	-	C-ISSI(3)	C-ISSI(3)	C-ISSI(3)
Priority	Request	-	-	-	M-ISSI	-	-
Record-Route	Request	NO-ISSI	NO-ISSI	NO-ISSI	C-ISSI(4)	-	NO-ISSI
Record-Route	2xx, 18x	-	NO-ISSI	NO-ISSI	C-ISSI(4)	-	NO-ISSI
Route	Request	C-RFC(5)	NO-ISSI(5)	NO-ISSI(5)	C-RFC(5)	NO-ISSI(5)	NO-ISSI
To	Copied(1)	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC
Via	Copied	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC
Warning	Response	-	NO-ISSI	NO-ISSI	C-ISSI(6)	C-ISSI(6)	C-ISSI(6)

- (1): Refer to Section 5
 (2): Refer to Section 3.5.7
 (3): Refer to Section 3.5.12

- (4): Refer to Section 3.5.14
- (5): Refer to Section 3.5.15
- (6): Refer to Section 3.5.18
- (7): Refer to Section 3.5.5

If a compact form exists for a header field it SHOULD be used (Refer to Section 20 of **[RFC3261]**). However, examples are given in the extended form for more clarity in this document.

An RFSS SHALL be able to decode both the compact and extended forms for any SIP header field.

SIP Header Encoding for the MESSAGE method for Supplementary Data is described in **[102BACD]**.

3.5.1 Accept

*[Replace existing text in this section of **[102BACA]** with the text below.]*

This header field indicates the acceptable bodies in the response:

- SDP, P25-ISSI, and P25-SNDP bodies³, as described in Sections 3.7, 3.8, and 3.9, respectively.
- Bodies for Supplementary Data, are defined in **[102BACD]**.
- Manufacturer-specific body, if acceptable, are as described in Section 3.5.7.

The Accept header field carries a list of body type descriptors, following the syntax and semantics defined in **[RFC2616]**, and as referenced in sections 7, 20.1, and 25.1 of **[RFC3261]**.

Referring to the ABNF specifications for the SIP protocol in Section 25.1 of **[RFC3261]**, the following basic parsing construct for the Accept header field SHALL be supported:

```
Accept = "Accept" HCOLON [ accept-range *(COMMA accept-range) ]
accept-range = media-range
media-range = m-type SLASH m-subtype
m-type = discrete-type
discrete-type = "application"
```

³ The phrases "SDP", "P25-ISSI" and "P25-SNDP bodies", used throughout this document are simply convenient shorthand abbreviations for the literal terms "application/sdp;level=1", "application/x-tia-p25-issi" and "application/x-tia-p25-sndcp". Any use of these convenient shorthand abbreviations, or similarly structured shorthand abbreviations throughout this document does not indicate such use has been approved or mandated by the APCO/NASTD/FED Project 25 Steering Committee, nor does such appearance indicate such use is limited to Project 25 implementations exclusively.

m-subtype = extension-token
extension-token = ietf-token / x-token
ietf-token = "sdp;level=1"
x-token = issi-token / supdat-token / sndcp-token / mfid-token
issi-token = "x-tia-p25-issi"
supdat-token = "x-tia-p25-supdat"
sndcp-token = "x-tia-p25-sndcp"
mfid-token = "x-tia-p25-issi-" mfid
mfid = 2HEX DIGIT

The **Accept** header field SHALL always contain the following body type descriptors (i.e., **accept-range** values in the aforementioned basic parsing construct):

application/sdp;level=1
application/x-tia-p25-issi

In the event that the RFSS supports TIA-102 Supplementary Data Services, this header field SHALL also contain the following body type descriptor:

`application/x-tia-p25-supdat`

In the event that the RFSS supports TIA-102 Packet Data Services, this header field SHALL also contain the following body type descriptor:

`application/x-tia-p25-sndcp`

In the event that manufacturer-specific bodies are acceptable, this header field MAY also contain the following body type descriptor:

`application/x-tia-p25-issi-<mfid>`

where mfid is the manufacturer's identifier as specified in [102BAAC].

~~mfid = 2HEX-DIGIT~~

Examples:

Accept: application/sdp;level=1, application/x-tia-p25-issi

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-sndcp

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-issi-44

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-sndcp, application/x-tia-p25-issi-44

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-supdat

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-supdat, application/x-tia-p25-issi-44

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-sndcp, application/x-tia-p25-supdat

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-sndcp, application/x-tia-p25-supdat,
application/x-tia-p25-issi-44

3.5.2 Allow

[Replace existing text in this section of [102BACA] with the text below.]

The Allow header field (with ABNF notation as specified in Section 25.1 of [RFC3261]) indicates all SIP methods supported by the server for ISSI.

This header field SHALL include following SIP methods:

ACK, BYE, CANCEL, INVITE, REGISTER

In the event that the RFSS supports TIA-102 Supplementary Data Services, this header field SHALL also include:

MESSAGE

In the event that the RFSS supports the SIP OPTIONS method, this header field SHALL also include:

OPTIONS

Example:

Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE,
OPTIONS

3.5.3 Call-ID

[Replace existing text in this section of [102BACA] with the text below.]

As stated in [RFC3261], all SIP UAs need a means of guaranteeing that the Call-ID header fields they produce are not inadvertently generated by any other UA. Any ISSI UA SHOULD NOT generate a Call-ID already generated by itself or by any other UA, unless overridden by method-specific (e.g., INVITE Method specific) behavior, as is the case for SIP call control over the ISSI. Indeed, for any given SU-to-SU call or group call over the ISSI, Call-IDs for the various call segments MAY or MAY NOT be the same.

Call-IDs are case-sensitive and are simply compared byte-by-byte, in accordance with sections 8.1.1.4 and 20.8 of [RFC3261].

The construction of the Call-ID is manufacturer-dependent.

The following is an example construction of the Call-ID, provided here for information only:

Call-ID	=	("Call-ID:" / "i:") SP callid CRLF
callid	=	timestamp "." id-counter "@p25dr"
timestamp	=	8HEX-DIGIT
id-counter	=	8HEX-DIGIT

where

- timestamp is the 8 hexadecimal characters of the 32 LSB bits of the 64 bits timestamp. The 64 bits timestamp is the host setup time relative to "January, 1st 1970 – 0h" expressed in seconds.
- id-counter MAY be chosen using a manufacturer-dependent algorithm. For instance, this field MAY be the 8 hexadecimal characters of the 32-bit integer value set to 0 at the host set-up and incremented by 1 modulo 2^{32} each time the RFSS needs a new CALL-ID identifier for generating a SIP REGISTER or a SIP INVITE.

Example:

Call-ID: 431ff9cd.0008c0f3@p25dr
 i: 431ff9cd.0008c0f3@p25dr (Compact form)

In the example above, the host has been reset to 01/01/1970-0h + 1 126 169 037 s (=431ff9cd h).

3.5.4 Contact

[No change from existing text in this section of [102BACA].]

3.5.5 Content-Disposition

[No change from existing text in this section of [102BACA].]

3.5.6 Content-Length

[No change from existing text in this section of [102BACA].]

3.5.7 Content-Type

[Replace existing text in this section of [102BACA] with the text below.]

If present (i.e. if a message body is present), this header field indicates the type of the body present in the message.

For SIP messages with multiple bodies, the Content-Type header field SHALL have the following construct:

Top-Level Content-Type
 :
Boundary Delimiter Line
Body-specific Content Type

Body Part
Boundary Delimiter Line
Body-specific Content Type
Body Part
 : (repeat the previous 3 components as needed)
Final Boundary Delimiter Line

For SIP messages with only one body, the Content-Type header field SHALL have the following construct:

Body-specific Content Type
 :
Body Part

ABNF notations for components of the above constructs are as follows:

Top-Level Content-Type

Referring to the ABNF specifications for the SIP protocol in Section 25.1 of [RFC3261], the basic parsing construct for the top-level Content-Type header field SHALL be as follows:

Content-Type = ("Content-Type" / "c") HCOLON media-type
 media-type = m-type SLASH m-subtype SEMI m-parameter
 m-type = composite-type
 composite-type = "multipart"
 m-subtype = extension-token
 extension-token = ietf-token
 ietf-token = "mixed"
 m-parameter = m-attribute EQUAL m-value
 m-attribute = "boundary"
 m-value = quoted-string
 quoted-string = SWS DQUOTE qdtext DQUOTE
 qdtext = "P25 ISSI body boundary"

Boundary Delimiter Line

The boundary delimiter line SHALL have syntax as specified in Section 5.1.1 of [RFC2046], using the following unique string to bind the bodies in all the ISSI messages:

P25 ISSI body boundary

Body-Specific Content-Type

Referring to the ABNF specifications for the SIP protocol in Section 25.1 of [RFC3261], the basic parsing construct for the body-specific Content-Type header field SHALL be as follows:

Content-Type = ("Content-Type" / "c") HCOLON media-type
 media-type = m-type SLASH m-subtype
 m-type = discrete-type
 discrete-type = "application"
 m-subtype = extension-token
 extension-token = ietf-token / x-token
 ietf-token = "sdp"
 x-token = issi-token / supdat-token / sndcp-token / mfid-token
 issi-token = "x-tia-p25-issi"
 supdat-token = "x-tia-p25-supdat"
 sndcp-token = "x-tia-p25-sndcp"
 mfid-token = "x-tia-p25-issi-" mfid
 mfid = 2HEX DIGIT

Body Part

For SDP body type, the ABNF notation SHALL be as specified in [RFC4566]

For the x-tia-p25-issi, x-tia-p25-supdat, and x-tia-p25-sndcp body types, the basic parsing construct SHALL be as follows:

Body Part = 1*(m-parameter CRLF)
 m-parameter = m-attribute ":" m-value
 m-attribute = token
 m-value = token

Final Boundary Delimiter Line

The final boundary delimiter line following the last body part SHALL be identical to the previous boundary delimiter lines, with the addition of "--" after the character string, in accordance with Section 5.1.1 of [RFC2046].

A TIA-102 ISSI supports the SDP body type, the x-tia-p25-issi body type, the x-tia-p25-supdat body type, and the x-tia-p25-sndcp body type. See [102BACD]

for the definition and use of the x-tia-p25-supdat body type when using the MESSAGE method for Supplementary Data.

Each manufacturer MAY add its own body with a name composed of the radical name "x-tia-p25-issi-" followed by the two hexadecimal digits of the manufacturer's <mfid>. The <mfid> values are defined in [102BAAC].

~~man-body = "x-tia-p25-issi-" mfid~~
~~mfid = 2HEX-DIGIT~~

If an RFSS receives a manufacturer-specific body with an mfid that it does not support, then the RFSS SHOULD ignore the received manufacturer-specific body, and SHOULD return a GEN_IGNORE_MANUF_BODY 2xx response (refer to Annex A of this document).

All these bodies MAY be present in the same message. In this case, the top-level Content-type header field in the message SHALL be set as follows in order to specify the literal string, which is used as a boundary delimiter line between the bodies:

~~Content-Type: multipart/mixed; boundary="P25 ISSI body boundary"~~
~~c: multipart/mixed; boundary="P25 ISSI body boundary" (Compact form)~~

~~The boundary delimiter line can be any character string (1-70 characters) beginning with " " ([RFC2046]). The string SHALL NOT appear in any of the ISSI SIP message bodies but can be different from one message to another. However, in this specification, a unique string is used to bind the bodies in all the ISSI messages:~~

~~P25 ISSI body boundary~~

~~The boundary delimiter line following the last body part is identical to the previous boundary delimiter lines, with the addition of " " after the character string.~~

A Content type indicating a SDP body if present SHALL be set as follows:

~~Content-Type: application/sdp~~
~~c: application/sdp (Compact form)~~

A Content type indicating a x-tia-p25-supdat body type if present SHALL be set as follows:

~~Content-Type: application/x-tia-p25-supdat~~
~~c: application/x-tia-p25-supdat (Compact form)~~

A Content type indicating a x-tia-p25-sndcp body type if present SHALL be set as follows:

```
Content-Type: application/x-tia-p25-sndcp
c: application/x-tia-p25-sndcp (Compact form)
```

A Content type indicating a x-tia-p25-issi body type if present SHALL be set as follows:

```
Content-Type: application/x-tia-p25-issi
c: application/x-tia-p25-issi (Compact form)
```

A Content type indicating a manufacturer-specific body for an mfid of %x07, if present is set as follows:

```
Content-Type: application/x-tia-p25-issi-07
c: application/x-tia-p25-issi-07 (Compact form)
```

Table 15 shows the Content header settings for various SIP message types over the ISSI, for SU, SG and SN Context mobility objects.

Table 15 - Content header field used for ISSI

Message	Direction	Presence of the bodies
REGISTER message for an SU or SG with force or confirm action required	Home RFSS => Serving RFSS	P25-ISSI body
REGISTER message for an SU registration request	Serving RFSS => Home RFSS	P25-ISSI body
200 OK for a REGISTER message requesting SU service profile parameters	Home RFSS => Serving RFSS	P25-ISSI body and up to 14 P25-SNDCP bodies
200 OK for a REGISTER message requesting SG service profile parameters	Home RFSS => Serving RFSS	P25-ISSI body
REGISTER message for a PD SN Context Registration Request (with "SU Presence" parameter value set to "Confirmed")	Serving RFSS => Home RFSS	P25-ISSI body P25-SNDCP body
200 OK for a PD SN Context Registration Request ((with "SU Presence" parameter value set to "Confirmed")	Home RFSS => Serving RFSS	P25-SNDCP body
200 OK for an SU Serving Query Request	Home RFSS => Serving RFSS	P25-ISSI body and up to 14 P25-SNDCP bodies
200 OK for a PD SN Context Serving Query Request	Home RFSS => Serving RFSS	P25-SNDCP body
REGISTER message for a PD SN Context Home Query request (with "force" qualifier)	Home RFSS => Serving RFSS	P25-ISSI body P25-SNDCP body
200 OK for a PD SN Context Home Query request	Serving RFSS => Home RFSS	P25-SNDCP body
INVITE Request for an SU-to-SU Call	Any	SDP and P25-ISSI bodies
200 OK acknowledging an INVITE Request for an SU-to-SU Call	Any	SDP and P25-ISSI bodies
INVITE/Re-INVITE Request for a confirmed group call	Any	SDP and P25-ISSI bodies
INVITE/Re-INVITE Request for an unconfirmed group call	Any	SDP body and optionally P25-ISSI body
200 OK acknowledging an INVITE/Re-INVITE Request for a confirmed group call	Any	SDP and P25-ISSI bodies
200 OK acknowledging an INVITE/Re-INVITE Request for an unconfirmed group call	Any	SDP body and optionally P25-ISSI body
OPTIONS request for an RFSS Service Capability Polling procedure	Polling RFSS => Polled RFSS	P25-ISSI body
200 OK acknowledging an OPTIONS Request for an RFSS Service Capability Polling procedure	Polled RFSS => Polling RFSS	P25-ISSI body

For all the other SIP messages, if there is no manufacturer-specific body, the Content header field is not present.

Example:

```
Content-Type: multipart/mixed; boundary="P25 ISSI body boundary"

--P25 ISSI body boundary
Content-Type: application/sdp
...
--P25 ISSI body boundary
Content-Type: application/x-tia-p25-issi
...
--P25 ISSI body boundary
Content-Type: application/x-tia-p25-issi-18
...
--P25 ISSI body boundary--
```

3.5.8 Cseq

[Replace existing text in this section of [102BACA] with the text below.]

This header field is defined in [RFC3261]. It SHALL be present in all SIP messages.

The method part of CSeq is case-sensitive, in accordance with Section 20.16 of [RFC3261].

Example:

```
Cseq: 153 INVITE
```

3.5.9 Expires

[No change from existing text in this section of [102BACA].]

3.5.10 From

[No change from existing text in this section of [102BACA].]

3.5.11 Max-Forwards

[No change from existing text in this section of [102BACA].]

3.5.12 MIME-Version

[No change from existing text in this section of [102BACA].]

3.5.13 Priority

[No change from existing text in this section of [102BACA].]

3.5.14 Record-Route

[No change from existing text in this section of [102BACA].]

3.5.15 Route

[No change from existing text in this section of [102BACA].]

3.5.16 To

[No change from existing text in this section of [102BACA], except for Subsection 3.5.16.1 below.]

3.5.16.1 To Header Field for REGISTER

[No change from existing text in this section of [102BACA], except for selected subsections thereof, as shown below.]

3.5.16.1.1 To field for SUID (a(ii)) and SGID registration

[Modify the title of this section of [102BACA] as shown above, and replace existing text in this section with the text below.]

A Serving RFSS registering an SUID or SGID SHALL use form:

To = ("To:" / "t:") SP sd-to-spec [tag] CRLF
 sd-to-spec = "<sip:" g-or-u ">"
 tag = tag-param
 g-or-u = grp / unit
 grp = sg-id "@p25dr;user=TIA-P25-SG"
 unit = su-id "@p25dr;user=TIA-P25-SU"

For example, for SUID 00001.002.000003 and no tag the field would appear as:

t: <sip:00001002000003@p25dr;user=TIA-P25-SU>

and for SGID 00001.002.0003 and no tag, the field would appear as:

t: <sip:000010020003@p25dr;user=TIA-P25-SG >

3.5.16.1.2 To field for Group Supplementary Data Interest ~~(a(i))~~
[Modify the title of this section of [102BACA] as shown above.]

3.5.16.1.3 To field for FNE Group Supplementary Data Interest ~~(e(i))~~
[Modify the title of this section of [102BACA] as shown above.]

3.5.16.1.4 To field for FNE Unit Supplementary Data Interest ~~(e(ii), e(iii))~~
[Modify the title of this section of [102BACA] as shown above.]

3.5.16.1.5 To field for Group SD Monitoring Interest ~~(b(i.1))~~
[Modify the title of this section of [102BACA] as shown above.]

3.5.16.1.6 To field for FNE Group SD Monitoring Interest ~~(b(i.2))~~
[Modify the title of this section of [102BACA] as shown above.]

3.5.16.1.7 To field for Unit SD Monitoring Interest ~~(b(ii), b(iii))~~
[Modify the title of this section of [102BACA] as shown above.]

3.5.16.1.8 To field for PD SN Context Registration
[No change from existing text in this section of [102BACA].]

3.5.16.2 To Header Field for INVITE, CANCEL, ACK, and BYE
[No change from existing text in this section of [102BACA].]

3.5.16.3 To Header Field for OPTIONS
[No change from existing text in this section of [102BACA].]

3.5.16.4 To Header Field for MESSAGE
[No change from existing text in this section of [102BACA].]

3.5.17 Via
[No change from existing text in this section of [102BACA].]

3.5.18 Warning

[Replace existing text in this section of [102BACA] with the text below.]

This optional header field is used in SIP-responses to carry specific P25 Condition Codes for the ISSI application.

The Warning header field contains a 3-digit code, a host name and a warning text. The 1xx and 2xx warnings are used for HTTP, whereas 3xx warnings are SIP-specific. The 399 warning code specifies a "Miscellaneous warning". This code is used in the ISSI application to transmit the specific P25 Condition Codes. The warning text contains both the specific P25 Condition Code and its associated text. The host name is filled with the rfss name where the ISSI condition occurs.

Warning is set as follows for the 399 code:

Warning	=	"Warning: 399" SP warning-content CRLF
warning-content	=	rfss-name "p25dr" SP warning-role
warning-role	=	DQUOTE p25-warning [":" SP role] DQUOTE
p25-warning	=	p25-warn [":" SP p25-condition-text]
p25-warn	=	"TIA-P25-ISSI-" p25-condition-code
p25-condition-code	=	2HEX-DIGIT
p25-condition-text	=	qdtex

Where

rfss-name is defined in Section 3.4.1.1

qdtex is as defined in **[RFC3261]**, except that it SHALL NOT contain any colon.

role = "Home" / "Serving" / "CalledHome" / "CalledServ" / "CallingHome" /
 "CallingServ" / "GroupHome" / "GroupServ" / "FNEhome" /
 "FNEserv" / "Listening" / "Unknown"

The "role" field is supplied by the RFSS generating the ISSI condition, and reflects its understanding of its role in the subject transaction. The "role" SHALL be included in Warnings generated for Supplementary Data transactions, and MAY be included in all ISSI transactions.

Example:

Warning: 399 5E.7AC.003A1.p25dr "TIA-P25-ISSI-28: Colliding Call: CalledHome"

In addition to the generic SIP responses specified in Annex A of this document, Table 24 summarizes the SIP responses, associated P25 Condition Codes, with mnemonic references used throughout this document, and their specific usage in voice and mobility management services over the ISSI. Note that P25 Condition Codes for Supplementary Data are reserved here, and are defined in **[102BACD]**.

Each mnemonic has a 3-character prefix indicating the applicable TIA-102 services for the referenced SIP response P25 Condition Code:

- MME : TIA-102 Mobility Management Services
- PDE : TIA-102 Packet Data Services
- VCE : TIA-102 Voice Services

When a manufacturer wishes to report an ISSI condition that is not covered by the standard, they SHOULD employ the GEN_MANUF_DISCRETION warning as defined in Annex A of this document and include appropriate text to describe the ISSI condition.

If a P25 Condition Code is Mandatory (according to the “Inclusion of P25 Condition Code in SIP Response” column in Table 24), then it SHALL be included for the specified SIP response. If a P25 Condition Code is Optional, then it MAY be included for the specified SIP response. If a manufacturer chooses to include an optional P25 Condition Code for any specified SIP response, the P25 Condition Code as specified in Table 24 SHALL be used.

IMPORTANT NOTE: Before assigning a new P25 Condition Code to any SIP response, check the list of P25 Condition Codes already defined in Table 24 (see below) and Table 57 (Generic SIP Responses from an RFSS over the ISSI) in this document. This is to avoid duplicate and/or conflicting assignment of P25 Condition Codes.

Table 24 - P25 Condition Code Summary

P25-condition-code	Mnemonic	P25-condition-text	SIP Response	Usage	Inclusion of P25 Condition Code in SIP Response
%x0A	VCE_NO_RTP	"No RTP Resource"	606 NOT ACCEPTABLE	U2U Call: The Unit to Unit Call MAY NOT be set-up due to a lack of RTP resource	Optional
%x0B	VCE_NO_RF	"No RF Resource"	606 NOT ACCEPTABLE	U2U Call: The Unit to Unit Call MAY NOT be set up due to a lack of RF resource	Optional
%x0C	VCE_VOCODER_MODES_UNACCEPTED	"Vocoder modes unaccepted"	606 NOT ACCEPTABLE	U2U or Group Call: None of the vocoder modes offered by the adjacent RFSS is accepted by this RFSS	Mandatory
%x13	VCE_CALLING_SU_INHIBITED	"Calling SU radio inhibited"	403 FORBIDDEN	U2U Call: The Calling Home RFSS detects that the Calling SU is radio inhibited.	Optional
%x14	VCE_CALLING_SU_NOT_REG	"Calling SU not registered"	403 FORBIDDEN	U2U Call: The Calling Home RFSS detects that the Calling Serving RFSS is not registered for the Calling SUID	Mandatory
%x15	VCE_CALLED_SU_NOT_REG	"Called SU not registered"	403 FORBIDDEN	U2U Call: The Called Home RFSS detects that the Called SUID is not registered	Optional
%x16	VCE_CALLING_SU_NOT_AUTHOR	"Calling SU not authorized"	403 FORBIDDEN	U2U Call: The Calling Home RFSS detects that the Calling SU is not authorized to make this call	Optional
%x17	VCE_CALLED_SU_NOT_AUTHOR	"Called SU not authorized"	403 FORBIDDEN	U2U Call: The Called Home RFSS detects that the Called SU is not authorized to respond to this call	Optional
%x18	MME_SU_REG_DISALLOWED	"SU not authorized"	403 FORBIDDEN	Unit registration: The Home RFSS of the SU disallows the registration of this unit	Optional
%x19	VCE_SG_NOT_REG	"SG not registered"	403 FORBIDDEN	Group Call: SG's Home RFSS detects that the Serving RFSS is not registered for the SGID	Mandatory
%x1A	MME_SG_REG_DISALLOWED	"SG not authorized"	403 FORBIDDEN	Group Registration: The SG's Home RFSS disallows the registration of this group	Optional

P25-condition-code	Mnemonic	P25-condition-text	SIP Response	Usage	Inclusion of P25 Condition Code in SIP Response
%x1B	MME_MUST_CONFIRM_SU_PRESENCE	"SU presence confirmation required"	403 FORBIDDEN	Unit Registration: The Home RFSS of the SU denies the "SU Presence Not Confirmed" registration because the SU is currently registered at a different Serving RFSS. SN Context registration: The Home RFSS of the SU detects that the Serving RFSS is not registered for the SUID	Optional for unit registration Mandatory for SN Context registration
%x1D	VCE_CALLED_SU_INHIBITED	"Called SU radio inhibited"	403 FORBIDDEN	U2U Call: The Called Home RFSS detects that the Called SU is radio inhibited.	Optional
%x1E	VCE_CALLING_SU_UNKNOWN	"Unknown Calling SU"	404 NOT FOUND	U2U Call: the Calling Home RFSS detects the Calling SU is unknown	Optional
%x1F	VCE_CALLED_SU_UNKNOWN	"Unknown Called SU"	404 NOT FOUND	U2U Call: the Called Home RFSS detects the Called SU is unknown	Optional
%x20	VCE_CALLED_HOME_UNKNOWN	"Unknown Called Home RFSS ID"	404 NOT FOUND	U2U Call: The Calling Home RFSS detects that the RFSS ID of the Called Home RFSS is not known	Optional
%x21	MME_SG_UNKNOWN	"Unknown target group"	404 NOT FOUND	Group registration: the Home or the Serving RFSS detects that the SG is unknown	Mandatory only when this ISSI condition is originated from the Serving RFSS.
%x21	VCE_SG_UNKNOWN	"Unknown target group"	404 NOT FOUND	Group Call: the Home or the Serving RFSS detects that the SG is unknown	Mandatory
%x22	MME_SU_UNKNOWN	"Unknown SU"	404 NOT FOUND	Unit Registration: the Home or the Serving RFSS detects that the SU is unknown SN Context registration: the Home or the Serving RFSS detects that the SU is unknown	Mandatory only when this ISSI condition is originated from the Serving RFSS.
%x23	VCE_SU_NOT_REG_BY_CALLED_SERV	"Called Serving RFSS has not registered Called SUID with Called Home RFSS"	404 NOT FOUND	U2U Call: The Called Serving RFSS detects that it has not registered the called SUID with the Called Home RFSS.	Mandatory

P25-condition-code	Mnemonic	P25-condition-text	SIP Response	Usage	Inclusion of P25 Condition Code in SIP Response
%x28	VCE_COLLIDING_CALL	"Colliding Call"	603 DECLINE	U2U Call: The Call is rejected because the same call is proceeding in the opposite direction Group Call: Home RFSS (resp. Serving RFSS) rejects the INVITE request from the Serving RFSS (resp. Home RFSS) because it is already inviting the Serving RFSS (resp. Home RFSS)	Optional
%x29	VCE_CALLED_SU_BUSY	"Called SU is busy"	486 BUSY HERE	U2U Call: The Call is rejected because the Called SU is busy or does not respond.	Optional
%x2A	VCE_CALLED_SU_REFUSE_CALL	"Called SU refuses the call"	486 BUSY HERE	U2U Call: The Call is rejected because the Called SU has cancelled the call while it was waiting for RF resource availability.	Optional
%x2B	VCE_CANNOT_PROTECT_AUDIO	"Protected audio call not granted"	403 FORBIDDEN	U2U or Group Call: The Called Home or SG Home RFSS detects that a request for a protected audio call cannot be granted.	Optional
%x2C	VCE_CANNOT_UNPROTECT_AUDIO	"Unprotected audio call not granted"	403 FORBIDDEN	U2U or Group Call: The Called Home or SG Home RFSS detects that a request for an unprotected audio call cannot be granted.	Optional
%x2D	MME_OBJECT_UNKNOWN	"Unknown Mobility Object"	404 NOT FOUND	Registration of Mobility Object: the Home or the Serving RFSS detects that the mobility object is unknown	Mandatory only when this ISSI condition is originated from the Serving RFSS.
%x2E	MME_OBJECT_REG_DISALLOWED	"Registration of mobility object disallowed"	403 FORBIDDEN	Registration of Mobility Object: Home RFSS of the mobility object disallows registration of this object	Optional
%x2F	MME_SU_AUTHEN_REQUIRED	"SU Requires Authentication"	403 FORBIDDEN	Unit Registration: The Home RFSS of the SU denies the registration because the SU requires Subscriber Authentication	Mandatory
%x50	VCE_NO_RF_AT_CALLED_SERV	"RF resource not available"	183 SESSION PROGRESS	U2U call: the called RFSS has no RF resource	Optional
%x55	MME_PROCESSING_HOME_QUERY	"Query processing in progress"	183 SESSION PROGRESS	Home Query: to inform the Home RFSS that the Serving RFSS is attempting to fulfill the received query request.	Optional

P25-condition-code	Mnemonic	P25-condition-text	SIP Response	Usage	Inclusion of P25 Condition Code in SIP Response
%x56	MME_SU_INFO_NOT_AUTHOR	"SU info not authorized"	403 FORBIDDEN	Serving Query: to inform the Serving RFSS that the Serving RFSS is not authorized to receive information about the subject SU.	Optional
%x57	MME_OBJ_NOT_REG_AT_SERV	"Mobility object is not registered at this Serving RFSS"	403 FORBIDDEN	Serving Deregistration: Home RFSS detects that the mobility object is not registered at this Serving RFSS.	Optional
%x70-7F		(Reserved for Supplementary Data use, see [102BACD])			Optional
%x82	PDE_SU_NOT_AUTHOR	"SU not authorized to perform packet data services"	403 FORBIDDEN	SN Context registration: The Home RFSS of the SU detects that the SU is not authorized to perform packet data services	Optional
%x83	PDE_STAT_IP_ADDR_INCORRECT	"Incorrect Static IP address"	403 FORBIDDEN	SN Context registration: The Home RFSS of the SU detects that the static IP address provided by the SU is incorrect	Optional
%x84	PDE_STAT_IP_ADDR_UNSUPPORTED	"Static IP address not supported"	403 FORBIDDEN	SN Context registration: The Home RFSS of the SU detects that static IP address assignment is not supported	Optional
%x85	PDE_DYN_IP_ADDR_POOL_EMPTY	"Dynamic IP address pool empty"	403 FORBIDDEN	SN Context registration: The Home RFSS of the SU detects that there are no more available IP addresses for a dynamic IP address assignment	Optional
%x86	PDE_DYN_IP_ADDR_UNSUPPORTED	"Dynamic IP address not supported"	403 FORBIDDEN	SN Context registration: The Home RFSS of the SU detects that dynamic IP address assignment is not supported	Optional

3.6 Status-Code and Reason-Phrase

[No change from existing text in this section of [102BACA].]

3.7 SDP Announcements

[No change from existing text in this section of [102BACA], except for subsections 3.7.2 and 3.7.3 below.]

3.7.1 SDP offer/answer usage

[No change from existing text in this section of [102BACA].]

Limited Use Copy

3.7.2 SDP parameters encoding

[No change from existing text in this section of [102BACA], except for subsections 3.7.2.1 and 3.7.2.2 below.]

3.7.2.1 Session-level part

[No change from existing text in this section of [102BACA], except for Subsection 3.7.2.1.2 below.]

3.7.2.1.1 The protocol version field

[No change from existing text in this section of [102BACA].]

3.7.2.1.2 The origin field

[Replace existing text in this section of [102BACA] with the text below.]

This field gives the originator of the media session. It is expressed as follows:

origin-field = "o=" username SP session-id SP session-version SP
network-type SP address-type SP address CRLF

[RFC4566] requires that (username, session-id, network-type, address-type, address) forms a globally unique identifier for the SDP session. As (network-type, address-type, address) identifies the host where the session is created, (username, session-id) has to identify the session within the host. A void identifier is used for the username, whereas the session-id distinguishes the sessions.

then

username	SHOULD be set to "-"
session-id	is a numeric field which identifies the session. To ensure uniqueness, a timestamp or an id-counter is suggested.
session-version	the version of the announcement. [RFC4566] suggests increasing the version when a modification is made to the session data. A Network Time Protocol (NTP) format timestamp is recommended in [RFC4566] , a timestamp as used in the session-id is suggested.
network-type	is set to "IN"
address-type	is set to "IP4"
address	is the IP address

username	=	"-"
session-id	=	1*DIGIT

session-version	=	1*DIGIT
network-type	=	"IN"
address-type	=	("IP4")
address	=	IP4-address

The IP4-address is described in **[RFC4566]**.

Example:

o-- 43258265 43258376 IN IP4 135.24.98.126

3.7.2.1.3 The session name field

*[No change from existing text in this section of **[102BACA]**.]*

3.7.2.1.4 The time field

*[No change from existing text in this section of **[102BACA]**.]*

3.7.2.1.5 The connection field

*[No change from existing text in this section of **[102BACA]**.]*

3.7.2.2 Media description part

*[No change from existing text in this section of **[102BACA]**, except for Section 3.7.2.2.2 below.]*

3.7.2.2.1 The media announcement field

*[No change from existing text in this section of **[102BACA]**.]*

3.7.2.2.2 Attributes fields

*[Replace existing text in this section of **[102BACA]** as shown below.]*

The general form of an rtpmap attribute field is:

attribute-field	=	"a=" attribute CRLF
attribute	=	"rtpmap:" payload-type SP encoding-name "/" clock-rate
payload-type	=	"100" / "101" / "102"
encoding-name	=	"X-TIA-P25-IMBE" / "X-TIA-P25-FullRate" / "X-TIA-P25-HalfRate" / "X-TIA-P25-Native"
clock-rate	=	"8000"

The clock rate parameter is set to 8000.

No encoding parameter is defined.

The payload-type and encoding-name in the rtpmap attribute of the media description SHALL be set as follows:

Table 26 - payload-type and encoding-name of the rtpmap attribute

ISSI Vocoder Mode	payload-type	encoding-name
Full Rate	100	X-TIA-P25-FullRate
Half Rate	101	X-TIA-P25-HalfRate
Native	102	X-TIA-P25-Native

Thus

For Full Rate: a=rtpmap:100 X-TIA-P25-FullRate/8000
 For Half Rate: a=rtpmap:101 X-TIA-P25-HalfRate/8000
 For Native: a=rtpmap:102 X-TIA-P25-Native/8000

NOTE: In older RFSS implementations that only support the Full Rate vocoder mode (payload-type of value 100), the encoding-name of the rtpmap attribute is set to "X-TIA-P25-IMBE".

Thus:

a=rtpmap:100 X-TIA-P25-IMBE/8000

3.7.3 SDP examples

[No change from existing text in this section of [102BACA], except for subsections 3.7.3.1 and 3.7.3.2 below.]

3.7.3.1 SDP Offer with ~~Multiple~~ Multiple Vocoder Modes Being Offered

[Modify the title of this section of [102BACA] as shown above, and replace existing text in this section with the text below.]

This example describes a session with the following parameters:

- the session is set up for a group call
- the setup timestamp is 43278265
~~— this is a new session, and it is the first version of the session~~
- IPv4 is used, and the host 192.53.18.122 create the session and handles the data connection on port 49298
- Native, Half Rate, and Full Rate vocoder modes are offered

then the SDP part is set as follows:

v=0

```

o=- 43278265 43278376 IN IP4 192.53.18.122
s=TIA-P25-GroupCall
c=IN IP4 192.53.18.122
t=0 0
m=audio 49298 RTP/AVP 102 101 100
a=rtpmap:102 X-TIA-P25-Native/8000
a=rtpmap:101 X-TIA-P25-HalfRate/8000
a=rtpmap:100 X-TIA-P25-FullRate/8000

```

3.7.3.2 SDP Offer in Older RFSS Implementations

[Replace existing text in this section of [102BACA] with the text below.]

This example describes a session with the following parameters:

- the session is set up for a group call
- the setup timestamp is 43278265
- ~~— this is a new session, and it is the first version of the session~~
- IPv4 is used, and the host 192.53.18.122 create the session and handles the data connection on port 49298
- Full Rate vocoder mode is offered

then the SDP part is set as follows:

```

v=0
o=- 43278265 43278376 IN IP4 192.53.18.122
s=TIA-P25-GroupCall
c=IN IP4 192.53.18.122
t=0 0
m=audio 49298 RTP/AVP 100
a=rtpmap:100 X-TIA-P25-IMBE/8000

```

3.8 X-TIA-P25-ISSI Body Parameters

[Replace existing text in this section of [102BACA], and selected subsections thereof, as shown below.]

This section describes the X-TIA-P25-ISSI application used to carry the User Service Profile (USP) information, Group Service Profile (GSP) information, RFSS service capability information, ISSI specific register parameters information, and ISSI specific call parameters information.

In any transaction, a given X-TIA-P25-ISSI application MAY **only** carry the USP information, or the GSP information, or the RFSS service capability information.

or the ISSI specific register parameters information, or the ISSI specific call parameters information, or a combination of the aforesaid information, as appropriate. The RFSS SHALL simply discard any additional, irrelevant profile or parameter information received, and it SHALL proceed with the transaction. Depending on the implementation, the RFSS MAY generate an error log to locally report this anomaly.

The application/x-tia-p25-issi media type is defined by the following information:

Media type name: application
 Media subtype name: x-tia-p25-issi
 Required parameters: version
 'Version' gives the version of the X-TIA-P25-ISSI protocol. The initial protocol version is set to: "version = 1.0"
 Optional parameters: None

The SIP Content-type header field is set as defined in Section 3.5.7. The X-TIA-P25-ISSI descriptions are entirely textual. The X-TIA-P25-ISSI description consists in one or more lines of text. The order of these lines has no special meaning. The names and parameters of the fields are case-insensitive strings.

Handling of HEX-DIGIT based parameters is as specified in Section 3.1 of this document.

3.8.1 Group Service Profile attributes

[Replace existing text in this section of [102BACA], and selected subsections thereof, as shown below.]

Table 27 summarizes the available attributes for groups, which are detailed in the following subsections.

Table 27 - Group Service Profile encoding in the ISSI body

Name	Field	Compact form	Meaning	Default
Access Permission	g-access	g-as	Defines whether or not the group is allowed to make non-emergency calls.	Non-emergency Calls enabled.
Announcement Group	g-agroup	g-ag	Defines the announcement subscriber group for the group.	No Default
Priority	g-pri	g-p	Defines the priority of the group.	Subject to Local Policy
Emergency Capable	g-ecap	g-ec	Defines whether or not emergency calls MAY be made on the group.	Emergency Calls Enabled.
Emergency Preemption	g-epreempt	g-ep	Defines whether ruthless or non-ruthless pre-emption is used for emergency calls.	Non-ruthless (next available resource)
RF Hang Time	g-rfhangt	g-h	Defines the hang-timer for the group.	0 tenths of seconds (Transmission trunked)
Confirmed Call Setup Time	g-ccsetupT	g-c	Defines the confirmed call setup timer for the group.	0 tenths of seconds (Unconfirmed calls)
Interrupt Mode	g-intmode	g-in	Defines under what conditions talk-group members MAY interrupt each other during a call	Never allowed
Security	g-sec	g-s	Defines the allowable security levels of the call	Clear or Secure
Interconnect Flag	g-ic	g-ic	Defines whether or not the group MAY have PSTN sourced calls	Interconnect calls disallowed
Interconnect Security	g-icsecstart	g-ics	Defines the required security level at the beginning of an interconnect call.	Clear
Group Alias	g-alias	g-als	Defines an alias for the group in order to support interoperable console alias display for CSSI-connected consoles in a multi-vendor deployment..	None
Manufacturer Specific Parameter	g-man<mfid>...	g-m<mfid>..	Allows for manufacturer specific attributes	None

All the Group Service Profile lines begin with “g-“ .

3.8.1.1 Access Permission (g-access)

[Replace existing text in this section of [102BACA] with the text below.]

The Group Access Permission attribute defines whether or not the group is allowed to make non-emergency calls at the Serving RFSS. It has the following format in ABNF notation:

`access-permission = ("g-access:" / "g-as:") boolean CRLF`

where “0” is used to indicate that non-emergency calls on the group MAY NOT be placed from the Serving RFSS, and “1” is used to indicate that non-emergency calls MAY be placed.

If the Access Permission attribute is not present, non-emergency calls MAY be placed from the Serving RFSS.

3.8.1.2 Announcement Group (g-agroup)

[Replace existing text in this section of [102BACA] with the text below.]

The Announcement Group attribute defines the announcement subscriber group for the group. It has the following format in ABNF notation:

`announcement-group = ("g-agroup:" / "g-ag:") 12HEX-DIGIT CRLF`

where the 12HEX-DIGIT represent the fully-qualified hexadecimal form of the AGID.

If no announcement group attribute is included, then the group has no announcement subscriber group.

3.8.1.3 Priority (g-pri)

[Replace existing text in this section of [102BACA] with the text below.]

The Priority attribute defines the suggested priority of the group when queuing group calls on a busy system. The Serving RFSS MAY independently determine the relative priorities of competing groups homed to different systems. However, it SHOULD maintain the relative priorities provisioned by the Home RFSS for competing calls homed to the same system.

NOTE - Operators need to be aware that failing to maintain relative priorities could cause interoperability issues.

The Priority attribute has the following format, in ABNF notation:

group-priority = ("g-pri:" / "g-p:") priority CRLF

priority = "1" / "2" / "3" / "4" / "5" / "6" / "7" / "8" / "9" / "10"

where the priority is restricted to the integer values 1 through 10. Group calls with higher priority values are to be de-queued before group calls with lower priority values.

1 is the lowest priority value.

If no Priority attribute is provided by the Home RFSS, the Serving RFSS MAY assign a priority to the group in accordance with local policy and programming.

3.8.1.4 Emergency Capable (g-ecap)

[Replace existing text in this section of [102BACA] with the text below.]

The Emergency Capable attribute defines whether or not emergency calls can be made on the group. It has the following format, in ABNF notation:

emergency-capable = ("g-ecap:" / "g-ec:") boolean CRLF

where a boolean "0" is used to indicate that emergency calls are not allowed, and a "1" is used to indicate that emergency calls are allowed. If the emergency capable attribute is not provided, emergency calls are allowed from the Serving RFSS.

3.8.1.5 Emergency Preemption (g-eprempt)

[Replace existing text in this section of [102BACA] with the text below.]

The Emergency Preemption attribute indicates the type of pre-emption to be applied for emergency calls. Its ABNF format is as follows:

emergency-preemption = ("g-epreempt:" / "g-ep:") boolean CRLF

where a boolean "0" indicates that non-ruthless preemption is to be used (next available resource) and a boolean "1" indicates that ruthless preemption is to be used. If the emergency preemption attribute is not provided, non-ruthless preemption is to be used.

3.8.1.6 RF Hang Time (g-rfhangt)

[Replace existing text in this section of [102BACA] with the text below.]

The RF Hang Time attribute specifies the RF hang-timer for the group. It has the following format in ABNF notation:

hang-time = ("g-rfhangt:" / "g-h:") rfhangt CRLF

rfhangt = int

where the integer value is the RF hang-timer in tenths of seconds and is limited to the range 0 – 32767. If the RF hang timer attribute is not specified, the RF hang timer is 0 (Transmission Trunked).

The RF hang time has no infinity value.

3.8.1.7 Confirmed Call Setup Time (g-ccsetupT)

[Replace existing text in this section of [102BACA] with the text below.]

The Confirmed Call setup attribute specifies the maximum setup time for a confirmed call on the group. It has the following format in ABNF notation:

cc-setup = ("g-ccsetupT:" / "g-c:") ccT CRLF

ccT = int

This attribute SHALL be used to indicate whether calls on the group are confirmed or unconfirmed, as detailed in Section 3.8.5.6, with integer values of the attribute defined as follows:

- A value of 0 indicates that the call is unconfirmed.
- A value in the range 1-32766 specifies the setup time in tenths of seconds and indicates that the call is confirmed.

- A value of 32767 indicates an infinite setup time, and indicates that the call is confirmed.

This attribute MAY also be used for other purposes in accordance with local policy.

3.8.1.8 Interrupt Mode (g-intmode)

[Replace existing text in this section of [102BACA] with the text below.]

The Interrupt Mode attribute specifies the audio interrupt mode of the voice group. It has the following format in ABNF notation:

interrupt-mode = ("g-intmode:" / "g-in:") imode CRLF

imode = "0" / "1" / "2"

where imode = "0" indicates that audio interruption by SUs within a talk group call is never allowed, "1" indicates that it is allowed based on SU priority, and "2" indicates that it is always allowed. If the Interrupt Mode is not specified, the Interrupt Mode is 0.

3.8.1.9 Group Security Level (g-sec)

[Replace existing text in this section of [102BACA] with the text below.]

The Group Security Level attribute indicates the allowable security levels for the group. It has the following ABNF format:

security-level = ("g-sec:" / "g-s:") security CRLF

security = "1" / "2" / "3"

where

"1"

indicates clear mode calls only are allowed

"2"

indicates that secure mode calls only are allowed, and

"3"

indicates that both clear and secure mode calls are allowed.

If the group security attribute is not provided, both clear and secure mode calls are to be allowed.

3.8.1.10 Interconnect Flag (g-ic)

[Replace existing text in this section of [102BACA] with the text below.]

The Interconnect Flag attribute defines whether or not the group MAY have PSTN sourced calls. It has the following format in ABNF notation:

interconnect = ("g-ic:" / "g-ic:") boolean CRLF

where "0" indicates that PSTN source calls are not allowed, and "1" indicates that they are. If the interconnect flag is not present, interconnect calls are not allowed.

3.8.1.11 Interconnect Security (g-icsecstart)

[Replace existing text in this section of [102BACA] with the text below.]

The Interconnect Security parameter indicates whether PSTN interconnect calls are to start in the clear or in a secure mode. Its ABNF format is as follows:

Interconnect-security = ("g-icsecstart:" / "g-ics:") boolean CRLF

where "0" indicates that interconnect calls are to start in the clear, and "1" indicates that they are to start secure. If the interconnect security attribute is not provided, calls are to be started in the clear.

3.8.1.12 Group Alias (g-alias)

[Add this new section to [102BACA] with the text below.]

The Group Alias attribute defines an alias for the group in order to support interoperable console alias display for CSSI-connected consoles in a multi-vendor deployment. It has the following format in ABNF notation:

group-alias = ("g-alias:" / "g-als:") DQUOTE galias DQUOTE CRLF
galias = UTF8-octets

In the context of the above ABNF notation:

1. UTF8-octets, as defined in Section 4 of **[RFC3629]**, SHALL be a string of UTF-8 characters that constitute the Group Alias.
2. UTF8-octets SHALL not exceed 48 octets in total length.
3. Each UTF-8 character SHALL represent, using a sequence of 1 to 4 octets, a character from the Universal Character Set (UCS) by encoding the UCS character as specified in Section 3 of **[RFC3629]**.
4. Decoding a UTF-8 character SHALL be as specified in Section 3 of **[RFC3629]**.
5. An RFSS SHALL handle Group Aliases that contain non-printable characters in accordance with local policy.

Example:

g-alias:"Precinct37-GroupB2"

3.8.1.13 Manufacturer Specific (g-man<mfid>)

*[Replace existing text in this section of **[102BACA]** with the text below.]*

The Manufacturer Specific attribute allows manufacturer defined profile information that MAY be used for private or experimental purposes. It SHALL have the following radical with the following ABNF notation:

man-grpprof-radical = ("g-man" / "g-m") mfid "-" octet-string ":" pvalue CRLF
 mfid = 2HEX-DIGIT

where pvalue is defined in Section 25.1 of **[RFC3261]**.

The mfid values are defined in **[102BAAC]**.

Manufacturer specific attributes that are not understood by an RFSS SHOULD be ignored.

Example:

g-manA4-myparameter:1

3.8.1.14 Example Group Service Profiles

[No change from existing text in this section of [102BACA], except for Subsection 3.8.1.14.1 below.]

3.8.1.14.1 Fully Specified Example

[Replace existing text in this section of [102BACA] with the text below.]

The following example illustrates a fully specified group service profile using default values (where appropriate) or typical values (where no default is specified).

```
g-access:1
g-agroup:0B4561A27271
g-pri:2
g-ecap:1
g-epreempt:0
g-rfhangt:0
g-ccsetupt:0
g-intmode:0
g-sec:1
g-ic:0
g-icsecstart:0
g-alias:"Precinct37-GroupB2"
```

The same list in compact form:

```
g-as:1
g-ag:0B4561A27271
g-p:2
g-ec:1
g-ep:0
g-h:0
g-c:0
g-in:0
g-s:1
g-ic:0
g-ics:0
g-als:"Precinct37-GroupB2"
```

3.8.1.14.2 Minimum Group Service Profile

The minimum group service profile has no attributes, since all attributes are optional. In this case, the Serving RFSS SHALL use implicit default values for all attributes, except those without default values.

3.8.2 User Service Profile attributes

[Replace existing text in this section of [102BACA], and selected subsections thereof, as shown below.]

Table 28 summarizes the available attributes for users, which are detailed in the following subsections.

Limited Use Copy

Table 28 - User Service Profile encoding in the ISSI body

Name	Field	Compact form	Meaning	Default
System Access Permission	u-access	u-as	Indicates the type of access allowed to the unit on the Serving RFSS.	Full Access
Duplexity	u-dup	u-d	Indicates whether the unit is full or half-duplex.	Half-duplex
Secure Capable	u-sec	u-s	Indicates whether or not the SU is capable of secure calls.	No
Group Call Capability	u-gcall	u-g	Indicates the types of group calls that the SU MAY make.	All
U2U Call Permissions	u-ucall	u-uc	Indicates the SU's permissions for unit-to-unit calls.	None
U2U Call Priority	u-upri	u-p	Indicates SU's priority in unit to unit calls.	Subject to local policy
Interconnect Permission	u-iccall	u-icc	Indicates the types of interconnect calls the SU MAY participate in.	None
Interconnect Secure	u-icsec	u-ics	Indicates acceptable security modes for interconnect calls	Clear or Secure
Interconnect Priority	u-icpri	u-icp	Priority level when involved in interconnect calls	Subject to local policy
Authentication Type/Capability	u-authtype	u-at	Authentication Type	Unit does NOT support Authentication
Authentication Policy	u-authpol	u-apo	Authentication Policy	Serving RFSS MAY authenticate the unit.
Authentication Parameters	u-authparm	u-apa	Authentication Parameters	None
Availability Check	u-availcheck	u-ac	Indicates the ability of the SU to perform availability check	Both Availability Check and Direct Call
Call Set-Up preference	u-prefsetup	u-prf	Indicates the calling SU preference to make an availability check or a direct call of the Called SU when it sets up an SU-to-SU call.	Availability Check
Radio Inhibit	u-radioinhib	u-rdinh	Indicates whether or not the SU is radio inhibited [102AABD].	Not inhibited

Name	Field	Compact form	Meaning	Default
Packet Data Capable	u-packetdata	u-pd	Indicates whether the SU is TIA-102 Packet Data capable	No
SU Alias	u-alias	u-als	Defines an alias for the SU in order to support interoperable console alias display for CSSI-connected consoles in a multi-vendor deployment.	None
Manufacturer Specific Parameter	u-man<mfid>...	u-m<mfid>...	Allows for manufacturer specific attributes	None

All the User Service Profile lines begin with “u-“.

3.8.2.1 System Access Permission (u-access)

[Replace existing text in this section of [102BACA] with the text below.]

The System Access Permission attribute defines the access permissions for the SU on the Serving RFSS. It has the following format:

system-access = (“u-access:” / ”u-as:”) access CRLF

access = “0” / “1” / “2”

where access of

“0”

“1”

“2”

indicates no access is allowed,
indicates that full access is allowed, and
indicates that only emergency access is allowed.

If no system access permission is specified, full access is allowed subject to the local policies and programming of the Serving RFSS.

3.8.2.2 Duplexity (u-dup)

[Replace existing text in this section of [102BACA] with the text below.]

The Duplexity attribute indicates whether or not the unit is full duplex. It has the following ABNF format:

duplexity = ("u-dup:" / "u-d:") boolean CRLF

where a boolean value of "0" indicates a half-duplex SU, and a value of "1" indicates a full-duplex SU. If the duplexity attribute is not specified, the unit is half-duplex.

3.8.2.3 Secure Capable (u-sec)

[Replace existing text in this section of [102BACA] with the text below.]

The Secure Capable attribute indicates whether or not the unit is capable of secure (encrypted) operation. It has the following ABNF format:

secure = ("u-sec:" / "u-s:") boolean CRLF

where a boolean value of "0" indicates no secure capability, and a value of "1" indicates an encryption capability. If the secure capable attribute is not specified, the unit is not capable of secure operation.

3.8.2.4 Group Call Permissions (u-gcall)

[Replace existing text in this section of [102BACA] with the text below.]

The Group Call Capability attribute specifies the types of group calls allowed that the given SU MAY initiate. Its format is as follows:

group-permissions = ("u-gcall:" / "u-g:") gcap CRLF

gcap = "0" / "1" / "2" / "3"

where gcap of

"0"

indicates no group call permission

"1"

indicates Non-emergency group call permission

"2"

indicates Emergency group call permission, and

"3"

indicates both emergency and non-emergency permission.

If the capability is not specified, the unit is given both emergency and non-emergency permissions.

3.8.2.5 Unit-to-unit Call Permissions (u-ucall)

[Replace existing text in this section of [102BACA] with the text below.]

The Unit-to-unit Call Permissions (u-ucall) indicates the types of unit-to-unit calls in which the SU MAY participate. Its format is:

unitcall-permissions = ("u-ucall:" / "u-uc:") ucap CRLF

ucap = "0" / "1" / "2" / "3"

where ucap of

"0"

indicates no unit-to-unit call permission

"1"

indicates receive permission

"2"

indicates permission to initiate a unit-to-unit call

"3"

indicates permission to both receive and initiate a unit-to-unit call.

If no unit-to-unit call permission is provided, the unit has no permission to participate in unit-to-unit calls.

3.8.2.6 Unit-to-unit Call Priority (u-upri)

[Replace existing text in this section of [102BACA] with the text below.]

The unit-to-unit Call Priority attribute specifies the priority to use for the individual unit in unit-to-unit calls. Its format is:

unit-priority = ("u-upri:" / "u-p:") priority CRLF

priority = "1" / "2" / "3" / "4" / "5" / "6" / "7" / "8" / "9" / "10"

where the priority is restricted to the integer values 1 through 10. Units with higher priority are more important than units with lower priority values. 1 is the lowest priority value.

If no Priority attribute is provided by the Home RFSS, the Serving RFSS MAY assign a priority to the unit in accordance with local policy and programming.

3.8.2.7 Interconnect Permissions (u-icall)

[Replace existing text in this section of [102BACA] with the text below.]

The Interconnect Permissions attribute specifies the permissions of the unit vis-à-vis interconnect (PSTN) calls. It has the following format:

interconnect-permission = ("u-iccall:" / "u-icc:") icperm CRLF

icperm = "0" / "1" / "2" / "3"

where icperm of

"0"	indicates that the unit does not have permission to participate in interconnect calls,
"1"	indicates that the unit has permission to be the recipient of an interconnect call,
"2"	indicates that the unit has permission to be the initiator of an interconnect call, and
"3"	indicates that the unit has permission to both initiate and receive interconnect calls.

If no interconnect permission attribute is specified, the unit does not have permission to participate in interconnect calls.

3.8.2.8 Interconnect Security (u-icsec)

[Replace existing text in this section of [102BACA] with the text below.]

The Interconnect Security attribute specifies the acceptable security modes of interconnect calls. Its format is as follows:

ic-security = ("u-icsec:" / "u-ics:") icsec CRLF

icsec = "1" / "2" / "3"

where icsec of

"1"	indicates clear mode calls only are allowed
"2"	indicates that secure mode calls only are allowed, and
"3"	indicates that both clear and secure mode calls are allowed.

If no interconnect security attribute is specified, both secure and clear calls are acceptable.

3.8.2.9 Interconnect Call Priority (u-icpri)

[Replace existing text in this section of [102BACA] with the text below.]

The Interconnect Call Priority attribute specifies the priority to use for the individual in interconnect calls. Its format is:

unit-priority = ("u-icpri:" / "u-icp:") priority CRLF

priority = "1" / "2" / "3" / "4" / "5" / "6" / "7" / "8" / "9" / "10"

where the priority is restricted to the integer values 1 through 10. Units with higher priority are more important than units with lower priority values. 1 is the lowest priority value.

If no Priority attribute is provided by the Home RFSS, the Serving RFSS MAY assign a priority to the unit in accordance with local policy and programming.

3.8.2.10 Authentication Type/Capability (u-authtype)

[Replace existing text in this section of [102BACA] with the text below.]

The Authentication Type/Capability attribute specifies the type of authentication to be applied to the unit. The ABNF format of the attribute is:

authentication_type = ("u-authtype:" / "u-at:") atype CRLF

atype = "0" / "1"

where atype of

"0"

indicates that the unit does not support authentication, and,

"1"

indicates that the unit SHALL be authenticated using link layer authentication as specified in [102AACE], subject to the authentication policy specified in Section 3.8.2.11.

If no authentication type is specified, the unit does not support authentication.

3.8.2.11 Authentication Policy (u-authpol)

[Replace existing text in this section of [102BACA] with the text below.]

The Authentication Policy attribute specifies the authentication policy to be applied to the unit. Its ABNF format is:

authentication_policy = ("u-authpol:" / "u-apo:") apol CRLF

apol = "0" / "1" / "2"

where apol of

“0”

indicates that the Serving RFSS MAY authenticate the unit, depending on the availability of the authentication parameters (u-authparm) and the Serving RFSS local policy.

“1”

indicates that the Serving RFSS SHALL authenticate the unit when the home is reachable. If the home is unreachable, the Serving RFSS MAY authenticate the unit, depending on the availability of the authentication parameters (u-authparm) and the Serving RFSS local policy. *If the Home RFSS is reachable, but the Serving RFSS does not have authentication parameters for the unit and is unable to retrieve them from the Home RFSS, the Serving RFSS MAY allow the unit to register in accordance with local policy.*

“2”

indicates that the Serving RFSS SHALL always authenticate the unit. If the Serving RFSS does not have authentication parameters (u-authparm) for the unit and is unable to retrieve them from the home, the Serving RFSS SHALL NOT allow the unit to register.

If no authentication policy is specified and the Authentication Type/Capability parameter is present, then the Serving RFSS MAY authenticate the unit, depending on the availability of the authentication parameters (u-authparm) and the Serving RFSS local policy. The Authentication Policy parameter SHALL be ignored if the Authentication Type/Capability attribute indicates that the unit does not support authentication (i.e. has a value of "0" or is not present).

3.8.2.12 Authentication Parameters (u-authparm)

*[Replace existing text in this section of **[102BACA]** with the text below.]*

The Authentication Parameters attribute specifies the authentication parameters associated with the authentication type specified in the Authentication Type attribute. Its format is:

authentication-parms = ("u-authparm:" / "u-apa:") rs "," ks "," ksprime
CRLF

rs = 10 (SP 2HEX-DIGIT)

ks = 16 (SP 2HEX-DIGIT)

ksprime = 16 (SP 2HEX-DIGIT)

where

rs	is the RS parameter described in [102AACE]
ks	is the KS parameter described in [102AACE], and
ksprime	is the KS prime parameter described in [102AACE].

Due to the rs definition, a space is required after the colon.

rs, ks and ksprime are interpreted octet-wise from left to right. For example, in

u-authparm: 11 12 13 14 15 16 17 18 19 1a, 30 31 32 33 34 35 36 37 38 39 3a
3b 3c 3d 3e 3f, 40 41 42 43 44 45 46 47 48 49 4a 4b 4c 4d 4e 4f

RS[0] = 11 and RS[9] = 1a. Likewise, KS[0] = 30 and KS[15] = 3f, and KS'[0] = 40 and KS'[15] = 4f.

If the authentication type attribute is non-zero, the authentication parameter attribute SHALL be supplied.

3.8.2.13 Availability Check Attribute (u-availcheck)

[Replace existing text in this section of [102BACA] with the text below.]

The Availability Check attribute specifies the ability of the SU to support an Availability Check, a direct call or both. It has the following ABNF notation:

availability-check = ("u-availcheck:" / "u-ac:") type CRLF

type = "1" / "2" / "3"

where

- 1 indicates that the SU supports only Availability Check
- 2 indicates that the SU supports only Direct Call
- 3 indicates that the SU supports both Availability Check and Direct Call

If the “Availability Check” parameter is missing, the Serving RFSS SHALL assume that the calling SU supports both Availability Check and Direct Call.

3.8.2.14 Call Set-Up Preference Attribute (u-prefsetup)

[Replace existing text in this section of [102BACA] with the text below.]

The Call Set-up Preference attribute indicates the calling SU preference to make an availability check or a direct call of the Called SU when it sets up an SU-to-SU call. It has the following ABNF notation:

call-set-up-preference = (“u-prefsetup:” / ”u-prf:”) type CRLF

type = “0” / “1”

where

0 indicates that the Calling SU prefers Availability Check for the Called SU
1 indicates that the Calling SU prefers Direct Call for the Called SU

If “call setup preference” is not provided by the Home RFSS, the Serving RFSS SHALL assume that the calling SU prefers availability check.

3.8.2.15 Radio Inhibit (u-radinhib)

[Replace existing text in this section of [102BACA] with the text below.]

The Radio Inhibit attribute indicates whether the SU is radio inhibited or not [102AABD]. It has the following ABNF notation:

radio-inhibit = (“u-radioinhib:” / ” u-rdinh:”) type CRLF

type = “0” / “1”

where

0 indicates that the SU is not radio inhibited
1 indicates that the SU is radio inhibited

Upon receipt of a “Radio Inhibit” attribute value “1” as part of an SU’s User Service Profile from the Home RFSS, the Serving RFSS SHOULD:

- Consider the SU radio inhibited, and
- Process service requests from the SU in accordance with procedures in [102AABD], and
- Process service requests targeting the SU in accordance with procedures in [102AABD].

If the “Radio Inhibit” attribute is not provided by the Home RFSS, the Serving RFSS SHALL assume that the SU is not radio inhibited.

3.8.2.16 Packet Data Capable (u-packetdata)

[Replace existing text in this section of [102BACA] with the text below.]

The Packet Data Capable attribute indicates whether or not the unit is capable of TIA-102 packet data operation. It has the following ABNF format:

`packetdata = ("u-packetdata:" / "u-pd:") boolean CRLF`

where a boolean value of “0” indicates no TIA-102 packet data capability, and a value of “1” indicates TIA-102 packet data capability. If the Packet Data Capable attribute is not specified, the unit is not capable of TIA-102 packet data operation.

3.8.2.17 SU Alias (u-alias)

[Add this new section to [102BACA] with the text below.]

The SU Alias attribute defines an alias for the SU in order to support interoperable console alias display for CSSI-connected consoles in a multi-vendor deployment. It has the following format in ABNF notation:

`su-alias = ("u-alias:" / "u-als:") DQUOTE ualias DQUOTE CRLF`

`ualias = UTF8-octets`

In the context of the above ABNF notation:

1. UTF8-octets, as defined in Section 4 of **[RFC3629]**, SHALL be a string of UTF-8 characters that constitute the SU Alias.
2. UTF8-octets SHALL not exceed 48 octets in total length.
3. Each UTF-8 character SHALL represent, using a sequence of 1 to 4 octets, a character from the Universal Character Set (UCS) by encoding the UCS character as specified in Section 3 of **[RFC3629]**.
4. Decoding a UTF-8 character SHALL be as specified in Section 3 of **[RFC3629]**.
5. An RFSS SHALL handle SU Aliases that contain non-printable characters in accordance with local policy.

Example:

```
u-alias:"Precinct26-ELsu-H94"
```

3.8.2.18 Manufacturer Specific Attributes (u-man<mfid>)

*[Replace existing text in this section of **[102BACA]** with the text below.]*

The Manufacturer Specific attribute allows manufacturer defined profile information that MAY be used for private or experimental purposes. It SHALL have the following radical with the following ABNF notation:

```
man-usrprof-radical = ("u-man" / "u-m") mfid "-" octet-string ":"
mfid                 = 2HEX-DIGIT
```

The mfid values are defined in **[102BAAC]**.

Manufacturer specific attributes that are not understood by an RFSS SHOULD be ignored.

3.8.2.19 Example User Service Profiles

*[No change from existing text in this section of **[102BACA]**, except for Subsection 3.8.2.19.1 below.]*

3.8.2.19.1 Fully Specified Profile

*[Replace existing text in this section of **[102BACA]** with the text below.]*

The following example illustrates a fully specified user service profile using default values (where appropriate) or typical values (where no default is specified). Note that the unit-to-unit call priority attribute (u-upri) is specified, but would be ignored because unit-to-unit calls are not allowed (u-ucal). Likewise, the interconnect call security (u-icsec) and priority (u-icpri) attributes are specified but would be ignored because interconnect calls are not allowed (u-iccall).

```

u-access:1
u-dup:0
u-sec:0
u-gcall:3
u-ucall:0
u-upri:10
u-iccall:0
u-icsec:1
u-icpri:10
u-authtype:0
u-availcheck:1
u-prefsetup:0
u-radioinhib:0
u-packetdata:0
u-alias:"Precinct26-ELsu-H94"

```

The same list in compact form:

```

u-as:1
u-d:0
u-s:0
u-g:3
u-uc:0
u-p:10
u-icc:0
u-ics:1
u-icp:10
u-at:0
u-ac:1
u-prf:0
u-rdinh:0
u-pd:0
u-als:"Precinct26-ELsu-H94"

```

3.8.2.19.2 Minimal User Service Profile (With Authentication)

The following example illustrates a minimal user service profile for a unit that requires authentication. Note that all attributes can be defaulted except for the authentication fields.

```
u-authtype:1
u-authpol:2
u-authparm: 11 12 13 14 15 16 17 18 19 1a, 30 31 32 33
34 35 36 37 38 39 3a 3b 3c 3d 3e 3f, 40 41 42 43 44 45 46
47 48 49 4a 4b 4c 4d 4e 4f
```

Or in compact form:

```
u-at:1
u-apo:2
u-apa: 11 12 13 14 15 16 17 18 19 1a, 30 31 32 33 34
35 36 37 38 39 3a 3b 3c 3d 3e 3f, 40 41 42 43 44 45 46 47
48 49 4a 4b 4c 4d 4e 4f
```

3.8.3 RFSS Service **Capability** Profile attributes

[Modify the title of this section of [102BACA] as shown above, and replace existing text in selected subsections thereof, as shown below.]

3.8.3.1 Encrypted SU-to-SU Call Capability (r-ucall-sec)

[Replace existing text in this section of [102BACA] with the text below.]

The Encrypted SU-to-SU Call Capability attribute defines whether or not the RFSS is capable of encrypted unit to unit calls at this time. It has the following format in ABNF notation:

encrypted-u2u-call-capability = ("r-ucall-sec:" / "r-us:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of secure SU-to-SU calls, and "1" is used to indicate that the RFSS is capable of secure SU-to-SU calls.

3.8.3.2 Clear SU-to-SU Call Capability (r-ucall-clear)

[Replace existing text in this section of [102BACA] with the text below.]

The Clear SU-to-SU Call Capability attribute defines whether or not the RFSS is capable of SU-to-SU calls in the clear at this time. It has the following format in ABNF notation:

clear-u2u-call-capability = ("r-ucall-clear:" / "r-uc:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of clear SU-to-SU calls, and "1" is used to indicate that the RFSS is capable of clear SU-to-SU calls.

3.8.3.3 Encrypted Group Call Capability (r-gcall-sec)

[Replace existing text in this section of [102BACA] with the text below.]

The Encrypted Group Call Capability attribute defines whether or not the RFSS is capable of encrypted unconfirmed group calls at this time. It has the following format in ABNF notation:

encrypted-group-call-capability = ("r-gcall-sec:" / "r-gs:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of encrypted unconfirmed group calls, and "1" is used to indicate that the RFSS is capable of encrypted unconfirmed group calls.

3.8.3.4 Clear Group Call Capability (r-gcall-clear)

[Replace existing text in this section of [102BACA] with the text below.]

The Clear Group Call Capability attribute defines whether or not the RFSS is capable of clear unconfirmed group calls at this time. It has the following format in ABNF notation:

clear-group-call-capability = ("r-gcall-clear:" / "r-gc:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of clear unconfirmed group calls, and "1" is used to indicate that the RFSS is capable of clear unconfirmed group calls.

3.8.3.5 Encrypted Confirmed Group Call Capability (r-gcall-econf)

[Replace existing text in this section of [102BACA] with the text below.]

The Encrypted Confirmed Group Call Capability attribute defines whether or not the RFSS is capable of encrypted confirmed group calls at this time. It has the following format in ABNF notation:

enc-conf-group-call-capability = ("r-gcall-econf:" / "r-ecg:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of encrypted confirmed group calls, and "1" is used to indicate that the RFSS is capable of encrypted confirmed group calls.

3.8.3.6 Clear Confirmed Group Call Capability (r-gcall-cconf)

[Replace existing text in this section of [102BACA] with the text below.]

The Clear Confirmed Group Call Capability attribute defines whether or not the RFSS is capable of clear confirmed group calls at this time. It has the following format in ABNF notation:

clr-conf-group-call-capability = ("r-gcall-cconf:" / "r-ccg:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of clear confirmed group calls, and "1" is used to indicate that the RFSS is capable of clear confirmed group calls.

3.8.3.7 SD Emergency Alarm Capability (r-sd-ealarm)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Emergency Alarm Capability attribute defines whether or not the RFSS is capable of the SD Emergency Alarm function at this time. It has the following format in ABNF notation:

sd-emergency-alarm-capability = ("r-sd-ealarm:" / "r-sd-ea:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Emergency Alarm function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.8 SD Emergency Alarm Cancel Capability (r-sd-ealarmc)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Emergency Alarm Cancel Capability attribute defines whether or not the RFSS is capable of the SD Emergency Alarm Cancel function at this time. It has the following format in ABNF notation:

sd-emergency-alarm-cancel-capability = ("r-sd-ealarmc:" / "r-sd-ec:")
boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Emergency Alarm Cancel function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.9 SD Group Emergency Cancel Capability (r-sd-egroupc)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Group Emergency Cancel Capability attribute defines whether or not the RFSS is capable of the SD Group Emergency Cancel function at this time. It has the following format in ABNF notation:

sd-group-emergency-cancel capability = ("r-sd-egroupc:" / "r-sd-gc:")
boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Group Emergency Cancel function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.10 SD Call Alert Capability (r-sd-calert)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Call Alert Capability attribute defines whether or not the RFSS is capable of the SD Call Alert function at this time. It has the following format in ABNF notation:

sd-call-alert-capability = ("r-sd-calert:" / "r-sd-ca:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Call Alert function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.11 SD Short Message Capability (r-sd-smsg)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Short Message Capability attribute defines whether or not the RFSS is capable of the SD Short Message function at this time. It has the following format in ABNF notation:

sd-short-message-capability = ("r-sd-smsg:" / "r-sd-sm:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Short Message function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.12 SD Status Query Capability (r-sd-squery)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Status Query Capability attribute defines whether or not the RFSS is capable of the SD Status Query function at this time. It has the following format in ABNF notation:

sd-status-query-capability = ("r-sd-squery:" / "r-sd-sq:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Status Query function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.13 SD Status Update Capability (r-sd-supdate)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Status Update Capability attribute defines whether or not the RFSS is capable of the SD Status Update function at this time. It has the following format in ABNF notation:

sd-status-update-capability = ("r-sd-supdate:" / "r-sd-su:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Status Update function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.14 SD Radio Unit Monitor Capability (r-sd-rum)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Radio Unit Monitor Capability attribute defines whether or not the RFSS is capable of the SD Radio Unit Monitor function at this time. It has the following format in ABNF notation:

sd-radio-unit-monitor-capability = ("r-sd-rum:" / "r-sd-rm:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Radio Unit Monitor function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.15 SD Radio Unit Monitor Enhanced Capability (r-sd-rume)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Radio Unit Monitor Capability attribute defines whether or not the RFSS is capable of the SD Radio Unit Monitor Enhanced function at this time. It has the following format in ABNF notation:

sd-radio-unit-monitor-enhanced-capability = ("r-sd-rume:" / "r-sd-re:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Radio Unit Monitor Enhanced function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.16 SD Radio Check Capability (r-sd-rcheck)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Radio Check Capability attribute defines whether or not the RFSS is capable of the SD Radio Check function at this time. It has the following format in ABNF notation:

sd-radio-check-capability = ("r-sd-rcheck:" / "r-sd-rc:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Radio Check function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.17 SD Radio Detach Capability (r-sd-rdetach)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Radio Detach Capability attribute defines whether or not the RFSS is capable of the SD Radio Detach function at this time. It has the following format in ABNF notation:

sd-radio-detach-capability = ("r-sd-rdetach:" / "r-sd-rd:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Radio Detach function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.18 SD Radio Inhibit Capability (r-sd-rinhibit)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Radio Inhibit Capability attribute defines whether or not the RFSS is capable of the SD Radio Inhibit function at this time. It has the following format in ABNF notation:

sd-radio-inhibit-capability = ("r-sd-rinhibit:" / "r-sd-ri:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Radio Inhibit function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.19 SD Radio Uninhibit Capability (r-sd-runinhibit)

[Replace existing text in this section of [102BACA] with the text below.]

The SD Radio Uninhibit Capability attribute defines whether or not the RFSS is capable of the SD Radio Uninhibit function at this time. It has the following format in ABNF notation:

sd-radio-uninhibit-capability = ("r-sd-runinhibit:" / "r-sd-ru:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Radio Uninhibit function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.20 Packet Data Capability (r-pdata)

[Replace existing text in this section of [102BACA] with the text below.]

The Packet Data Capability attribute defines whether or not the RFSS is capable of supporting Packet Data at this time. It has the following format in ABNF notation:

sd-packet-data-capability = ("r-pdata:" / "r-pd:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of supporting packet data, and "1" is used to indicate that the RFSS is capable of supporting packet data.

3.8.3.21 Adjacent Site (r-adjsite)

[Replace existing text in this section of [102BACA] with the text below.]

The Adjacent Site attribute defines a site in an RFSS that is adjacent to a site in the other RFSS. It has the following format in ABNF notation:

Adjacent-site = ("r-adjsite:" / "r-as:") lra SP wacn-id SP sys-id SP
rfss-id SP site-id SP
channel-T SP channel-R SP
serv-class SP signalling CRLF

where the parameters represent the values in the ADJ_STS_BCST air interface message as defined in [102AABC] and are defined as follows:

lra	= 2HEX-DIGIT
wacn-id	= 5HEX-DIGIT
sys-id	= 3HEX-DIGIT
rfss-id	= 2HEX-DIGIT
site-id	= 2HEX-DIGIT
channel-T	= 4HEX-DIGIT
channel-R	= 4HEX-DIGIT
serv-class	= 2HEX-DIGIT
signalling	= 1HEX-DIGIT

If an Adjacent Site attribute is included in the profile it SHALL contain all of the parameters. If the channel-R field indicates a value of 0xFFFF, then the indication of the channel-R field SHALL be ignored, and the channel-T field SHALL be used as an implicitly defined channel. Note that "signalling" represents the "CFVA" bits in the ADJ_STS_BCST message.

Zero, one or many of the Adjacent Site attributes MAY be present in the RFSS service profile.

Note that the channels to be sent in the Adjacent Site attributes and the bandplan

information needed at the receiving RFSS are coordinated and provisioned by system administrators.

3.8.3.22 Unit Roaming Capability (r-roam-unit)

[Replace existing text in this section of [102BACA] with the text below.]

The Unit Roaming Capability attribute defines whether or not the RFSS is capable of supporting roaming of its SUs to the other RFSS at this time. It has the following format in ABNF notation:

unit-roaming-capability = ("r-roam-unit:" / "r-ru:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of supporting SU roaming to the other RFSS, and "1" is used to indicate that the RFSS is capable of supporting SU roaming to the other RFSS.

3.8.3.23 Group Roaming Capability (r-roam-grp)

[Replace existing text in this section of [102BACA] with the text below.]

The Group Roaming Capability attribute defines whether or not the RFSS is capable of supporting roaming of its Groups to the other RFSS at this time. It has the following format in ABNF notation:

group-roaming-capability = ("r-roam-grp:" / "r-rg:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of supporting Group roaming to the other RFSS, and "1" is used to indicate that the RFSS is capable of supporting Group roaming to the other RFSS.

3.8.3.24 RFSS Known Capability (r-rfss-known)

[Replace existing text in this section of [102BACA] with the text below.]

The RFSS Known Capability attribute defines whether or not the RFSS is provisioned for the other RFSS at this time. It has the following format in ABNF notation:

RFSS-known-capability = ("r-rfss-known:" / "r-rk:") boolean CRLF

where "0" is used to indicate that the RFSS is NOT provisioned for the other RFSS, and "1" is used to indicate that the RFSS is provisioned for the other RFSS.

3.8.3.25 Vocoder Mode Capability (r-vmodecap)

[Replace existing text in this section of [102BACA] with the text below.]

The Vocoder Mode Capability attribute defines the vocoder modes that the RFSS is capable of. It has the following format in ABNF notation:

vocoder-mode-capability = ("r-vmodecap:" / "r-vmc:") 1*(voc-mode SP) CRLF
voc-mode = "0" / "1" / "2"

where:

- 0 - indicates that the RFSS is capable of Full Rate vocoder mode
- 1 - indicates that the RFSS is capable of Half Rate vocoder mode
- 2 - indicates that the RFSS is capable of Native vocoder mode

This definition requires a space before the CRLF.

3.8.3.26 Manufacturer Specific Attribute (r-man<mfid>)

[Replace existing text in this section of [102BACA] with the text below.]

The Manufacturer Specific attribute allows manufacturer defined profile information that MAY be used for private or experimental purposes. It SHALL have the following radical with the following ABNF notation:

man-rfss-radical = ("r-man" / "r-m") mfid "-" octet-string ":" pvalue CRLF
mfid = 2HEX-DIGIT

where pvalue is defined in Section 25.1 of RFC 3261.

The mfid values are defined in [102BAAC]

Manufacturer specific attributes that are not understood by an RFSS SHOULD be ignored.

Example:

r-manA4-myparameter:1

3.8.3.27 Example RFSS Service Profiles

[No change from existing text in this section of [102BACA].]

3.8.4 ISSI specific register parameter attributes

[No change from existing text in this section of [102BACA], except for subsections 3.8.4.7 and 3.8.4.8 below.]

3.8.4.1 Force Register

[No change from existing text in this section of [102BACA].]

3.8.4.2 Confirm Register

[No change from existing text in this section of [102BACA].]

3.8.4.3 SU Presence

[No change from existing text in this section of [102BACA].]

3.8.4.4 SU Roaming

[No change from existing text in this section of [102BACA].]

3.8.4.5 SU Authentication

[No change from existing text in this section of [102BACA].]

3.8.4.6 Emergency Mode Register

[No change from existing text in this section of [102BACA].]

3.8.4.7 SNC Change

[Replace existing text in this section of [102BACA] with the text below.]

This parameter is used by the Serving RFSS in the PD SN Context registration request to inform the Home RFSS whether the procedure is pertaining to initial SMDCP context activation request, or pertaining to updated SMDCP context parameter values subsequently determined by the Serving RFSS and accepted by the SU.

This parameter has the following ABNF format:

snc-change = ("snc-change:"/"snc-c:") Boolean CRLF

0: No Change, Request pertains to initial SMDCP context activation request (Default)

1: Change, Request pertains to updated SMDCP context parameter values subsequently determined by the Serving RFSS and accepted by the SU

Example:

snc-change:0
snc-c:1

3.8.4.8 Supported Vocoder Modes

[Replace existing text in this section of [102BACA] with the text below.]

This parameter is used in the Group Registration procedure. It is used by a Serving RFSS in group registration requests, and by a Group Home RFSS in group registration responses. The "Supported Vocoder Modes" qualifier indicates the vocoder mode(s) supported by the RFSS for this group, listed in the order of preference, with the first vocoder mode listed being preferred. See Section 5 for further details.

This parameter has the following ABNF format:

supported-vm = ("r-vocodermode:" / "r-vm:") 1*(voc-mode SP) CRLF

voc-mode = "0" / "1" / "2"

0: Full Rate vocoder mode

1: Half Rate vocoder mode

2: Native vocoder mode

This definition requires a space before the CRLF.

If this parameter is not present in a group registration request or response, no assumption can be made on the RFSS's supported vocoder mode.

Examples:

r-vocodermode:2 1 0

r-vm:1

3.8.5 ISSI specific call parameter attributes

[No change from existing text in this section of [102BACA].]

3.9 X-TIA-P25-SNDP Body Parameters

[Replace existing text in this section of [102BACA] with the text below.]

This section describes the X-TIA-P25-SNDP application used to carry the SNDP context parameters.

Only one X-TIA-P25-SNDP body MAY be carried in a request or a 200 OK response pertaining to a PD SN Context mobility management procedure.

Up to fourteen X-TIA-P25-SNDP bodies MAY be carried in the 200 OK response to a unit registration request also used for SNDP context transfer.

The application/x-tia-p25-sndp media type is defined by the following information:

Media type name: application

Media subtype name: x-tia-p25-sndp

Required parameters: version

'Version' gives the version of the X-TIA-P25-SNDP protocol. The initial protocol version is set to: "version = 1.0"

Optional parameters: None

The SIP Content-type header field is set as defined in section 3.5.7.

The X-TIA-P25-SNDP descriptions are entirely textual. The X-TIA-P25-SNDP description consists in one or more lines of text. The order of these lines has no special meaning. The names and parameters of the fields are case-insensitive strings.

Handling of HEX-DIGIT based parameters is as specified in Section 3.1 of this document.

Table 34 summarizes the available attributes specific to PD SN Context mobility management.

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Table 34 - ISSI-specific SN Context Mobility Management Parameters⁴

SNDP Data Field [102BAEB] [102BACF]	Field	Compact form	Meaning related to the considered SNDP Context	Default value
SNDP Version	r-snprotv	r-snv	SNDP protocol version	3: SNDP version 3
NSAPI	r-snnsapi	r-sns	SNDP nsapi	1: nsapi 1
NAT	r-snnat	r-snn	Network Address Type	1: IPv4 Dynamic Address
IPv4 Address	r-snipv4ad	r-snipa	IPv4 address	None
DSUT	r-sndsut	r-snds	Data Subscriber Unit Type	1: Alternating Trunked Voice & Data MRC
MDPCO	r-snmcpco	r-snmco	MDP Configuration options	0: PPP
Data Access Control (DAC)	r-sndac	r-snd	PDCH Access Rights	None
PDUPM	r-snpdupm	r-snpp	PDU priority maximum	0: Lowest priority
Ready Timer	r-snreadyt	r-snrt	Ready timer	None
MTU	r-snmtu	r-snm	MTU	4: 1500 bytes
Standby Timer	r-snstandbyt	r-snst	Standby timer	None
PCO Config Protocol	r-snpcoc	r-snpc	PCO Configuration Protocol	0: PPP
PCO Protocol Identity	r-snpcoi	r-snpi	PCO Protocol Identity	\$C023: PAP
PCO Protocol Identity content length	r-snpcoicl	r-snpicl	PCO Protocol Identity content length	0: No Protocol Identity content
PCO Protocol Identity content	r-snpcoic	r-snpic	PCO Protocol Identity content	None
APN Length	r-snapnl	r-snal	Format of the APN coding	0: No APN defined
APN index	r-snapni	r-snai	Value of the APN octet	None
APN string	r-snapns	r-snas	Value of the APN string	None
MSO Length	r-snmsol	r-snml	Length of the MSO Value field	0: MSO Value field not present
MSO Value	r-snmsov	r-snmv	Contents of the MSO Value field	None
Home PDIG Address	r-snhpdigad	r-snhpda	IP address of Home PDIG for an SU	None

⁴ Fields related to compression of SNDP headers and payload are left out of this table because SNDP-based compression is being handled by the Serving RFSS and the SU, but not the Home RFSS.

Table 35 summarizes which Sndcp Context fields are present in the various SN Context mobility Management messages. 1xx provisional messages do not contain any specific SN Context mobility management parameter.

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Table 35 – SNDCP Fields in SN Context Mobility Management Messages⁵

SIP Header Field Direction Serving(S) => Home (H) Home (H) => Serving (S)	Where	SU Register-Register S=>H (SNDCP Context Transfer)	SNC Register-Register S => H (with “SU Presence” parameter value of “Confirmed”)	SNC Register-Home Query (with “Force” qualifier) H => S	SNC Register-Home Query (without any qualifier) H => S	SNC Register-Serving Query (without any qualifier) S => H
SNDCP Version	Request	-	M-ISSI	M-ISSI	-	-
SNDCP Version	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
nsapi	Request	-	M-ISSI	M-ISSI	-	-
nsapi	200	T-ISSI	-	-	-	-
NAT	Request	-	M-ISSI	M-ISSI	-	-
NAT	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
IPv4 Address	Request	-	C-ISSI	C-ISSI	-	-
IPv4 Address	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
DSUT	Request	-	M-ISSI	-	-	-
DSUT	200	T-ISSI	-	-	M-ISSI	M-ISSI
MDPCO	Request	-	M-ISSI	-	-	-
MDPCO	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
Data Access Control (DAC)	Request	-	M-ISSI	-	-	-
Data Access Control (DAC)	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
PDUPM	Request	-	M-ISSI	-	-	-
PDUPM	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
Ready Timer	Request	-	M-ISSI	-	-	-
Ready Timer	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
MTU	Request	-	M-ISSI	-	-	-
MTU	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
Standby Timer	Request	-	M-ISSI	-	-	-
Standby Timer	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
PCO Config Protocol	Request	-	C-ISSI	-	-	-

⁵ Fields related to compression of SNDCP headers and payload are left out of this table because SNDCP-based compression is being handled by the Serving RFSS and the SU, but not the Home RFSS.

SIP Header Field Direction Serving(S) => Home (H) Home (H) => Serving (S)	Where	SU Register-Register S=>H (SND CP Context Transfer)	SNC Register-Register S => H (with "SU Presence" parameter value of "Confirmed")	SNC Register-Home Query (with "Force" qualifier) H => S	SNC Register-Home Query (without any qualifier) H => S	SNC Register-Serving Query (without any qualifier) S => H
PCO Config Protocol	200	-	C-ISSI	-	-	-
PCO Protocol Identity	Request	-	C-ISSI	-	-	-
PCO Protocol Identity	200	-	C-ISSI	-	-	-
PCO Protocol Identity content length	Request	-	C-ISSI	-	-	-
PCO Protocol Identity content length	200	-	C-ISSI	-	-	-
PCO Protocol Identity content	Request	-	C-ISSI	-	-	-
PCO Protocol Identity content	200	-	C-ISSI	-	-	-
APN Length	Request	-	C-ISSI	C-ISSI	-	-
APN Length	200	T-ISSI	C-ISSI	-	C-ISSI	C-ISSI
APN index	Request	-	C-ISSI	C-ISSI	-	-
APN index	200	T-ISSI	C-ISSI	-	C-ISSI	C-ISSI
APN string	Request	-	C-ISSI	C-ISSI	-	-
APN string	200	T-ISSI	C-ISSI	-	C-ISSI	C-ISSI
MSO Length	Request	-	C-ISSI	C-ISSI	-	-
MSO Length	200	T-ISSI	C-ISSI	-	C-ISSI	C-ISSI
MSO Value	Request	-	C-ISSI	C-ISSI	-	-
MSO Value	200	T-ISSI	C-ISSI	-	C-ISSI	C-ISSI
Home PDIG Address	200	-	M-ISSI	-	-	-

T-ISSI means that if an SNDCP context is to be transferred, the Home RFSS provides in the 200 OK response to the new Serving RFSS's SU Register-Register request, the most current parameter value known to the Home RFSS and known to the SU, or no information if the parameter was not present during SNDCP context activation. C-ISSI and M-ISSI are as defined in Section 1.1 of this document.

The value of each SNDCP parameter depends on the message that carries the parameter:

- In an initial SNC register-register request (i.e. snc-change set to 0), the SNDCP parameter value is either provided by the SU in its SNDCP context activation request if the parameter exists in the request, or by the Serving RFSS.
- In subsequent SNC register-register requests (i.e. snc-change set to 1), the SNDCP parameter value is the one ultimately determined by the Serving RFSS and accepted by the SU.
- In an SNC Register-query with force qualifier, the SNDCP parameter value is the one to be applied in the SNDCP context activation command
- In the 200 OK response to an SNC Home query with no qualifier, the SNDCP parameter value is the last one known to the Serving RFSS that has been accepted by the SU.
- In the 200 OK response to an SNC Serving query, the SNDCP parameter value is the last one known to the Home RFSS that has been accepted by the SU.

The default value for each SNDCP parameter is as shown in Table 34.

[No change from existing text in all subsections 3.9.x of [102BACA].]

3.10 Examples

[No change from existing text in this section of [102BACA], except for Subsection 3.10.2 below.]

3.10.1 Mobility Management Examples [Informative]

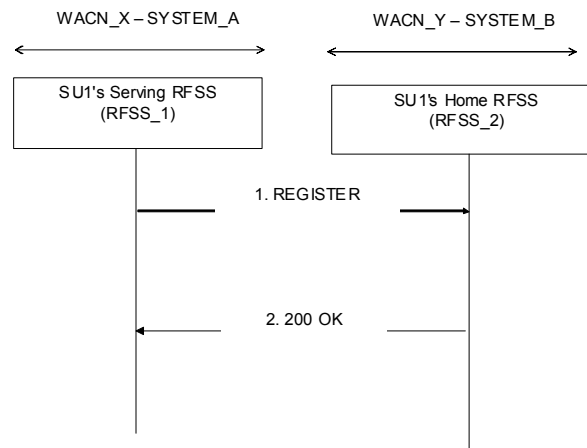
[No change from existing text in this section of [102BACA].]

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3.10.2 Call Processing Examples [Informative]

[Replace existing text in this section of [102BACA] with the text below.]

Example 1 - SU Registration addressing example (see Figure 36):



Variable are used in the example for better clarity but SHALL be replaced by the hexadecimal values in the real case	
variable	example
WACN_X	9EF60
WACN_Y	568FA
SYSTEM_A	570
SYSTEM_B	FA1
RFSS_1	6F
RFSS_2	B0
SUID1	568FAFA190FA41

Figure 36 - SU Registration Example

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1. REGISTER from SU1's Serving RFSS to the SU1's home system <pre>REGISTER sip:RFSS_2.SYSTEM_B.WACN_Y.p25dr SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK15ADC0.1 Max-Forwards: 70 To: <sip:SUID1@p25dr;user=TIA-P25-SU> From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=16879 Call-ID: 4398c1c2.0008c0f3@p25dr CSeq: 1 REGISTER Contact: <sip:SUID1@RFSS_1.SYSTEM_A.WACN_X.p25dr;user=TIA-P25-S U> Expires: 86400 Accept: application/sdp;level=1, application/x-tia-p25-issi, application/x-tia-p25-issi-18 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0</pre> Notes: 1. the SIP URI of the To and From header fields indicate that the SUID1 is related to the SU's numbering plan. This explains the "@p25dr" and "user=TIA-P25-SU" elements. 2. the values SYSTEM_B and WACN_Y are obtained from the SUID1 value analysis. 3. the value RFSS_2 has been provisioned previously in the Serving RFSS_1.	2. OK response from SU1's home system to SU1's Serving RFSS <pre>SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK15ADC0.1 To: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=7A901 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=16879 Call-ID: 4398c1c2.0008c0f3@p25dr CSeq: 1 REGISTER Contact: <sip:SUID1@RFSS_2.SYSTEM_B.WACN_Y.p25dr;user=TIA-P25-SU > Expires: 4294967295 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS MIME-Version: 1.0 Content-Disposition: session; handling=required Content-Length: ...</pre>
--	---

Example 2 - SU-to-SU Call addressing example (see Figure 37):

The TRYING and RINGING provisional responses are not shown.

Variables are used in the example for better clarity but SHALL be replaced by the hexadecimal values in the real case	
variable	example
WACN_X	9EF60
WACN_Y	568FA
WACN_Z	AF501
WACN_T	FED99
SYSTEM_A	570
SYSTEM_B	FA1
SYSTEM_C	0F2
SYSTEM_D	991
RFSS_1	6F
RFSS_2	B0
RFSS_3	BB
RFSS_4	E1
SUID1	568FAFA190FA41
SUID2	AF5010F201FF4E

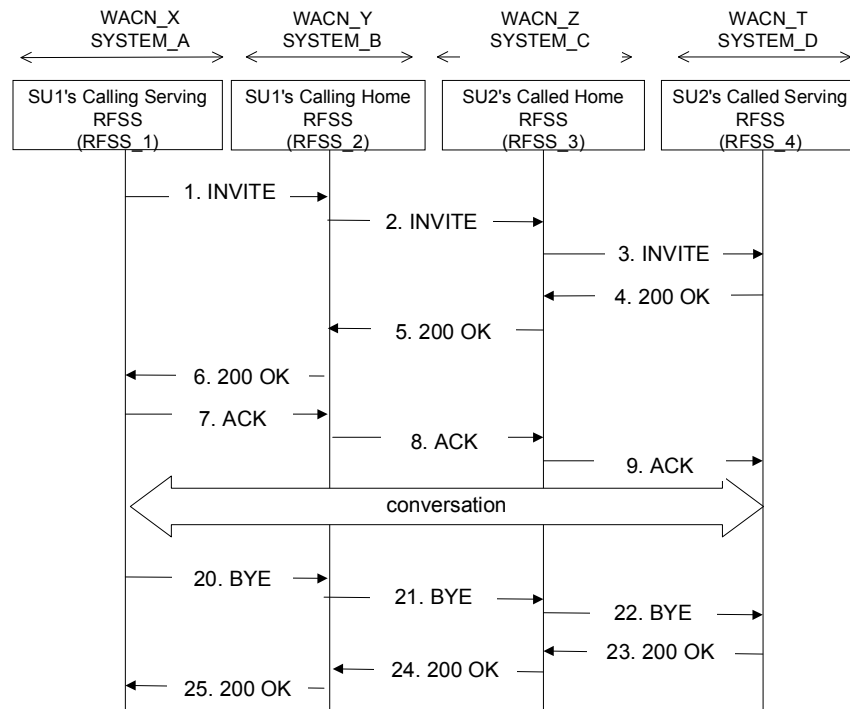


Figure 37 - SU-to-SU Call example

1. INVITE from SU1's Calling Serving RFSS to SU1's Calling Home RFSS

INVITE sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0
 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKac456326.1
 Max-Forwards: 70
 Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr>
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=00016879
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 1 INVITE
 Contact: <sip:SUID1@RFSS_1.SYSTEM_A.WACN_X.p25dr;user=TIA-P25-SU>
 Accept: application/sdp;level=1, application/x-tia-p25-issi,
 application/x-tia-p25-issi-18
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Priority: 5;a
 MIME-Version: 1.0
 Content-Disposition: session; handling=required

Notes:

1. This message is sent to the address RFSS_2.SYSTEM_B.WACN_Y.p25dr as provisioned in the RFSS.
2. The Route header field with TIA-P25-U2Uorig tells the RFSS to behave as the "Calling Home RFSS ". The calling SU is identified in the From header field.
3. The Calling Home RFSS forwards the INVITE to the Called Home RFSS on a new SIP dialog.
- 4; The MIME-Version header field denotes the presence of a P25 body and/or a manufacturer-specific body.

2. INVITE from Calling Home RFSS of SU1 to Called Home RFSS of SU2

INVITE sip:SUID2@ RFSS_3.SYSTEM_C.WACN_Z.p25dr;user=TIA-P25-SU SIP/2.0
 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK78f365.1
 Max-Forwards: 70
 Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr>
 Record-Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr>
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=1113258f
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 2 INVITE
 Contact: <sip:SUID1@RFSS_1.SYSTEM_A.WACN_X.p25dr;user=TIA-P25-SU>
 Accept: application/sdp;level=1, application/x-tia-p25-issi,
 application/x-tia-p25-issi-18
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Priority: 5;a
 MIME-Version: 1.0
 Content-Disposition: session; handling=required

Notes:

1. The Route header field referring to the RFSS identified by RFSS_2 is removed. The Record-Route header field referring to the RFSS identified by RFSS_2 is added. The SIP URI provided in the Record-Route will be used later by the SU1's **Calling** Serving RFSS.
2. The Route header field referring to the RFSS identified by RFSS_3 is added. The Route header field with TIA-P25-U2Udest tells the RFSS to behave as the "Called Home RFSS ". The Called SU is identified in the To header field
3. This message is sent to the Home System of SU2.

3. INVITE from SU2's Called Home RFSS to SU2's Called Serving RFSS

INVITE sip:SUID2@RFSS_4.SYSTEM_D.WACN_T.p25dr;user=TIA-P25-SU SIP/2.0
 Via: SIP/2.0/UDP RFSS_3.SYSTEM_C.WACN_Z.p25dr;branch=z9hG4bK25fea7.1
 Max-Forwards: 70
 Record-Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr>
 Record-Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr>
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=226fab57
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 2 INVITE
 Contact: <sip:SUID1@RFSS_1.SYSTEM_A.WACN_X.p25dr;user=TIA-P25-SU>
 Accept: application/sdp;level=1, application/x-tia-p25-issi,
 application/x-tia-p25-issi-18
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Priority: 5;a
 MIME-Version: 1.0
 Content-Disposition: session; handling=required

Note:

1. The Route header field referring to the RFSS identified by RFSS_3 is removed. The Record-Route header field referring to the RFSS identified by RFSS_3 is added. The SIP URI provided in the Record-Route will be used later by the SU2's Serving RFSS.
2. SU2's Home RFSS retrieves the contact address of SU2's Serving RFSS from its location service, and then sends an INVITE to this RFSS.
3. As no Route header field is present and the radical name of the Request URI is equal to the radical name of the "To" field, the receiving RFSS knows it has to behave as a Called Serving RFSS.

. . .	
4. OK response from SU2's Called Serving RFSS to SU2's Called Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_3.SYSTEM_C.WACN_Z.p25dr;branch=z9hG4bK25fea7.1 Record-Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr> Record-Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr> To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=2269ea87 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=226fab57 Call-ID: 4300f57f.0008f0ac@p25dr CSeq: 2 INVITE Contact:<sip:SUID2@RFSS_4.SYSTEM_D.WACN_T.p25dr;user=TIA-P25-SU> Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS MIME-Version: 1.0 Content-Disposition: session; handling=required . . . 5. OK response from SU2's Called Home RFSS to SU1's Calling Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK78f365.1 Record-Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr> Record-Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr> To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=1165fabe From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=1113258f Call-ID: 4300f57f.0008f0ac@p25dr CSeq: 2 INVITE Contact:<sip:SUID2@RFSS_4.SYSTEM_D.WACN_T.p25dr;user=TIA-P25-SU> Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS MIME-Version: 1.0 Content-Disposition: session; handling=required . . . 6. OK response from SU1's Calling Home RFSS to SU1's Calling Serving RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKac456326.1 Record-Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr> Record-Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr> To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=000978879 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=00016879 Call-ID: 4300f57f.0008f0ac@p25dr CSeq: 1 INVITE Contact:<sip:SUID2@RFSS_4.SYSTEM_D.WACN_T.p25dr;user=TIA-P25-SU> Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS	7. ACK from SU1's Calling Serving RFSS to SU1's Calling Home RFSS ACK sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKeb362754.1 Max-Forwards: 70 Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr> Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr> To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=00978879 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=00016879 Call-ID: 4300f57f.0008f0ac@p25dr CSeq: 1 ACK Content-Length: 0 8. ACK from SU1's Calling Home RFSS to SU2's Called Home RFSS ACK sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bKf37568.1 Max-Forwards: 70 Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr> To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=1165fabe From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=1113258f Call-ID: 4300f57f.0008f0ac@p25dr CSeq: 1 ACK Content-Length: 0 9. ACK to SU2's Called Serving RFSS ACK sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0 Via: SIP/2.0/UDP RFSS_3.SYSTEM_C.WACN_Z.p25dr;branch=z9hG4bKaf572e.1 Max-Forwards: 70 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=2269ea87 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=2226fab57 Call-ID: 4300f57f.0008f0ac@p25dr CSeq: 1 ACK Content-Length: 0

```
MIME-Version: 1.0
Content-Disposition: session; handling=required
. . .
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20. BYE from SU1's Calling Serving RFSS to SU1's Calling Home RFSS

BYE sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0
 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK4652c23a.1
 Max-Forwards: 70
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=00978879
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=00016879
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 2 BYE
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Content-Length: 0

21. BYE from SU1's Calling Home RFSS to SU2's Called Home RFSS

BYE sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0
 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK86573e.1
 Max-Forwards: 70
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=1165fabe
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=1113258f
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 2 BYE
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Content-Length: 0

22. BYE from SU2's Called Home RFSS to SU2's Called Serving RFSS

BYE sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0
 Via: SIP/2.0/UDP RFSS_3.SYSTEM_C.WACN_Z.p25dr;branch=z9hG4bK572a8c.1
 Max-Forwards: 70
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=2269ea87
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=2226fab57
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 2 BYE
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Content-Length: 0

23. OK response from SU2's Called Serving RFSS to SU2's Called Home RFSS

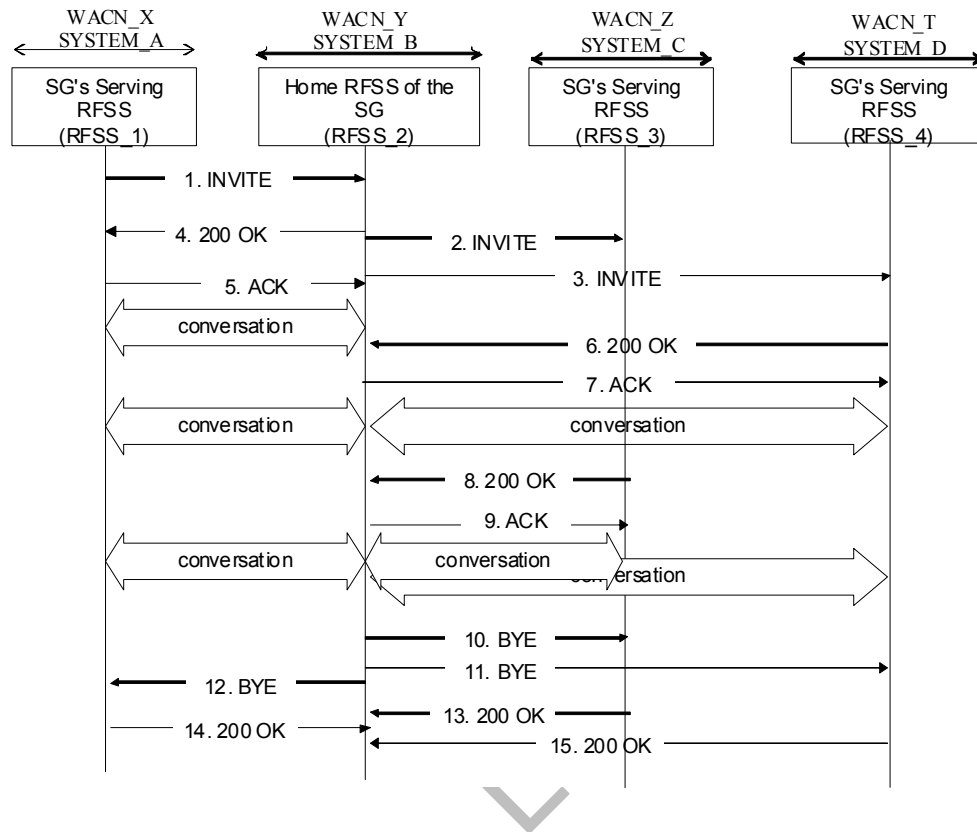
SIP/2.0 200 OK
 Via: SIP/2.0/UDP RFSS_3.SYSTEM_C.WACN_Z.p25dr;branch=z9hG4bK25fea7.1
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=2269ea87
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=2226fab57
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 2 BYE
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Content-Length: 0

24. OK response from the SU2's Called Home RFSS to SU1's Calling Home RFSS

SIP/2.0 200 OK
 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK78f365.1
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=1165fabe
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=1113258f
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 2 BYE
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Content-Length: 0

25. OK response from SU1's Calling Home RFSS to SU1's Calling Serving RFSS

SIP/2.0 200 OK
 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKac456326.1
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=00978879
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=00016879
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 2 BYE
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Content-Length: 0

Example 3 - Unconfirmed Group Call addressing example (see Figure 38):**Figure 38 - Unconfirmed Group Call example**

Variable are used in the example for better clarity but SHALL be replaced by the hexadecimal values in the real case

variable	example
WACN_X	9EF60
WACN_Y	568FA
WACN_Z	AF501
WACN_T	FED99
SYSTEM_A	570
SYSTEM_B	FA1
SYSTEM_C	0F2
SYSTEM_D	991
RFSS_1	6F
RFSS_2	B0
RFSS_3	BB
RFSS_4	E1
SGID	568FAFA190FA

1. INVITE from SG's Serving RFSS_1 to SG's Home RFSS

INVITE sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0
 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKac456326.1
 Max-Forwards: 70
 Route: <sip:TIA-P25-Groupcall@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr>
 To: <sip:SGID@p25dr;user=TIA-P25-SG>
 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=00016879
 Call-ID: 431ff9cd.0008f0ac@p25dr
 CSeq: 1 INVITE
 Contact: <sip:SGID@RFSS_1.SYSTEM_A.WACN_X.p25dr;user=TIA-P25-SG>
 Accept: application/sdp;level=1, application/x-tia-p25-issi,
 application/x-tia-p25-issi-l8
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Priority: 8;a
 Content-Disposition: session; handling=required

Notes:

1. This message is sent to the address RFSS_2.SYSTEM_B.WACN_Y.p25dr as provisioned in the RFSS
2. The Route header field tells the RFSS to behave as the "Home RFSS of the SG". Addressing a Home RFSS general rule described in the addressing section is applied.
3. During the SG registration process the Home RFSS has stored all the RFSS to contact for this SGID
4. As there is no MIME-Version header field, there is only a SDP body in this message

2. INVITE from SG's Home RFSS to the Serving RFSS_3

INVITE sip:SGID@RFSS_3.SYSTEM_C.WACN_Z;user=TIA-P25-SG SIP/2.0
 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK0612357.1
 Max-Forwards: 70
 To: <sip:SGID@p25dr;user=TIA-P25-SG>
 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=1113258f
 Call-ID: 431ff9cd.00089a43@p25dr
 CSeq: 1 INVITE
 Contact: <sip:SGID@RFSS_3.SYSTEM_B.WACN_Y.p25dr;user=TIA-P25-SG>
 Accept: application/sdp;level=1, application/x-tia-p25-issi,
 application/x-tia-p25-issi-l8
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Priority: 8;a
 Content-Disposition: session; handling=required

Notes:

1. The Route header field referring to the RFSS identified by RFSS_2 is removed.
2. As no Route header field mentioning the "TIA-P25-Groupcall" element is present, the RFSS receiving this knows it has to behave as a "Serving RFSS of the SG". The SG is identified in the To and From header fields.

3. INVITE from SG's Home RFSS to the Serving RFSS_4

INVITE sip:SGID@RFSS_4.SYSTEM_D.WACN_T;user=TIA-P25-SG SIP/2.0
 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK05478967.1
 Max-Forwards: 70
 To: <sip:SGID@p25dr;user=TIA-P25-SG>
 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=22254abd
 Call-ID: 431ff9cd.00089a44@p25dr
 CSeq: 1 INVITE
 Contact: <sip:SGID@RFSS_4.SYSTEM_C.WACN_Z.p25dr;user=TIA-P25-SG>
 Accept: application/sdp;level=1, application/x-tia-p25-issi,
 application/x-tia-p25-issi-l8
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Priority: 8;a
 Content-Disposition: session; handling=required

4. OK response from SG's Home RFSS to SG's Serving RFSS_1

SIP/2.0 200 OK
 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKac456326.1
 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=0069ea87
 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=00016879
 Call-ID: 431ff9cd.0008f0ac@p25dr
 CSeq: 1 INVITE
 Contact: <sip:SGID@RFSS_2.SYSTEM_B.WACN_Y.p25dr;user=TIA-P25-SG>
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Content-Disposition: session; handling=required

5. ACK from SG's Serving RFSS_1 to SG's Home RFSS ACK sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKeb362754.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=0069ea87 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=00016879 Call-ID: 431ff9cd.0008f0ac@p25dr CSeq: 1 ACK Content-Length: 0 6. OK response from the SG's Serving RFSS_4 to SG's Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK05478967.1 To: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=2265fabe From: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=22254abd Call-ID: 431ff9cd.00089a44@p25dr CSeq: 1 INVITE Contact: <sip:SGID@RFSS_4.SYSTEM_D.WACN_T.p25dr;user=TIA-P25-SG> Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS MIME-Version: 1.0 Content-Disposition: session; handling=required . . . 7. ACK from SG's Home RFSS to SG's Serving RFSS_4 ACK sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bKf37568.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=2265fabe From: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=22254abd Call-ID: 431ff9cd.00089a44@p25dr CSeq: 1 ACK Content-Length: 0	8. OK response from the SG's Serving RFSS_3 to SG's Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK0612357.1 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=11978879 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=1113258f Call-ID: 431ff9cd.00089a43@p25dr CSeq: 1 INVITE Contact: <sip:SGID@RFSS_3.SYSTEM_C.WACN_Z.p25dr;user=TIA-P25-SG> Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS MIME-Version: 1.0 Content-Disposition: session; handling=required . . . 9. ACK from SG's Home RFSS to SG's Serving RFSS_3 ACK sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKaf572e.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=11978879 From: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=1113258f Call-ID: 431ff9cd.00089a43@p25dr CSeq: 1 ACK Content-Length: 0

10. BYE from SG's Home RFSS to SG's Serving RFSS_3 BYE sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK4652c23a.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=11978879 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=1113258f Call-ID: 431ff9cd.00089a43@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0 Note: The Home RFSS maintains its own Cseq numbering space, which in this example begins with 3. 11. BYE from SG's Home RFSS to SG's Serving RFSS_4 BYE sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK86573e.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=1165fabe From: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=1113258f Call-ID: 431ff9cd.00089a44@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0 12. BYE from SG's Home RFSS to SG's Serving RFSS_1 BYE sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK572a8c.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=0069ea87 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=00016879 Call-ID: 431ff9cd.0008f0ac@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0	13. OK response from SG's Serving RFSS_3 to SG's Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK4652c23a.1 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=11978879 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=1113258f Call-ID: 431ff9cd.00089a43@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0 14 OK response from SG's Serving RFSS_1 to SG's Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK05478967.1 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=0069ea87 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=00016879 Call-ID: 431ff9cd.0008f0ac@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0 15. OK response from SG's Serving RFSS_4 to SG's Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK1578956.1 To: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=1165fabe From: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=1113258f Call-ID: 431ff9cd.00089a44@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0
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3.10.3 RFSS Service Capability Polling Examples [Informative]

*[No change from existing text in this section of **[102BACA]**.]*

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4. RTP Message Vocabulary for ISSI

[No change from existing text in this section of [102BACA], except for selected subsections thereof, as shown below.]

4.1 RTP Fixed Header Field Definitions

[No change from existing text in this section of [102BACA].]

4.2 ISSI Payload Header

[No change from existing text in this section of [102BACA], except for Subsection 4.2.2 below.]

4.2.1 Control Octet Field Definitions

[No change from existing text in this section of [102BACA].]

4.2.2 Block Header Field Definitions

[Replace existing text in this section of [102BACA] with the text below.]

The block headers describe the payload data of the packet. Each block header is a 32-bit word as shown in Figure 43.



Figure 43 - Block Header

Each block header SHALL contain a bit (E) to identify whether the payload type is a standard IANA type (E set to 0) or a profile specific type (E set to 1). For voice services over the ISSI, the E bit SHALL always be set to 1 (profile specific type).

Each block header SHALL also contain a 7 bit block type. The block type is taken from Table 37 and identifies the format and content of the corresponding block in the sequence of ISSI blocks.

Table 37 - ISSI Block Types

Block Type	Meaning
0	Full Rate Voice
1	Packet Type
2,3,4	Reserved for Fixed Station Interface
5	Full Rate ISSI Header Word
6 - 10	Reserved for Fixed Station Interface and Conventional ISSI
11	RF PTT Control Word
12 - 14	Reserved for Fixed Station Interface
15	Console PTT Control Word
16 - 30	Reserved for Conventional ISSI
31 - 32	Reserved for Fixed Station Interface
33	Half Rate ISSI Header Word
34	Half Rate Voice
35	TalkSpurt Vocoder Mode
36	User Alias
37 - 62	Reserved for Future Expansion
63 - 127	Manufacturer Specific

Each block header SHALL contain a 14-bit timestamp offset. This SHALL be in the same units as the RTP header timestamp (125 microseconds). The sum of the RTP timestamp and the offset is the timestamp of each block.

1. For all non-voice, non-manufacturer-specific ISSI blocks (i.e., with Block Type not being 0 or 34 or 63-127), since they are never positioned after a voice block in any given RTP packet, the Timestamp Offset SHALL be set to 0.
2. In the event that a single RTP packet carries one or more Full Rate Voice Blocks (Block Type = 0), per Section 4.4.2 of this document:
 - a. The Timestamp Offset for the first Full Rate Voice Block SHALL be set to 0.
 - b. The Timestamp Offset for the second Full Rate Voice Block, if present, SHOULD be set to 160.
 - c. The Timestamp Offset for the third Full Rate Voice Block, if present, SHOULD be set to 320.

3. For a Half Rate Voice Block (Block Type = 34), the Timestamp Offset SHALL be set to 0.

Each block header SHALL contain a 10-bit length indicator. This SHALL be the number of octets contained in the corresponding block.

For Full Rate Voice, the voice data transmitted over the ISSI between RFSSs comprises:

- Caller Identification and optionally User Alias
- Information from the CAI Header Data Unit that has been both Golay and Reed-Solomon decoded
- Full Rate voice data that has been both Golay and Hamming decoded
- Encryption Synchronization information that has been Hamming decoded but not Reed-Solomon Decoded, and
- Low Speed Data that has been cyclically decoded

For Half Rate Voice, the voice data transmitted over the ISSI between RFSSs comprises:

- Caller Identification and optionally User Alias
- Information from the MAC_PTT PDU that has been both Golay and Reed-Solomon decoded
- Half Rate voice data that has been Golay decoded
- Encryption Synchronization information

The following sections define the formats and content of each of the ISSI payload blocks, and the rules for constructing ISSI RTP packets.

4.3 Definition of ISSI Blocks

[No change from existing text in this section of [102BACA], except for selected subsections thereof, as shown below.]

4.3.1 Full Rate Voice (Block Type = 0)

[Replace existing text in this section of [102BACA] with the text below.]

Full Rate Voice Blocks (shown in Figure 44) carry TIA-102 Common Air Interface (CAI) Full Rate **[102BABA]** voice data representing 20 msec. of speech, along with a variety of additional information that varies with the Frame Type.

0											1											2											3										
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1												
Frame Type									U0 (b11-0)											U1 (b11-0)																							
U2 (b11-0)									U3 (b11-0)											U4 (b10-3)																							
U4		U5(b10-0)									U6(b10-0)									U7 (b6-0)																							
Et		Er		M	L	E4	E1		SF		rsvd		Additional Frame Type Specific Data (Variable Length)																														
Additional Frame Type Specific Data (Variable Length)																																											

Figure 44 - Full Rate Voice Data

Table 38 summarizes the Full Rate frame types supported by the ISSI, the contents of which are described in the following sub-sections.

Table 38 - Full Rate Frame Types

Name	Frame Type	Description
FR1 (formerly IMBE1)	\$C2	Full Rate Voice 1
FR2 (formerly IMBE2)	\$C3	Full Rate Voice 2
FR3 (formerly IMBE3)	\$C4	Full Rate Voice 3
FR4 (formerly IMBE4)	\$C5	Full Rate Voice 4
FR5 (formerly IMBE5)	\$C6	Full Rate Voice 5
FR6 (formerly IMBE6)	\$C7	Full Rate Voice 6
FR7 (formerly IMBE7)	\$C8	Full Rate Voice 7
FR8 (formerly IMBE8)	\$C9	Full Rate Voice 8
FRFRAME9 (formerly IMBEFRAME9)	\$6A	Full Rate Voice 9 + Low Speed Data
FR10 (formerly IMBE10)	\$CB	Full Rate Voice 10
FR11 (formerly IMBE11)	\$CC	Full Rate Voice 11
FRFRAME12 (formerly IMBEFRAME12)	\$6D	Full Rate Voice 12 + Encryption Sync
FRFRAME13 (formerly IMBEFRAME13)	\$6E	Full Rate Voice 13 + Encryption Sync
FRFRAME14 (formerly IMBEFRAME14)	\$6F	Full Rate Voice 14 + Encryption Sync
FRFRAME15 (formerly IMBEFRAME15)	\$70	Full Rate Voice 15 + Encryption Sync
FRFRAME16 (formerly IMBEFRAME16)	\$71	Full Rate Voice 16 + Encryption Sync
FRFRAME17 (formerly IMBEFRAME17)	\$72	Full Rate Voice 17 + Encryption Sync
FRFRAME18 (formerly IMBEFRAME18)	\$73	Full Rate Voice 18 + Low Speed Data

The Full Rate data is divided into 8 decoded message vectors (U0-U7). The message vectors U0-U6 have been decoded (error correction code removed). Vector U7 is unprotected.

The report count fields SHALL reflect the number of decoding errors detected when the over the air frame was recovered at a receiver and the error correcting code (Golay or Hamming) was decoded. These MAY be used for error mitigation when the voice is decoded. If the number of errors detected is greater than can be indicated in the report field, the field SHALL be set to the maximum value.

Et SHALL be the total number of errors detected in the frame. This SHALL be the parameter ε_T described by equation 95 in reference **[102BABA]** limited to a maximum value of 7.

Er is a scaled version of ε_R described in equation 96 of reference **[102BABA]**. Er SHALL be equal to $400 \times \varepsilon_R$ rounded down to the nearest integer and limited to a maximum of 7.

M is the mute frame flag, and L is the lost frame flag. The originating RFSS MAY set M and/or L values for any given frame based upon errors described in sections 7.7 and 7.8 of **[102BABA]**, or in accordance with local policy. When M is set to 1, the audio output can be muted. When M is set to 0 and L is set to 1, the previous audio output can be repeated.

E4 SHALL be the number of errors detected in vector U4, limited to a maximum of 1.

E1 SHOULD be the number of errors detected in vector U1, limited to a maximum of 7. If not used, E1 SHOULD be set to 0.

SF indicates a 2-bit free running super-frame counter that is incremented on the first frame of each super-frame.

In addition to the above 14 octets present on all Full Rate frames, each frame type specifies zero or more additional octets that are appended to the end of the frame. The following subsections describe the additional information carried in each of the frame types.

4.3.1.1 Fully Decoded Voice Frames (\$C2 - \$C9, \$CB, \$CC)

*[No change from existing text in this section of **[102BACA]**.]*

4.3.1.2 Voice with Low Speed Data (\$6A, \$73)

*[No change from existing text in this section of **[102BACA]**.]*

4.3.1.3 Voice with Partially Decoded Encryption Synch (\$6D-\$72)

[No change from existing text in this section of [102BACA].]

4.3.2 ISSI Packet Type (Block Type = 1)

[Replace existing text in this section of [102BACA] with the text below.]

Figure 47 shows the ISSI Packet Type block payload.

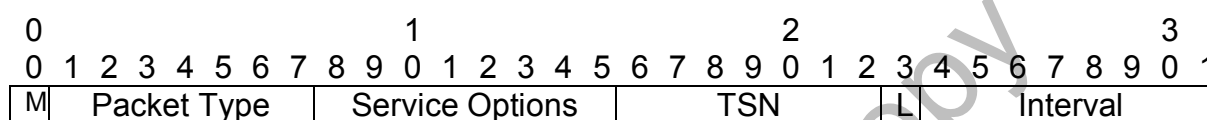


Figure 47 - ISSI Packet Type

For voice services over the ISSI, the RTP Packet Type fields SHALL be set as follows (all numbers are in decimal):

- M (1 bit): The M bit indicates the mute status of a particular talk spurt. The M bit SHALL be processed as described in Section 7.
- Packet Type (7 bits): Identifies the RTP packet type. The following values are defined. All other values are reserved. Procedures related to the use of the various RTP packets are given in Section 7.
- 0 - PTT Transmit Request
 - 1 - PTT Transmit Grant,
 - 2 - PTT Transmit Progress
 - 3 - PTT Transmit End
 - 4 - PTT Transmit Start
 - 5 - PTT Transmit Mute
 - 6 - PTT Transmit Unmute
 - 7 - PTT Transmit Wait
 - 8 - PTT Transmit Deny
 - 9 - PTT Heartbeat
 - 10 - PTT Heartbeat Query

Service Options (8 bits):	The service options are taken from the “Service Options” octet (as defined in [102AABC]) in the Unit to Unit Voice Service Request message or Group Voice Service Request message that the RFSS receives from the requesting SU. If unknown, assume decimal 4.
TSN (7 bits):	The Transmission Sequence Number identifies a particular talk spurt in the RTP stream. Use of the TSN is described in Section 7.
L (1-bit):	When set, the Losing Audio Bit identifies a particular talk spurt as being less important (in the opinion of the transmitting RFSS) than some other talk spurt in the RTP stream. Use of the Losing Audio Bit is described in Section 7.
Interval:	Relevant only to the heartbeat and heartbeat query types, the Interval specifies the periodicity, in seconds, at which the sending RFSS wishes to receive heartbeats from the receiving RFSS. For all other Packet Types, the Interval parameter is ignored.

PTT Transmit Request is sent by an SMF to the MMF to initiate a talk spurt. It is sent until the talk spurt ends or until it is explicitly acknowledged with a *PTT Transmit Grant*, or *PTT Transmit Wait*, or *PTT Transmit Deny*, or until the Serving RFSS determines that it has to proceed, thereby switching to *PTT Transmit Progress* packets.

PTT Transmit Grant is sent by the MMF to the SMF in acknowledgment of *PTT Transmit Request*. It indicates the willingness of the MMF to receive a particular talk spurt from the SMF.

PTT Transmit Deny is sent by the MMF to the SMF in acknowledgment of *PTT Transmit Request*. It indicates that the MMF is unwilling to receive a particular talk spurt from the SMF.

PTT Transmit Wait is sent by the MMF to the SMF as a provisional acknowledgement of *PTT Transmit Request*. It indicates that the MMF has not yet determined whether or not to grant the SMF. Upon making its determination, the MMF sends either *PTT Transmit Grant* or *PTT Transmit Deny* as a final acknowledgement of *PTT Transmit Request*.

PTT Transmit Start is sent by a MMF to inform a SMF of the start of a talk spurt. *PTT Transmit Start* is not acknowledged.

PTT Transmit Progress carries voice content. It MAY be sent by either the MMF or the SMF to the other.

PTT Transmit End indicates to a receiving RFSS that a talk spurt is complete. It MAY be sent by either the MMF or the SMF to the other.

PTT Transmit Mute indicates to the receiving RFSS that the sending RFSS is currently unwilling to receive the voice payloads for the indicated talk spurt. It MAY be sent by either the MMF or the SMF to the other.

PTT Transmit Unmute indicates to the receiving RFSS that the sending RFSS wishes to resume receipt of voice payloads for the indicated talk spurt. It MAY be sent by either the MMF or the SMF to the other.

PTT Heartbeat is used to aid in the detection of failed links. Heartbeats are sent periodically by both the MMF and the SMF whether or not there is voice related RTP traffic for the session.

PTT Heartbeat Query provides a mechanism for an RFSS to request that its peer immediately send *PTT Heartbeat*.

4.3.3 Full Rate ISSI Header Word (Block Type = 5)

[No change from existing text in this section of [102BACA].]

4.3.4 RF PTT Control Word (Block Type = 11)

[Replace existing text in this section of [102BACA] with the text below.]

Voice services often need to authorize usage based on the source and allocate resources based on the priority of that source. This information needs to be carried across interfaces such as the Inter-RF Subsystem Interface (ISSI). The RF PTT Control Word carries this required information and is used for all non-console initiated voice services over the ISSI.

Figure 49 shows the RF PTT Control Word

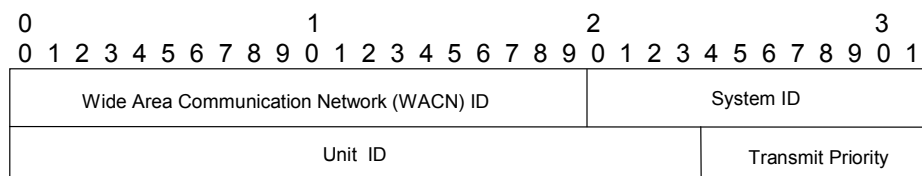


Figure 49 - RF PTT Control Word

For voice services over the ISSI, the PTT control fields SHALL be set as follows (all numbers are in decimal):

SU Identifier (56 bits): Fully qualified identifier (comprising the 20-bit WACN ID, 12-bit System ID, and 24-bit Unit ID) of an SU making a request for (or being granted) permission to transmit voice data (e.g., Full Rate frames), or of an SU that is currently transmitting voice data. In the event that the fully qualified SUID is unknown, all three fields (WACN, System and Unit) SHALL be set to 0.

NOTE: The SU Identifier could change during the talk spurt if the ISSI call was set up before the SU Identifier was known. In this case the default value of 0 is initially chosen and once the SU Identifier is known, the next PTT Transmit Progress's RF PTT Control Word is updated with the correct SU Identifier. The SU Identifier might not be initially known if the air interface is Message Trunked and the SU transmitted on the voice channel without first requesting to talk on the voice/control channel.

Transmit Priority (8 bits): Comprising a Priority Type (bits 7-4) sub-field and a Priority Level (bits 3-0) sub-field, as defined in **[T102]**. For trunked RFSSs, these sub-field values are set through various means, e.g., initially according to priority values in the Service Option field of the trunking control channel messages received from subtending SUs **[102AABC]**. The priority type is set to decimal 2 for preemptive priority talk spurts, 1 for priority talk spurts, and 0 for all other talk spurts. For example, for a Level 3 priority talk spurt, the priority type and priority level are set to 1 and 3, respectively. For conventional RFSSs, the priority type and priority level are set through other means than according to the values in the Service Option field, as there is no such field in the received conventional control messages **[102AABG]**.

4.3.5 Console PTT Control Word (Block Type = 15)

*[Replace existing text in this section of **[102BACA]** with the text below.]*

Voice services often need to authorize usage and allocate resources based on the type of source and the priority of the source's request. Also, end points such as consoles need to know the original source of the audio that a console is generating. This information needs to be carried across interfaces such as the Inter-RF Subsystem Interface (ISSI). The Console PTT Control Word enables these functions for console requested talk spurts.

Figure 50 shows the Console PTT Control Word

0										1										2										3									
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1								
Console SU Identifier (56 bit SUID)																																							
Console SU Identifier (continued)																								Transmit Priority															
Physical Console Identifier (56 bit SUID)																																							
Physical Console Identifier (continued)																								Transmission Descriptor															
Voice Source Identifier (56 bit SUID)																																							
Voice Source Identifier (continued)																								Console TX Priority															

Figure 50 - Console PTT Control Word

For voice services over the ISSI, the PTT control fields SHALL be set as follows (all numbers are in decimal):

Console SU Identifier (56 bit SUID): Fully qualified identifier (comprising the 20-bit WACN ID, 12-bit System ID, and 24-bit Unit ID) of a console SU making a request for (or being granted) permission to transmit voice data (e.g., Full Rate frames), or of a console SU that is currently transmitting voice data. In the event that the fully qualified SUID is unknown, all three fields (WACN, System and Unit) SHALL be set to 0. The default value of this field is the Physical Console Identifier field value.⁶

Transmit Priority (8 bits): Comprising a Priority Type (bits 7-4) sub-field and a Priority Level (bits 3-0) sub-field, as defined in [T102]. For console subsystems, these sub-field values are initially set according to local policy. The priority type is set to decimal 2 for preemptive priority talk spurts, 1 for priority talk spurts, and 0 for all other talk spurts. For example, for a Level 3 priority talk

⁶ Editor's Note: In the future, if this protocol is updated to support the conventional complexity 1 architecture, the Console SU Identifier field can be used to identify the console's assigned SUID for use on a particular conventional resource. In the complexity 1 architecture this field cannot be used to identify a physical console because the same console can be assigned different SUIDs for use on different conventional resources.

spurt, the priority type and priority level are set to 1 and 3, **respectively**.

Physical Console Identifier (56 bit SUID) This is the console's assigned Trunking SUID which is a fully qualified identifier (the console SUID comprising the 20-bit WACN ID, 12-bit System ID, and 24-bit Unit ID) of the console making a request for (or being granted) permission to transmit voice data (e.g., Full Rate frames), or of a console that is currently transmitting voice data. In the event that the fully qualified SUID is unknown or the transmission requestor or source is not a console, all three fields (WACN, System and Unit) SHALL be set to 0.

Transmission Descriptor (8 bits) Used to indicate various attributes associated with the console talk spurt.
 \$00: Indicates that the talk spurt is a trunking talk spurt⁷
 \$01: Indicates that the talk spurt is a conventional talk spurt⁸

Voice Source Identifier (56 bit SUID) A fully qualified identifier (an SUID comprising the 20-bit WACN ID, 12-bit System ID, and 24-bit Unit ID) of the talker that is the **original** source of the audio. In the event that the fully qualified SUID is unknown or, the source ID is not relevant in the domain of the talk spurt, all three fields (WACN, System and Unit) SHALL be set to 0. An example of where it would not be relevant in the domain of the talk spurt is when the console is patching a conventional talk spurt to a trunking talk group. The conventional SUID will not be relevant on the trunking talk group. The SUID is set to 0 to avoid duplicating an ID that already exists in the trunking system.

NOTE: The Voice Source Identifier could change during a talk spurt if the

⁷ Editor's Note: This setting is also used with conventional complexity 2 since the console does not know when trunking groups and units are mapped to conventional groups and units on a particular conventional fixed station.

⁸ This setting is for future use in the event that this protocol is updated to support the conventional complexity 1 architecture.

console is patching a call before the Voice Source Identifier is known. In this case the default value of 0 is initially chosen and once the Voice Source Identifier is known, the next PTT Transmit Progress's Console PTT Control Word is updated with the correct Voice Source Identifier. The Voice Source Identifier might not be initially known if the air interface is Message Trunked and the SU (whose transmission is being patched) transmitted on the voice channel without first requesting to talk on the voice/control channel.

Console TX Priority (8 bits) The priority associated with the console's transmission request. All values \$00 through \$FF are allowed, recommended values are:

- \$10: priority 1 (lowest and default value)
- \$30: priority 2
- \$50: priority 3
- \$70: priority 4
- \$90: priority 5
- \$B0: priority 6
- \$D0: priority 7
- \$F0: priority 8

4.3.5.1 General Usage of the Console PTT Control Word Block
[No change from existing text in this section of [102BACA].]

4.3.5.2 Transmission Descriptor Field Usage
[No change from existing text in this section of [102BACA].]

4.3.5.3 Physical Console Identifier Field Usage
[No change from existing text in this section of [102BACA].]

4.3.5.4 Console SU Identifier Field Usage
[No change from existing text in this section of [102BACA].]

4.3.5.5 Voice Source Identifier Field Usage

[No change from existing text in this section of **[102BACA]**.]

4.3.6 Half Rate ISSI Header Word (Block Type = 33)

[No change from existing text in this section of **[102BACA]**.]

4.3.7 Half Rate Voice (Block Type = 34)

[No change from existing text in this section of **[102BACA]**, except for subsections 4.3.7.1, 4.3.7.2 below]

4.3.7.1 4V Half Rate Voice Block (\$1 - \$5)

[Replace existing text in this section of **[102BACA]** with the text below.]

The 4V Half Rate Voice data block, corresponding to frame types \$1 to \$5, is as shown in Figure 52.

0										1										2										3									
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1								
Frame Type										U0-0 (b11-0)										U0-1 (b11-0)										U0-2 (b10-7)									
U0-2 (b6-0)										U0-3 (b13-0)										Et (0)			E1 (0)			M	L	U1-0 (b11-9)											
U1-0 (b8-0)										U1-1 (b11-0)										U1-2 (b10-0)																			
U1-3 (b13-0)										Et (1)			E1 (1)			M	L	U2-0 (b11-2)																					
U2-0 (b1-0)	U2-1 (b11-0)										U2-2 (b10-0)										U2-3 (b13-7)																		
U2-3 (b6-0)										Et (2)			E1 (2)			M	L	U3-0 (b11-0)										U3-1 (b11-7)											
U3-1 (b6-0)										U3-2 (b10-0)										U3-3 (b13-0)																			
Et (3)			E1 (3)			M	L	TS			ESS-B (b23-4)																												
ESS-B (b3-0)					Reserved					NOT SENT																													

Figure 52 – 4V Half Rate Voice Data

The 4V Half Rate Voice data block contains 4 voice frames.

Each voice frame, of frame number "n", where n= 0, 1, 2, 3, of the Half Rate Voice data is divided into 4 message vectors (Un-0, Un-1, Un-2, Un-3). These message vectors have been decoded (error correction code removed).

Each voice frame also includes error mitigation data as defined in sections C.3 and C.4 in Annex C of this document. The error count fields, E_t and E_1 , SHALL reflect the number of decoding errors detected when the over the air frame was recovered at a receiver and the error correcting code (2 Golay codes) was decoded. These MAY be used for error mitigation when the voice is decoded. If the number of errors detected is greater than can be indicated in the report field, the field SHALL be set to the maximum value.

$E_t(n)$ SHALL be the total number of errors detected in the n^{th} frame. This SHALL be the parameter ε_T described by equation 196 in reference [102BABA], limited to a maximum value of 7.

$E_1(n)$ SHALL be the number of errors detected in the second Golay codeword of the n^{th} frame, $\hat{c}_1$, described by equation 190 in reference [102BABA]. This SHALL be set to the value of ε_1 of equation 196 in reference [102BABA], limited to a maximum value of 7.

M is the mute frame flag, and L is the lost frame flag. The originating RFSS MAY set M and/or L values for any given frame based upon errors described in sections 14.6 and 14.7 of [102BABA], or in accordance with local policy. When M is set to 1, the audio output can be muted. When M is set to 0 and L is set to 1, the previous audio output can be repeated.

TS is a 4-bit field that indicates the TDMA timeslot in which the inbound 4V Half Rate Voice data block was received. The TS field value SHOULD be provided. If unknown, the TS field value SHALL be set to %xF (unknown).

ESS-B is a 24-bit field containing encoded Encryption Synchronization Signal (ESS) information associated with each 4V voice frame, as defined in [102BBAC].

4.3.7.2 2V Half Rate Voice Block (\$6)

[No change from existing text in this section of [102BACA], except for Subsection 4.3.7.2.1 below.]

4.3.7.2.1 Fully Decoded Encryption Synchronization Signalling Information

[Replace existing text in this section of [102BACA] with the text below.]

In this case (EXT = 0) the inbound ESS information has been successfully decoded by the RFSS.

Furthermore, the Fully Decoded ESS format (EXT=0) SHALL be used when the RFSS encodes a talk spurt's audio for transmission over the ISSI. For example, when a console within the RFSS is the originator of a talk spurt that is transmitted on the ISSI.

The 2V voice data block with fully decoded ESS information is as shown in Figure 53.

0											1											2											3										
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1												
Frame Type		U0-0 (b11-0)											U0-1 (b11-0)											U0-2 (b10-7)																			
U0-2 (b6-0)						U0-3 (b13-0)											Et (0)		E1 (0)		M	L	U1-0 (b11-9)																				
U1-0 (b8-0)						U1-1 (b11-0)											U1-2 (b10-0)																										
U1-3 (b13-0)											Et (1)		E1 (1)		M	L	TS		EXT		MI (b71-68)																						
MI (b67-36)																																											
MI (b35-4)																																											
MI (b3-0)		ALGID (b7-0)						KeyID (b15-0)																		Reserved																	

Figure 53 - 2V Half Rate Voice Data with Fully Decoded ESS

The Message Indicator (MI), Algorithm ID (AlgID), and Key ID (KeyID) are as defined in [102BBAC].

4.3.7.2.2 ESS-A Raw Parity Bits Information

In this case (EXT = 1) the inbound ESS information cannot be decoded by the RFSS. The 2V voice data block with ESS-A raw parity bits information is as shown in Figure 54.

0										1										2										3									
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1								
Frame Type					U0-0 (b11-0)															U0-1 (b11-0)															U0-2 (b10-7)				
U0-2 (b6-0)							U0-3 (b13-0)																		Et (0)			E1 (0)			M	L	U1-0 (b11-9)						
U1-0 (b8-0)										U1-1 (b11-0)															U1-2 (b10-0)														
U1-3 (b13-0)															Et (1)			E1 (1)			M	L		TS			EXT		ESS-A1 (b95-92)										
ESS-A1 (b91-60)																																							
ESS-A1 (b59-28)																																							
ESS-A1 (b27-0)																									ESS-A2 (b71-68)														
ESS-A2 (b67-36)																																							
ESS-A2 (b35-4)																																							
ESS-A2 (b3-0)					Reserved					NOT SENT																													

Figure 54 - 2V Half Rate Voice Data with ESS-A Raw Parity Bits

ESS-A is a 168-bit field, as defined in [102BBAC].

4.3.8 TalkSpurt Vocoder Mode (Block Type = 35)

[No change from existing text in this section of [102BACA].]

4.3.9 User Alias (Block Type = 36)

[Add this new section to [102BACA] with the text below.]

Equipment conforming to this standard can inform the receiving RFSS the SU Alias associated with the transmitting SUID. In the case of console requested talk spurts, the transmitting SUID would be the Voice Source Identifier in the Console PTT Control Word (see Section 4.3.5).

In the event that an RFSS supports transmission of SU Alias for any given talk spurt, it SHALL include the SU Alias, if available, in the PTT Transmit Request or PTT Transmit Start, and it SHOULD include the SU Alias in the PTT Transmit Progress at most every superframe. If an RFSS unmutes due to PTT Transmit Unmute, that constitutes a new talk spurt.

The Block Header defined in Section 4.2.2 above includes the length of the information within the block. The maximum length of the User Alias block is 48 octets as per Section 3.8.2.17 of this document. The block contains the SU Alias which is formatted per the guidelines of Section 3.8.2.17 above, with the leftmost character of the SU Alias as leading octet(s) in the block and the rightmost character as trailing octet(s) in the block. The general form of the block header and the block are shown below. In addition, an example block is provided.

Example:

```
u-alias:"Precinct26-ELsu-H94"
```

Block Header for User Alias Block Type =36																																							
012345678901234567890123456789012345678901																																							
012345678901234567890123456789012345678901																																							
E=1Block Type = 36TimeStamp Offset = 0Length ≤ 48																																							
General Form of User Alias Block Type = 36																																							
012345678901234567890123456789012345678901																																							
012345678901234567890123456789012345678901																																							
octet[0]										octet[1]										octet[2]										octet[3]									
octet[4]										octet[5]										octet[6]										octet[7]									
octet[8]										octet[9]										octet[10]										octet[11]									
octet[12]										octet[13]										octet[14]										octet[15]									
octet[16]										octet[17]										octet[18]										octet[19]									
octet[20]										octet[21]										octet[22]										octet[23]									
octet[24]										octet[25]										octet[26]										octet[27]									
octet[28]										octet[29]										octet[30]										octet[31]									
octet[32]										octet[33]										octet[34]										octet[35]									
octet[36]										octet[37]										octet[38]										octet[39]									
octet[40]										octet[41]										octet[42]										octet[43]									
octet[44]										octet[45]										octet[46]										octet[47]									
Example of User Alias Block Type = 36																																							
Precinct26-ELsu-H94																																							
12345678910111213141516171819																																							
50726563696e637432362d454c73752d483934																																							
Block Header for Example of User Alias Block Type =36																																							
012345678901234567890123456789012345678901																																							
012345678901234567890123456789012345678901																																							
E=1Block Type = 36TimeStamp Offset = 0Length =19																																							
Example User Alias Block																																							
012345678901234567890123456789012345678901																																							
012345678901234567890123456789012345678901																																							
50										72										65										63									
69										6e										63										74									
32										36										2d										45									
4c										73										75										2d									
48										39										34										Not Sent									

4.3.10 Manufacturer specific data (Block Types 63 – 127)

[No change from existing text in this section of [102BACA].]

4.4 Construction of RTP Packets over the ISSI

[No change from existing text in this section of [102BACA], except for selected subsections thereof, as shown below.]

4.4.1 Basic Construction for All Vocoder Modes

[Replace existing text in this section of [102BACA] with the text below.]

The following basic rules SHALL be applied in the construction of RTP packets transmitted over the ISSI.

1. An ISSI RTP packet SHALL include exactly one ISSI Packet Type block. In the following discussion, the Packet Type from the Packet Type Block is used to identify the packet type (e.g. a PTT Transmit Progress packet has a Packet Type of PTT Transmit Progress (2)).
2. The first ISSI Block in an RTP packet over the ISSI SHALL be an ISSI Packet Type block (Block Type = 1).
3. The second block in PTT Transmit Request, PTT Transmit Start and PTT Transmit Progress packets SHALL be the PTT Control Word Block (Block Type = 11 or 15). Console subsystems SHALL use Block Type 15 when a console is generating the talk spurt. Console subsystems SHALL use Block Type 11 for mobile requested talk spurts (when the console subsystem includes RF sites). All non-console RF subsystems SHALL use Block Type 11.
4. A PTT Transmit Request, or PTT Transmit Start, or PTT Transmit Progress packet MAY carry a User Alias Block pertaining to the transmitting SUID in the associated PTT Control Word Block.
5. Manufacturer Specific Blocks (Block Types 63 – 127) MAY appear in any RTP packet, and SHALL be the last sequential blocks in the packet.

4.4.2 Specific Construction for Full Rate Vocoder Mode

[Replace existing text in this section of [102BACA] with the text below.]

Figure 57 illustrates the various forms of ISSI RTP packets for Full Rate Vocoder Mode.

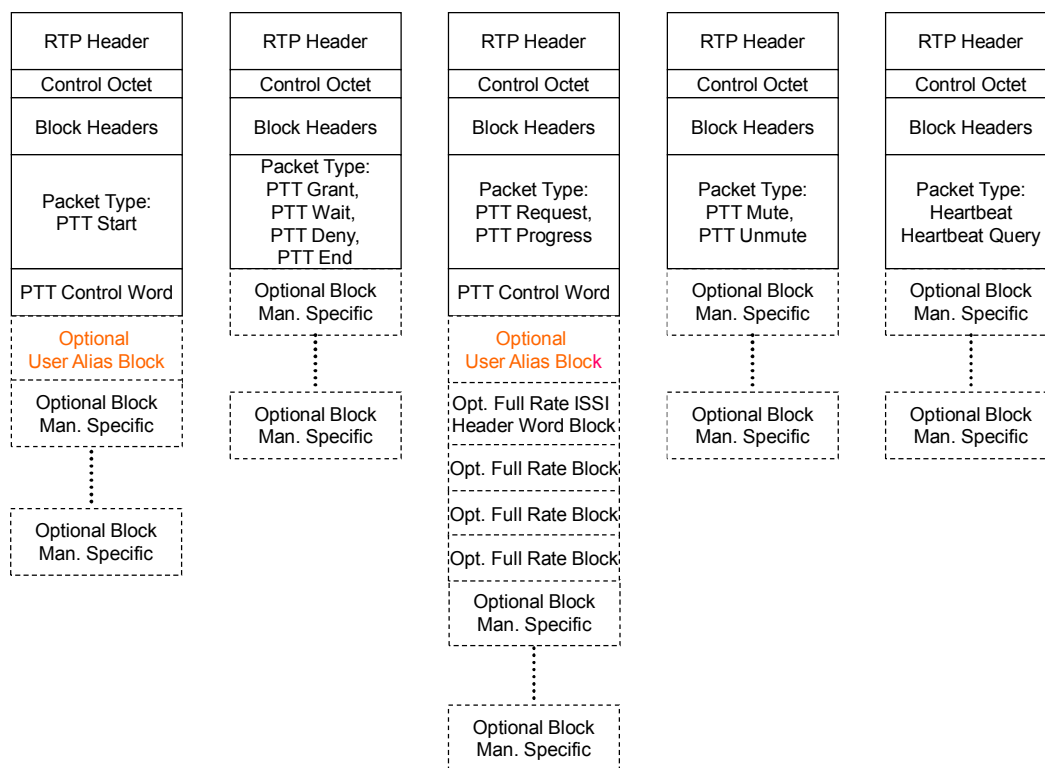


Figure 57 - ISSI RTP Packet Formats for Full Rate Vocoder Mode

In addition to the basic rules specified in Section 4.4.1, the following Full Rate Mode specific rules SHALL be applied in the construction of RTP packets transmitted over the ISSI:

1. Full Rate Voice Blocks (Block Type = 0) and Full Rate ISSI Header Word Blocks (Block Type 5) SHALL only be carried in PTT Transmit Request or PTT Transmit Progress packets.
2. Full Rate Voice Blocks and Full Rate ISSI Header Word Blocks SHALL appear in time order, with the earlier arriving blocks appearing before the later arriving blocks.
3. A single PTT Transmit Request or PTT Transmit Progress packet MAY carry zero to three Full Rate Voice Blocks, representing up to 60 msec. of speech. All of the Full Rate bearing RTP packets in a given talk spurt for a given TSN SHALL contain the same number of Full Rate blocks per

RTP packet, with the allowable exception of the last packet in the talk spurt. When multiple Full Rate blocks are bundled into each RTP packet, the RTP packet contents SHALL be aligned with the Full Rate superframe boundaries such that the FR1 (formerly IMBE1) block is always the first Full Rate block within any RTP packet which contains an FR1 (formerly IMBE1) block. I.e., FR18 and FR1 (formerly IMBE18 and IMBE1) SHALL never appear in the same RTP packet.

4.4.3 Specific Construction for Half Rate Vocoder Mode

[Replace existing text in this section of [102BACA] with the text below.]

Figure 58 illustrates the various forms of ISSI RTP packets for Half Rate Vocoder Mode.

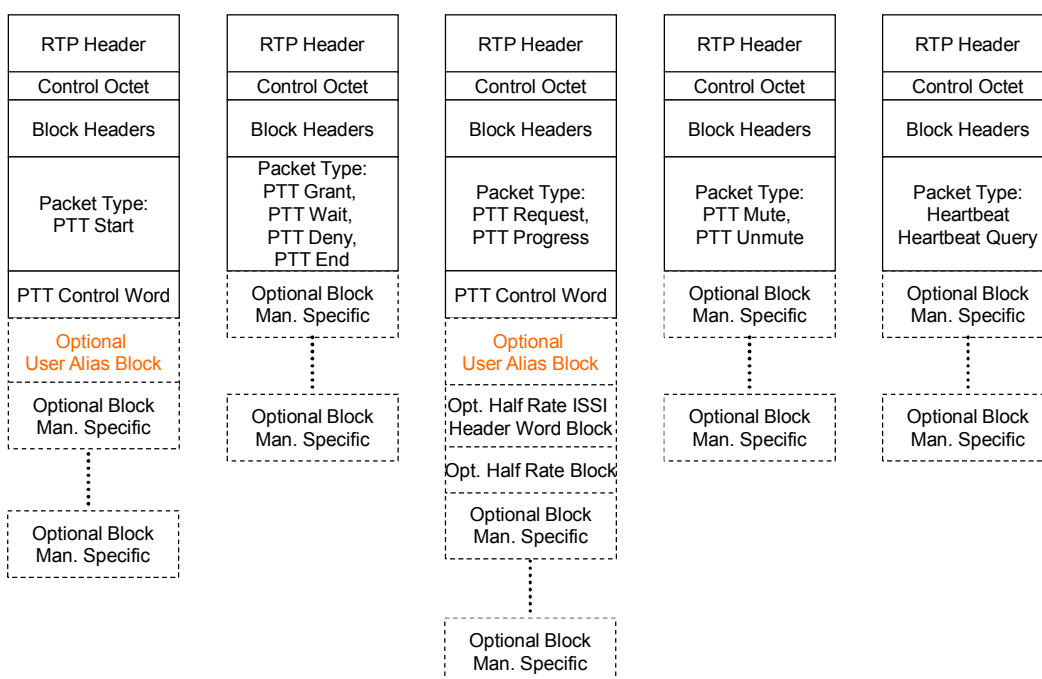


Figure 58 - ISSI RTP Packet Formats for Half Rate Vocoder Mode

In addition to the basic rules specified in Section 4.4.1, the following Half Rate

Mode specific rules SHALL be applied in the construction of RTP packets transmitted over the ISSI:

1. Half Rate ISSI Header Word Blocks (Block Type = 33) and Half Rate Voice Blocks (Block Type = 34) SHALL only be carried in PTT Transmit Request or PTT Transmit Progress packets.
2. Half Rate ISSI Header Word Blocks and Half Rate Voice Blocks SHALL appear in time order, with the earlier arriving blocks appearing before the later arriving blocks.
3. A single PTT Transmit Request or PTT Transmit Progress packet MAY carry a 4V Half Rate Voice Block, representing 80 msec. of speech plus ESS information, or a 2V Half Rate Voice Block, representing 40 msec. of speech plus ESS information.

4.4.4 Specific Construction for Native Vocoder Mode

[Replace existing text in this section of [102BACA] with the text below.]

Figure 59 illustrates the various forms of ISSI RTP packets for Native Vocoder Mode.

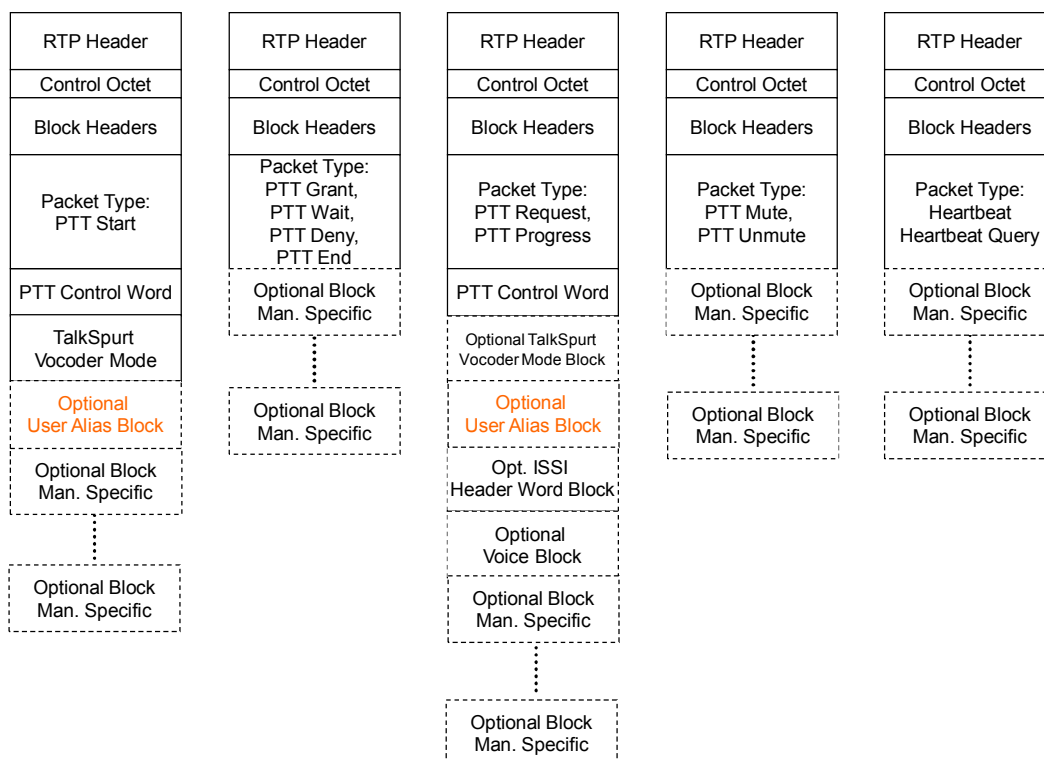


Figure 59 - ISSI RTP Packet Formats for Native Vocoder Mode

In addition to the basic rules specified in Section 4.4.1, the following Native Mode specific rules SHALL be applied in the construction of RTP packets transmitted over the ISSI:

1. The TalkSpurt Vocoder Mode Block (Block Type = 35), which indicates the vocoder mode of any given talk spurt, SHALL be carried in the PTT Transmit Start packets, and any PTT Transmit Request or PTT Transmit Progress packets that do not include voice blocks. Every PTT Transmit Request and PTT Transmit Progress packet SHALL include either one voice block type or a TalkSpurt Vocoder Mode Block.
2. For a Full Rate talk spurt, the Full Rate specific rules as specified in Section 4.4.2 SHALL apply.
3. For a Half Rate talk spurt, the Half Rate specific rules as specified in Section 4.4.3 SHALL apply.

5. Mobility Management

[No change from existing text in this section of [102BACA], except for Subsection 5.5 below.]

5.1 Definition of Mobility Objects

[No change from existing text in this section of [102BACA].]

5.2 Variations of the SIP REGISTER message

[No change from existing text in this section of [102BACA].]

5.3 MM-SAP/Message Modeling

[No change from existing text in this section of [102BACA].]

5.4 State Based Summary

[No change from existing text in this section of [102BACA].]

5.5 Mobility Management Procedures

[Replace existing text in this section of [102BACA], and selected subsections thereof, as shown below.]

Five Mobility Management procedures are defined herein:

1. **Serving Registration (SIP REGISTER-Register)** – the Serving Registration procedure is used by a Serving RFSS to register a mobility object (e.g., SU).

The Serving Registration mobility management procedure is applicable to all nine classes of mobility objects. Note that in all cases where multiple entities can express interest through Serving Registration in a single mobility object (Group addressed SD, General FNE addressed SD, and SD Monitoring), each Serving RFSS expresses interest once for all interested entities at that RFSS.

The Serving Registration procedure remains applicable to an SU (class 1) mobility object even when the object is inhibited.

When the Serving Registration procedure is used to register a Class 2 object (Group addressed voice) the Serving RFSS and the SG Home RFSS MAY use the procedure to inform each other of the vocoder mode(s) that they support for the group being registered. Possible

supported modes include Full Rate, Half Rate, and Native mode.

2. **Serving Query (SIP REGISTER-Query)** – the Serving Query procedure is used by a Serving RFSS to query a Home RFSS:

- a. For service profile information for an SU (class 1) mobility object even when the object is inhibited, or
- b. For the most current SMDCP context information (known to be agreed between the Serving RFSS and the SU) stored in the Home RFSS for an SN Context (class 9) mobility object.

Serving Query is not applicable to Group, supplementary data and supplementary data monitoring objects (classes 2-8).

3. **Home Query (SIP REGISTER-Query)** – the home query procedure is used by a Home RFSS:

- a. To query a Serving RFSS to determine whether or not it still has interest in an SU (class 1) or Group (class 2) mobility object. Home Query has two mutually exclusive qualifiers, for a total of three functions. The “force” qualifier requests that the Serving RFSS, either deny interest or, if it still has interest, re-register the mobility object via the Serving Registration procedure. In the case where the mobility object is an SU, the force qualifier requires that the SU be forced to re-register and re-affiliate with the Serving RFSS. The force qualifier MAY be used by a Home RFSS to, for example, withdraw permission for a mobility object to operate within a previously granted serving domain. The “confirm” qualifier requests that the Serving RFSS actively confirm its need for the subject mobility object.
- b. To initiate SMDCP context activation for an SN Context (class 9) mobility object, by sending a Home Query with the “force” qualifier to the Serving RFSS, causing it to send a Context Activation Command to the SU.
- c. To retrieve the most current SMDCP context information stored in the Serving RFSS for an SN Context (class 9) mobility object. Home Query without any qualifier is used in this case.

The Home Query procedure is not applicable to supplementary data and supplementary data monitoring objects (classes 3-8).

4. **Serving Deregistration (SIP REGISTER-Deregister)** – the serving deregistration procedure is initiated by a Serving RFSS to remove interest

in a mobility object.

The Serving Deregistration procedure is applicable to all nine classes of mobility objects.

5. **Home Deregistration (SIP REGISTER-Deregister)** – the home deregistration procedure is initiated by a Home RFSS:

- a. To inform a Serving RFSS that an SU (class 1) that was previously operating in its domain has moved to a different domain, or
- b. To deregister any of the nine classes of mobility objects from a Serving RFSS for reasons dependent upon local policy.

The “SU Roaming” qualifier is used to distinguish between these two types of home deregistration.

The deregistration of an SU via the home deregistration procedure MAY require that the Serving RFSS deregister groups that were affiliated with the SU if no other demand for the group exists within that RFSS.

SIP REGISTER messages SHALL be transmitted in accordance with the progressive timeout retransmission protocol for unreliable transport specified in **[RFC3261]** (Sections 17.1.2 and 17.2.2), subject to the following qualifications. The following “general requirements” apply to all five mobility management procedures specified in this section.

1. The value of T1 and T2 (the base timers) SHALL be established by administrative agreement between system operators responsible for the communicating RFSSs. The default value for the TIA-102 ISSI SHALL be as given in Table 58.
2. In the event that a Serving RFSS receives a SIP REGISTER request for an object not present at the Serving RFSS:
 - a. For an SU mobility object, the Serving RFSS SHALL return an MME_SU_UNKNOWN failure response.
 - b. For an SG mobility object, the Serving RFSS SHALL return an MME_SG_UNKNOWN failure response.
 - c. For any other mobility object, the Serving RFSS SHALL return an MME_OBJECT_UNKNOWN failure response.

3. In the event that a Home RFSS receives a SIP REGISTER request for an object not homed to the RFSS:
 - a. For an SU mobility object, the Home RFSS SHALL return an MME_SU_UNKNOWN failure response.
 - b. For an SG mobility object, the Home RFSS SHALL return an MME_SG_UNKNOWN failure response.
 - c. For any other mobility object, the Home RFSS SHALL return an MME_OBJECT_UNKNOWN failure response.
4. In the event that a Home RFSS receives a SIP REGISTER request pertaining to a Mobility Management procedure that it does not support, it SHALL return a GEN_PROC_UNSUPPORTED failure response, as specified in Annex A of this document.
5. In the event that a Serving RFSS receives a SIP REGISTER request pertaining to a Mobility Management procedure that it does not support, it SHALL return a GEN_PROC_UNSUPPORTED failure response, as specified in Annex A of this document.
6. Specific responses found in the following procedures SHALL be mandatory in preference to other responses implied or specified in **[RFC3261]**.
7. An RFSS MAY generate additional SIP responses of types 4xx, 5xx and 6xx, as defined in Annex A of this document, but never in preference to those specified herein.
8. With the exception of 183 Session Progress, no 1xx or 3xx responses SHOULD be generated in response to a SIP REGISTER request. When received, a 1xx response MAY be ignored, and a 3xx response SHOULD terminate the SIP transaction.
9. Contrary to **[RFC3261]** (Section 10.3, item 8), an RFSS responding to a SIP REGISTER request SHALL NOT enumerate all bindings for the given object in 200 OK responses. It SHALL, instead, only return the current binding between the RFSS initiating the request and the mobility object's Home RFSS.
10. When an RFSS responds to a SIP REGISTER request with a 200 OK, the contact header field in the 200 OK, if present, SHALL reflect the domain of the responding RFSS (see also Table 14). Thus:

- a. In the event that the responding RFSS is the Home RFSS of the mobility object, the "domain-name" in the contact header field of the 200 OK response SHALL be the domain-name representation of the Home RFSS.

NOTE: This is contrary to [RFC3261] (Section 10.3, item 8), which requires the 200 OK response returned by the registrar (i.e., the Home RFSS of the mobility object in ISSI context) to contain Contact header field value for the current binding (i.e., current contact address in the Serving RFSS's SIP REGISTER request, to which subsequent SIP requests for the mobility object should be forwarded), with the "domain-name" in the contact header field of the 200 OK response being the domain-name representation of the Serving RFSS, not that of the Home RFSS.

- b. In the event that the responding RFSS is the Serving RFSS of the mobility object, the "domain-name" in the contact header field of the 200 OK response SHALL be the domain-name representation of the Serving RFSS.

11. Following a timeout, a request SHALL NOT be retried for at least RetryTime seconds.

5.5.1 Serving Registration Procedure

[Replace existing text in this section of [102BACA] with the text below.]

The Serving Registration procedure is invoked by a Serving RFSS when it determines that it has interest in services associated with a mobility object. A typical trigger for a Serving RFSS to register an SU is the registration of the SU at a site within the serving domain of the RFSS. Group affiliation is a typical trigger for the Serving Registration of a group (although it should be noted that only the first affiliation to a particular group need be registered by any particular Serving RFSS). SNDCP context activation is a typical trigger for the Serving Registration of an SN Context. The Serving Registration procedure SHALL proceed as follows:

1. The Serving RFSS SHALL send a SIP REGISTER-Register message.
 - a. When the mobility object being registered is an SU or SN Context, the Serving RFSS MAY include an SU Presence parameter in the SIP REGISTER-Register message:
 - i. If the Serving Registration procedure is being invoked as a result of SU registration at the Serving RFSS, the Serving RFSS SHOULD include the SU Presence parameter with a value of "Confirmed" in the SIP REGISTER-Register message. A Serving RFSS SHALL NOT set the SU Presence parameter value to "Confirmed" if the Serving RFSS has not confirmed the presence of the SU being registered immediately prior to sending the SIP REGISTER-register message.
 - ii. If the Serving Registration procedure is being invoked as a result of an SU's initial SNDCP context activation request at the Serving RFSS, the Serving RFSS SHALL include in the SIP REGISTER-Register message:
 1. The SU Presence parameter with a value of "Confirmed", and
 2. The SNC Change parameter with a value of "No Change", and
 3. A P25-SNDCP body containing pertinent SNDCP context parameters received from the SU, as specified in Section 3.9.
 - iii. If the Serving Registration procedure is being invoked because the Serving RFSS wishes to update the Home RFSS on subsequent changes in SNDCP context parameter values for an existing SNDCP context, the Serving RFSS SHALL include in the SIP REGISTER-Register message:

1. the SU Presence parameter with a value of “Confirmed”, and
 2. The SNC Change parameter with a value of “Change”, and
 3. A P25-SNDCP body containing updated SNDCP context parameter values subsequently determined by the Serving RFSS and accepted by the SU, as specified in Section 3.9.
- iv. If the Serving Registration procedure is being invoked because the Serving RFSS wishes to refresh the SIP Register timer for an SN Context registration request, or in accordance with local policy, the Serving RFSS SHALL include the SU Presence parameter, with a value set to “Not Confirmed”, in the SIP REGISTER-Register message. The SIP REGISTER-Register message SHALL NOT include the SNC Change parameter.
- b. When the mobility object being registered is an individual SU, the Serving RFSS MAY include the SU Authentication parameter in the SIP REGISTER-Register message. If the Serving Registration procedure is being invoked as a result of SU registration with successful SU authentication at the Serving RFSS, the Serving RFSS SHALL include the SU Authentication parameter with a value of “SU Authentication Confirmed” in the SIP REGISTER-Register message. A Serving RFSS SHALL NOT set the SU Authentication parameter value to “SU Authentication Confirmed” if the Serving RFSS has not successfully authenticated the SU being registered prior to sending the SIP REGISTER-register message.
 - c. When the mobility object being registered is a group (SG), the Serving RFSS MAY include the Supported Vocoder Modes parameter in the SIP REGISTER-Register message. The supported vocoder modes are listed in order of preference as specified in Section 3.8.4.8.
2. Upon receipt of a SIP REGISTER-register the Home RFSS SHALL behave as follows:
 - a. In the event that the Home RFSS wishes to allow the object to be registered,

- i. it SHALL respond with a 200 OK message. The 200 OK message SHALL feature an "expires" header field indicating its registration expiration interval set by the Home RFSS, and the body SHALL include service profile information for the requested object.
 - 1. For SU registration, the Home RFSS SHOULD set the "Expires" header field to a value greater than 14400 (decimal).

NOTE: The value 14400 in seconds represents the SU's minimum (default) WUID validity timer value of 4 hours, per [AABD]. By setting the "Expires" header field to a value greater than 14400, the Home RFSS ensures that SIP registration for the SU over the ISSI expires later than the SU's minimum (default) WUID validity timer expiration. This helps minimize unconfirmed registration renewals from overriding SU-initiated registration in race conditions, while providing protection against excessive system loading.

- 2. For other mobility objects, the Home RFSS MAY set the "Expires" header field in accordance with local policy to facilitate timely database recovery in the event of a Home RFSS failure.
- ii. Furthermore, the Home RFSS SHALL update its location information for the mobility object based on the *contact* header field of the received SIP REGISTER message.
- iii. Furthermore, in the event that the object is a subscriber unit, and the subscriber was previously registered with a different Serving RFSS:
 - 1. The Home RFSS SHALL initiate a de-registration request procedure to the previous Serving RFSS, if any.
 - 2. The 200 OK returned to the New Serving RFSS SHALL include a P25-SNDCP body containing the context parameters for all active contexts. If there are no active contexts then the P25-SNDCP body is not present.

- iv. Furthermore, in the event that the object is a group (SG):
 - 1. If the received SIP REGISTER-Register message includes the Supported Vocoder Modes parameter, the Home RFSS SHOULD update its supported vocoder mode information associated with the Serving RFSS and the SG mobility object.
 - 2. The 200 OK message returned to the Serving RFSS MAY include the Supported Vocoder Modes parameter. The supported vocoder modes are listed in order of preference as specified in Section 3.8.4.8.
- v. Furthermore, in the event that the object is an SN Context, and the received SIP REGISTER-Register message includes the SNC Change parameter with a value of “No Change”, the Home RFSS SHALL verify the received parameters and determine the values of parameters, if any, to be returned to the Serving RFSS. If the received parameters are acceptable, the Home RFSS SHALL return a 200 OK message to the Serving RFSS. The 200 OK message SHALL include a P25-SNDCP body that contains a Home **Packet Data ISSI Gateway** (PDIG) Address and other pertinent SNDCP parameters as specified in Section 3.9.
- vi. Furthermore, in the event that the object is an SN Context, and the received SIP REGISTER-Register message includes the SNC Change parameter with a value of “Change”, the Home RFSS SHALL verify the received parameters. If the received parameters are acceptable, the Home RFSS SHALL update the stored parameter values for that object accordingly and return a 200 OK message to the Serving RFSS.
- vii. Following the Serving Registration of the object, INVITE messages for voice services SHALL be accepted from and propagated to the Serving RFSS in accordance with the call setup procedures.
- b. In the event that the Home RFSS wishes to disallow registration of the object at the Serving RFSS:
 - i. For an SU mobility object:

1. The Home RFSS SHALL return an MME_SU_AUTHEN_REQUIRED failure response in the event of the following:
 - a. The Home RFSS requires authentication of the SU, and
 - b. The SU Authentication parameter in the SIP REGISTER-register message from the Serving RFSS is set to "SU Authentication NOT performed" or the SU Authentication parameter is absent in the SIP REGISTER-register message from the Serving RFSS.
 2. The Home RFSS SHALL return an MME_MUST_CONFIRM_SU_PRESENCE failure response in the event of the following:
 - a. The SIP REGISTER-register message from the Serving RFSS does not contain the SU Presence parameter or that it contains the SU Presence parameter with a value of "Not Confirmed", and
 - b. The Home RFSS has detected that the SU is currently registered at a different Serving RFSS.
 3. In all other cases where the Home RFSS wishes to disallow Serving Registration of the SU, the Home RFSS SHALL return an MME_SU_REG_DISALLOWED failure response.
- ii. For an SG mobility object, the Home RFSS SHALL return an MME_SG_REG_DISALLOWED failure response.
 - iii. For an SN Context mobility object:
 1. In the event that the Home RFSS detects that the SU is not authorized to perform packet data services, it SHALL return a PDE_SU_NOT_AUTHOR failure response.

2. In the event that the Home RFSS detects that the static IP address provided by the SU is incorrect, it SHALL return a PDE_STAT_IP_ADDR_INCORRECT failure response.
 3. In the event that the Home RFSS detects that static IP address assignment is not supported, it SHALL return a PDE_STAT_IP_ADDR_UNSUPPORTED failure response.
 4. In the event that the Home RFSS detects that there are no more available IP addresses for a dynamic IP address assignment, it SHALL return a PDE_DYN_IP_ADDR_POOL_EMPTY failure response.
 5. In the event that the Home RFSS detects that dynamic IP address assignment is not supported, it SHALL return a PDE_DYN_IP_ADDR_UNSUPPORTED failure response.
 6. In all other cases where the Home RFSS wishes to disallow Serving Registration of the SN Context mobility object, the Home RFSS SHALL return an MME_OBJECT_REG_DISALLOWED failure response.
- iv. For any other mobility object, the Home RFSS SHALL return an MME_OBJECT_REG_DISALLOWED failure response.
3. Upon receipt of any response to the SIP REGISTER message, the Serving RFSS SHALL behave as follows:
 - a. In the event that the response is a 200 OK message, the object has been successfully registered. The Serving RFSS MAY schedule periodic re-registrations of the object to occur before the time specified in the *expires* header field of the 200 OK response has elapsed. If the 200 OK message is the response to a Serving Registration request for an SG mobility object, and if the 200 OK contains an ISSI body with a Group Service Profile which includes an Announcement Group attribute, then the Serving RFSS SHALL initiate a Serving Registration request for the announcement subscriber group indicated in the profile unless the Serving RFSS

already has registered interest in the announcement subscriber group. If the 200 OK message is the response to a Serving Registration request for an SG mobility object, and if the 200 OK contains the Supported Vocoder Modes parameter, then the Serving RFSS SHOULD update its supported vocoder mode information associated with the Home RFSS and the SG mobility object.

- b. In the event that the response is an MME_SU_UNKNOWN or MME_SG_UNKNOWN or MME_OBJECT_UNKNOWN failure response (as specified in the “general requirements” above), the Home RFSS is asserting that it is not the home for the given object. There is disagreement among system **databases** that needs to be resolved. The Serving RFSS SHOULD discontinue any attempt to register the object with the Home RFSS.
- c. In the event that the response is one of the following failure messages:
 - i. MME_MUST_CONFIRM_SU_PRESENCE
 - ii. MME_SU_REG_DISALLOWED
 - iii. MME_SG_REG_DISALLOWED
 - iv. MME_OBJECT_REG_DISALLOWED
 - v. PDE_SU_NOT_AUTHOR
 - vi. PDE_STAT_IP_ADDR_INCORRECT
 - vii. PDE_STAT_IP_ADDR_UNSUPPORTED
 - viii. PDE_DYN_IP_ADDR_POOL_EMPTY
 - ix. PDE_DYN_IP_ADDR_UNSUPPORTED

the Home RFSS has denied registration of the object at the Serving RFSS. The Serving RFSS SHALL discontinue any attempt to register the object until it receives a subsequent registration trigger. If appropriate, the Serving RFSS MAY notify the source of the registration trigger (e.g. an SU attempting to register or affiliate) of the error condition. Furthermore, the Serving RFSS SHOULD initiate de-registration procedures for other objects (such as groups) that were registered as a result of interest from the forbidden object. (For example, it would deregister any groups for which the SU was the only affiliated member.)

- d. In the event that the response is an MME_SU_AUTHEN_REQUIRED failure response, the Home RFSS has denied registration of the SU at the Serving RFSS due to failure of the Serving RFSS to authenticate the SU. The Serving

RFSS SHALL act in accordance with local policy to authenticate the SU or deny the SU registration.

- e. Other 4xx, 5xx and 6xx messages SHALL be interpreted in accordance with the SIP RFC and local policy.
- 4. If the SIP Serving Registration procedure terminates due to a SIP timeout (Section 17.1.2 of **[RFC3261]**), the Home RFSS is unreachable. The Serving RFSS SHALL take appropriate error recovery steps in accordance with its security and other policies, and MAY retry the request at a later time.

5.5.2 Serving Query

[Replace existing text in this section of [102BACA] with the text below.]

A Serving RFSS MAY query the Home RFSS of any applicable mobility object (as specified in Section 5.5) via the Serving Registration Query procedure at any time before or after Serving Registration of the mobility object with the Home RFSS.

A Serving RFSS that supports authentication and requires prior knowledge of the Service Profile information of a SU mobility object, SHALL query the Home RFSS using the Serving Registration Query procedure prior to authentication of the SU mobility object and subsequent Serving Registration of the SU mobility object with the Home RFSS.

A Serving RFSS that requires knowledge of the most current SMDCP context information stored in the Home RFSS for an SN Context (class 9) mobility object MAY query the Home RFSS of the object via the Serving Registration Query procedure. A typical trigger for a Serving Registration Query might be the need to re-synchronize databases in accordance with local policy.

The procedure SHALL proceed as follows:

1. The Serving RFSS SHALL send a SIP REGISTER-Query message to the Home RFSS of the mobility object.
2. A Home RFSS that receives an error free SIP REGISTER-Query message SHALL behave as follows:
 - a. In the event that the Serving RFSS is not authorized to receive information about the subject SU, return an MME_SU_INFO_NOT_AUTHOR failure response.

b. Otherwise:

- i. For a Serving Query pertaining to an SU (class 1) mobility object, respond with a 200 OK message containing the requested service profile information. The 200 OK message SHALL also include a P25-SNDCP body containing pertinent SNDCP context parameters (as specified in Section 3.9) for all currently registered SN Contexts.
- ii. For a Serving Query pertaining to an SN Context (class 9) mobility object, respond with a 200 OK message that includes a P25-SNDCP body containing the most current SNDCP context information (as specified in Section 3.9) stored in the Home RFSS for the object.

Moreover, the 200 OK responses SHALL contain the *contact* header field if the mobility object is currently registered with the querying Serving RFSS. If the mobility object is not registered with the querying Serving RFSS, the Home RFSS SHALL NOT include the *contact* header field.

- 3. In the event that the Serving RFSS fails to receive a response to the SIP REGISTER-Query message (i.e. experiences a SIP timeout in accordance with [RFC3261], Section 17.1.2), the Home RFSS is unreachable. The Serving RFSS MAY retry the request at a later time.
- 4. In the event that the Serving RFSS receives an MME_SU_INFO_NOT_AUTHOR failure response to the SIP REGISTER-Query message, it SHALL discontinue any attempt to register the object unless it receives a subsequent registration trigger, and, if appropriate, MAY notify the source of the registration trigger (e.g. an SU attempting to register) of the error condition.
- 5. In the event that the Serving RFSS receives a GEN_PROC_UNSUPPORTED failure response to the SIP REGISTER Query message, it MAY initiate Serving Registration by sending a SIP REGISTER-Register message to the Home RFSS and, for an SU mobility object, also indicating that the Serving RFSS has not successfully authenticated the object prior to sending the SIP REGISTER-Register message, in accordance with the Serving Registration procedure in Section 5.5.1 of this document.
- 6. In the event that the Serving RFSS receives an MME_SU_UNKNOWN or MME_OBJECT_UNKNOWN failure response (as specified in the “general requirements” above) to the SIP REGISTER-Query message, the Home RFSS is asserting that it is not the home for the given object. There is a

disagreement among system **databases** that needs to be resolved. The Serving RFSS SHOULD discontinue any attempt to register the object with the Home RFSS.

7. In the event that the Serving RFSS receives some other 4xx, 5xx or 6xx error response to the SIP REGISTER-Query message, it SHALL behave in accordance with local policy.

5.5.3 Home Query

[No change from existing text in this section of [102BACA].]

5.5.4 Serving Deregistration

[No change from existing text in this section of [102BACA].]

5.5.5 Home Deregistration

[No change from existing text in this section of [102BACA].]

6. Call Control

[No change from existing text in this section of [102BACA], except for selected subsections thereof, as shown below.]

6.1 CC-SAP

[No change from existing text in this section of [102BACA], except for Subsection 6.1.8.]

6.1.1 Answered VM

[No change from existing text in this section of [102BACA].]

6.1.2 Call Type

[No change from existing text in this section of [102BACA].]

6.1.3 Duplex Mode

[No change from existing text in this section of [102BACA].]

6.1.4 In-Call Roaming

[No change from existing text in this section of [102BACA].]

6.1.5 Offered VM

[No change from existing text in this section of [102BACA].]

6.1.6 Preference

[No change from existing text in this section of [102BACA].]

6.1.7 Priority

[No change from existing text in this section of [102BACA].]

6.1.8 RF State

[Replace existing text in this section of [102BACA] with the text below.]

The RF State parameter is related to the ISSI parameter conveyed in the INVITE, Re-INVITE, 200 OK, and 183 Session Progress messages (See RF Resource availability parameter in Section 3.8.5.2).

This parameter is ~~only present for confirmed group calls and is~~ significant in the following cases:

- For a CC_Setup.Request exerted by the Controlling Function at a Serving

RFSS

- For a CC_Setup.Indicate received at the Controlling Function of the Home RFSS
- For a CC_Setup.IndicateResponse(success, RTP OK) exerted by the Controlling Function at a Serving RFSS
- For a CC_Setup.Response(success, RTP OK) received at the Controlling Function of the Home RFSS
- For a CC_Modify.Request(RTP OK) exerted by the Controlling Function at a Serving RFSS
- For a CC_Modify.Indicate(RTP OK) received at the Controlling Function of the Home RFSS
- For a CC_Modify.IndicateResponse(success, RTP OK) exerted by the Controlling Function at a Serving RFSS
- For a CC_Modify.Response(success, RTP OK) received at the Controlling Function of the Home RFSS

This parameter indicates the availability of RF resources at the Serving RFSS.

6.1.9 RTP State

[No change from existing text in this section of [102BACA].]

6.1.10 Status_IR

[No change from existing text in this section of [102BACA].]

6.1.11 Status_IRC

[No change from existing text in this section of [102BACA].]

6.1.12 Status_R

[No change from existing text in this section of [102BACA].]

6.1.13 TX

[No change from existing text in this section of [102BACA].]

6.1.14 Protected

[No change from existing text in this section of [102BACA].]

6.2 SIP Transactions

[No change from existing text in this section of [102BACA].]

6.3 State Models for ISSI Voice Services

[No change from existing text in this section of [102BACA], except for Subsection 6.3.3 below.]

6.3.1 State Model: SU-to-SU Call - Originating RFSS

[No change from existing text in this section of [102BACA].]

6.3.2 State Model: SU-to-SU Call - Destination RFSS

[No change from existing text in this section of [102BACA].]

6.3.3 State Model: Group Call - Originating RFSS

[No change from existing text in this section of [102BACA], except for Subsection 6.3.3.3 below.]

6.3.3.1 Initial State

[No change from existing text in this section of [102BACA].]

6.3.3.2 Inviting State

[No change from existing text in this section of [102BACA].]

6.3.3.3 Wait Confirm State

[No change from existing text in this section of [102BACA], except for Subsection 6.3.3.3.3 below.]

6.3.3.3.1 CC_Setup.Confirm Event

[No change from existing text in this section of [102BACA].]

6.3.3.3.2 SIP BYE Event

[No change from existing text in this section of [102BACA].]

6.3.3.3.3 CC_Teardown.Request Event

[Replace existing text in this section of [102BACA] with the text below.]

The Controlling Function exerts a CC_Teardown.Request primitive when:

1. the RFSS decides to terminate the call for any reason, due to local policy.
2. the RFSS receives a MM_Deregistration.Indicate primitive that indicates that a group is no longer active on a Serving RFSS

Note that when the Controlling function exerts this primitive:

1. if the primitive was exerted due to local policy, the Controlling Function on the Home RFSS can, again according to local policy, try to re-establish the SIP session for this call segment by exerting a CC_Setup.Request primitive to the Call Control Protocol Element.
2. if the group is no longer active on the Serving RFSS, the Home RFSS Controlling Function releases RTP resources allocated for the call segment.
3. the Controlling Function for a Serving RFSS releases any allocated RTP resources and, when its RF Hang Timer expires, releases RF resources and tears down the call.

The Controlling Function SHOULD consider exertion of this primitive sufficient to tear down the call in the Call Control Protocol Element. The Call Control Protocol Element will not exert **any** response primitive in reply.

When this primitive is exerted, the Call Control Protocol Element will initiate a SIP BYE transaction to the Destination RFSS.

The Call Control Protocol Element terminates the call, moving to the *Final* state. The SIP Agent handles any SIP response to the SIP BYE message as described in Section 6.4.4.1.

6.3.3.4 RTP Session Open, No RTP resource at Local RFSS, and No RTP resource at Peer RFSS States

[No change from existing text in this section of [102BACA].]

6.3.3.5 Final State

[No change from existing text in this section of [102BACA].]

6.3.4 State Model: Group Call - Destination RFSS

[No change from existing text in this section of [102BACA].]

6.3.5 Sub-state Model: Group Call - RTP Session Open State

[No change from existing text in this section of [102BACA].]

6.3.6 Sub-state Model: Group Call - No RTP Resource at Peer RFSS State

[No change from existing text in this section of [102BACA].]

6.3.7 Sub-state Model: Group Call - No RTP Resource at Local RFSS State

[No change from existing text in this section of [102BACA].]

6.4 Basic ISSI SIP Control Procedures

[No change from existing text in this section of [102BACA].]

6.5 Call Control Procedures for ISSI SU-to-SU Voice Services

[No change from existing text in this section of [102BACA], except for subsections 6.5.3 and 6.5.4 below.]

6.5.1 General

[No change from existing text in this section of [102BACA].]

6.5.2 ISSI SU-to-SU Voice Call Setup

[No change from existing text in this section of [102BACA].]

6.5.3 ISSI Support for Roaming of the Called SU

[No change from existing text in this section of [102BACA], except for subsections 6.5.3.1 and 6.5.3.2 below.]

6.5.3.1 Actions Required at New Called Serving RFSS

[Replace existing text in this section of [102BACA] with the text below.]

In the event that the Called SU roams to a new Called Serving RFSS, the new Called Serving RFSS SHALL request the Called Home RFSS to register the Called SU, following procedures in Section 5 of this document.

The new Called Serving RFSS SHALL process the INVITE request from the Called Home RFSS according to procedures in Section 6.5.2.4 of this document, with the following qualifications:

1. In the event that the new Called Serving RFSS detects that none of the SDP media formats offered by the Called Home RFSS is accepted by the new Called Serving RFSS, it SHALL return a VCE_VOCODER_MODES_UNACCEPTED failure response.

NOTE: For older RFSS implementations that only support the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to "X-TIA-P25-IMBE" (see Section 3.7), a

GEN_SDP_PARAMS_UNACCEPTABLE failure response is returned instead.

2. The new Called Serving RFSS SHALL determine, from the “Initial Transmitter” parameter value in the received INVITE request, whether or not the called SU has been chosen by the Called Home RFSS to transmit first, and notify the called SU accordingly.
3. As the “p25c-incallroaming” parameter in the INVITE request is set to a value that indicates in-call roaming, **no 180 (RINGING)** OR 183 (SESSION PROGRESS) provisional response SHOULD be sent back to the Called Home RFSS.
4. When RF resource becomes available, the Called Serving RFSS SHALL return a 200 (OK) response to the Called Home RFSS. The 200 (OK) response SHALL include an “RF Resource Availability” parameter indicating that the Called Serving RFSS has available RF resource for the call. The Type 1 SDP answer in the SIP 200 (OK) response SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Called Home RFSS in its SIP INVITE request, has been accepted by the Called Serving RFSS.
5. In the event that RF resource, according to local policy, is not available within a reasonable timeframe (e.g., upon expiry of a local timer), the Called Serving RFSS SHALL return a VCE_NO_RF failure response.

6.5.3.2 Actions Required at Called Home RFSS

[Replace existing text in this section of [102BACA] with the text below.]

The Called Home RFSS SHALL register the Called SU as requested by the new Called Serving RFSS, following procedures in Section 5 of this document.

Upon successful registration of the Called SU in the new Called Serving RFSS:

1. In the event that the SU-to-SU call is set up and still in progress, the Called Home RFSS SHALL send an SIP INVITE request to the new Called Serving RFSS to establish a call segment between the two RFSSs, following the procedures in Section 6.4.1 of this document, with the following qualifications:
 - a. The SIP INVITE request SHALL include a Type 1 SDP offer (as defined in Section 3.7 of this document) that contains one or more media formats supported by the Called Home RFSS, listed in the

order of preference, with the first format listed being preferred, also taking into account the SDP media format that the Called Home RFSS currently uses for talk spurt transmission with the Calling Home RFSS.

- b. The SIP INVITE request SHALL include an "Initial Transmitter" parameter in its ISSI-specific body, as defined in Section 3.8.5.1 of this document, with value indicating its choice of whether the calling SU or called SU SHOULD transmit first.
 - c. The SIP INVITE request SHALL include a "p25c-incallroaming" parameter with value set to indicate in-call roaming,
2. In the event that the SIP INVITE request is denied by a 3xx, 4xx, 5xx, or 6xx failure response, then, subject to local policy:
- a. The Called Home RFSS MAY retry the INVITE request, modified as required per failure response received, or
 - b. The Called Home RFSS MAY send BYE requests to the Calling Home RFSS and previous Called Serving RFSS, according to the procedures in Section 6.4.4.1 of this document, to completely tear down the SU-to-SU call.
- ~~3. In the event that the SU-to-SU call set up is in progress, the Called Home RFSS SHALL terminate the previous SU-to-SU Call.~~

Upon successful establishment of a call segment with the new Called Serving RFSS, the Called Home RFSS SHALL bridge together the RTP sessions it has with the new Called Serving RFSS and with the Calling Home RFSS. The Called Home RFSS SHALL also send a SIP BYE request to the previous Called Serving RFSS to tear down the call segment between them, following the procedures in Section 6.4.4 of this document.

Upon successful teardown of the call segment between the Called Home RFSS and the previous Called Serving RFSS, the Called Home RFSS SHALL deregister the Called SU from the previous Called Serving RFSS, following procedures in Section 5 of this document.

6.5.3.3 Actions Required at Previous Called Serving RFSS

[No change from existing text in this section of [102BACA].]

6.5.4 ISSI Support for Roaming of the Calling SU

[Replace existing text in this section of [102BACA] with the text below.]

The procedures in Section 6.5.3 of this document apply, with the following qualifications:

1. “calling” and “called” are transposed.
2. ~~In the event that the SIP INVITE request is denied by a 3xx, 4xx, 5xx, or 6xx failure response, then, subject to local policy, the Calling Home RFSS MAY retry the INVITE request, modified as required per failure response received, or it MAY send BYE requests to the Calling Serving RFSS and Called Home RFSS to terminate the SU-to-SU call.~~

6.5.5 ISSI SU-to-SU Voice Call Release by Calling Serving RFSS

[No change from existing text in this section of [102BACA].]

6.5.6 ISSI SU-to-SU Voice Call Release by Calling Home RFSS

[No change from existing text in this section of [102BACA].]

6.5.7 ISSI SU-to-SU Voice Call Release by Called Home RFSS

[No change from existing text in this section of [102BACA].]

6.5.8 ISSI SU-to-SU Voice Call Release by Called Serving RFSS

[No change from existing text in this section of [102BACA].]

6.6 Call Control Procedures for ISSI Group Voice Services

[No change from existing text in this section of [102BACA], except for Subsection 6.6.1 below.]

6.6.1 ISSI Group Voice Call Setup

[No change from existing text in this section of [102BACA], except for subsections 6.6.1.2 and 6.6.1.3 below.]

6.6.1.1 General

[No change from existing text in this section of [102BACA].]

6.6.1.2 Group Voice Call Setup Initiated by Serving RFSS

[No change from existing text in this section of [102BACA], except for subsections 6.6.1.2.1 and 6.6.1.2.2 below.]

6.6.1.2.1 Actions Required at Serving RFSS

[Replace existing text in this section of [102BACA] with the text below.]

The Serving RFSS SHALL allocate RTP resources, and initiate a SIP INVITE request to the group's Home RFSS to ultimately establish an RTP session with the Home RFSS, according to the procedures in Section 6.4.1 of this document, with the following qualifications:

1. The SIP INVITE request SHALL include a Type 1 SDP offer (as defined in Section 3.7 of this document) that contains one or more media formats supported by the Serving RFSS, listed in the order of preference, with the first format listed being preferred, which MAY take into account information on the Home RFSS's set of supported SDP media formats that the Serving RFSS MAY have received during group registration over the ISSI (see Section 5).
2. The SIP INVITE request MAY include a "Group Call Type" parameter (as defined in Section 3.8.5.6) that indicates whether it is for a confirmed or unconfirmed group call.
3. The SIP INVITE request MAY include a "Protected" parameter (as defined in Section 3.8.5.7) that indicates whether the call's audio is requested to be protected.
4. For a confirmed group call, the SIP INVITE request SHALL include an "RF Resource Availability" parameter (as defined in Section 3.8.5.2) that indicates whether the Serving RFSS has available RF resource to support the confirmed group call. In the event that RF resource is currently unavailable, the Serving RFSS SHALL send a re-INVITE request, with appropriate "RF Resource Availability" parameter setting, to the Home RFSS when RF resource is later available, following the procedures in Section 6.4.3 of this document.
5. The Serving RFSS MAY cancel its SIP INVITE request, following procedures in Section 6.4.2 of this document.
6. Upon receipt of a VCE_SG_NOT_REG or VCE_SG_UNKNOWN failure response, the Serving RFSS SHALL consider itself not registered to the group and SHOULD, before attempting a new INVITE request, initiate

registration with the Home RFSS, following procedures in Section 5 of this document.

7. In the event that, while awaiting a response on the SIP INVITE request, the Serving RFSS receives a SIP INVITE or re-INVITE request from the Home RFSS, the Serving RFSS SHALL handle the received SIP INVITE or re-INVITE request, according to procedures in Section 6.6.3 of this document.
8. In the event that the “Protected” parameter, as specified in Section 3.8.5.7 of this document, is present in the received SIP 200 OK ISSI-specific body, or if the parameter is not present and takes on the implied default value of “unprotected”, then the Serving RFSS SHALL handle the call initially according to the received “Protected” value (be it protected or unprotected).
9. In the event that a SIP 3xx, 4xx, 5xx, or 6xx response for the SIP INVITE is received, the Serving RFSS MAY retry the SIP INVITE request, **modified as required per failure response received**, subject to local policy.

6.6.1.2.2 Actions Required at Home RFSS

[Replace existing text in this section of [102BACA] with the text below.]

The Home RFSS SHALL process the received SIP INVITE request, according to procedures in Section 6.4.1 of this document, with the following qualifications:

1. In the event that the Serving RFSS initiating the SIP INVITE request is not registered to the group, the Home RFSS SHALL return a VCE_SG_NOT_REG failure response.
2. In the event that the group is not known to the Home RFSS, the Home RFSS SHALL return a VCE_SG_UNKNOWN failure **response** response.
3. In the event that the Home RFSS detects that none of the SDP media formats offered by the Serving RFSS is accepted by the Home RFSS, it SHALL return a VCE_VOCODER_MODES_UNACCEPTED failure response.

NOTE: For older RFSS implementations that only support the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to “X-TIA-P25-IMBE” (see Section 3.7), a GEN_SDP_PARAMS_UNACCEPTABLE failure response is returned instead.

4. The Home RFSS SHALL process any received SIP CANCEL or BYE request following the procedures in Section 6.4.2.1 or 6.4.4.1 of this document.
5. In the event that the Home RFSS is able to allocate RTP resource to set up the RTP session with the originating RFSS, it SHALL return a SIP 200 (OK) response that carries a Type 1 SDP answer, and it SHALL proceed to invite other Serving RFSSs registered to the Home RFSS to join the call, according to the procedures in the following subsections, with the following qualification:
 - a. The Type 1 SDP answer in the SIP 200 (OK) response SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Serving RFSS in its SIP INVITE request, has been accepted by the Home RFSS. This MAY take into account information on the sets of SDP media formats supported by other Serving RFSSs in the same group that the Home RFSS MAY have received during their group registrations over ISSI (see Section 5).
6. In the event that the Home RFSS is unable to allocate RTP resource, it SHOULD return a SIP 200 (OK) response that carries a Type 2 SDP answer, and SHOULD send a re-INVITE request that carries a Type 1 SDP offer to the Serving RFSS when RTP resource is later available, following the procedures in Section 6.4.3 of this document. In this situation, and if the group call has not been already set-up with another Serving RFSS, the Home RFSS SHOULD NOT proceed to invite other Serving RFSSs registered with the Home RFSS for that group. The following qualification apply:
 - a. The Type 2 SDP answer in the SIP 200 (OK) response SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Serving RFSS in its SIP INVITE request, has been accepted by the Home RFSS. This MAY take into account information on the sets of SDP media formats supported by other Serving RFSSs in the same group that the Home RFSS MAY have received during their group registrations over ISSI (see Section 5).
7. In the event that the “Protected” parameter, as specified in Section 3.8.5.7 of this document, is present in the received INVITE request’s ISSI-specific body, or if the parameter is not present and takes on the implied default value of “unprotected”, the Home RFSS SHALL determine whether the SG supports the value of this parameter:

- a. In the event that the SG supports this parameter value, then the Home RFSS SHALL include the parameter value in its 200 OK response to the Serving RFSS, and in the SIP INVITE requests to the other Serving RFSSs for the SG. (see Section 6.6.1.3).
- b. In the event that the SG does not support this parameter value, then the Home RFSS SHALL return a VCE_CANNOT_PROTECT_AUDIO or VCE_CANNOT_UNPROTECT_AUDIO failure response, as appropriate, to the Serving RFSS.

6.6.1.3 Call Segment Setup Initiated by Home RFSS

[No change from existing text in this section of [102BACA], except for Subsection 6.6.1.3.1 below.]

6.6.1.3.1 Actions Required at Home RFSS

[Replace existing text in this section of [102BACA] with the text below.]

The Home RFSS SHALL allocate RTP resources for RTP sessions with other Serving RFSSs within the group that are successfully registered with the Home RFSS, but have not yet been invited to join the group call. Throughout the group call, the Home RFSS SHOULD, with allocated RTP resources, initiate SIP INVITE or re-INVITE requests to any Serving RFSS in the group that is not already participating in the call.

The Home RFSS SHALL then initiate SIP INVITE requests to these Serving RFSSs, to ultimately establish one-to-one RTP sessions with them, following the procedures in Section 6.4.1 of this document, with the following qualifications:

1. The SIP INVITE request SHALL include a Type 1 SDP offer (as defined in Section 3.7 of this document) that contains one or more media formats supported by the Home RFSS, listed in the order of preference, with the first format listed being preferred, taking into account SDP media format(s) that the Home RFSS MAY already be using for talk spurt transmissions with some other Serving RFSSs in the group. The Type 1 SDP offer MAY also take into account information on the sets of SDP media formats supported by this and other Serving RFSSs in the same group that the Home RFSS MAY have received during their group registrations over ISSI (see Section 5).

2. The SIP INVITE request MAY include a “Group Call Type” parameter (as defined in Section 3.8.5.6) that indicates whether it is for a confirmed or unconfirmed group call.
3. The SIP INVITE request MAY include a “Protected” parameter (as defined in Section 3.8.5.7) that indicates whether the call’s audio is requested to be protected. (see Section 6.6.1.2.2)
4. In the event that, while awaiting a final response on the SIP INVITE request to a given Serving RFSS, the Home RFSS receives a SIP INVITE or re-INVITE request from the Serving RFSS, the Home RFSS SHALL handle the received SIP INVITE or re-INVITE request, according to procedures in Section 6.6.3 of this document. In the event that the Home RFSS originated SIP INVITE request does not have precedence, the Home RFSS SHALL wait for a SIP final response from each remaining Serving RFSS in the group (i.e., any Serving RFSS in the group except the Serving RFSS whose SIP INVITE request has precedence) that has not yet returned any final response to the SIP INVITE from the Home RFSS, following the procedures in Section 6.4.1.1 of this document. Subsequently, the Home RFSS:
 - a. SHOULD send a SIP re-INVITE request to each remaining Serving RFSS in the group that has returned a 2xx final response to the SIP INVITE from the Home RFSS, following the procedures in Section 6.4.3.1 of this document, to update the session parameters to match those of the new call initiated by the given Serving RFSS.
 - b. SHALL initiate SIP INVITE requests to those Serving RFSSs in the group that have not yet joined the group call, following the procedures in Section 6.6.1.3 of this document.
5. Any time after receiving a SIP 1xx provisional response and before receiving a final response from the Serving RFSS, the Home RFSS MAY cancel its SIP INVITE request, following procedures in Section 6.4.2 of this document.
6. Upon receipt of a VCE_SG_UNKNOWN failure response, the Home RFSS SHALL consider the Serving RFSS no longer registered to the group and SHOULD remove this RFSS from its list of Serving RFSSs for the group.
7. In the event that a SIP 3xx, 4xx, 5xx, or 6xx response for the SIP INVITE is received, the Home RFSS MAY retry the SIP INVITE request, **modified as required per failure response received**, subject to local policy.

For an unconfirmed group call, talk spurt transmission MAY start as soon as the first SIP session and associated RTP session for a group call have been successfully established between the Home RFSS and a Serving RFSS.

For a confirmed group call, after successful setup of the first SIP session and RTP session, the Home RFSS SHALL control talk spurt transmission according to the following procedures:

1. Upon receipt of the first PTT Transmit Request from a Serving RFSS, the Home RFSS SHALL check whether sufficient number of RFSSs have RF resources available to support the confirmed group call.
2. In the event that sufficient RF resources are available, the Home RFSS SHALL grant the first PTT Transmit Request the permission to transmit talk spurts.
3. In the event that insufficient RF resources are available, the Home RFSS SHALL start a Tgchstartconfirmed timer.
 - a. In the event that sufficient RF resources later become available, the Home RFSS SHALL stop the Tgchstartconfirmed timer and grant one of the received PTT Transmit Requests the permission to transmit, using the selection criteria as specified in Section 7.9.1 of this document.
 - b. Upon expiry of the Tgchstartconfirmed timer, the Home RFSS SHOULD grant one of the received PTT Transmit Requests the permission to transmit talk spurts, using the selection criteria as specified in Section 7.9.1 of this document.
4. Depending on local policy, the Home RFSS MAY or MAY NOT check RF resource availability, by repeating the above steps, before granting each subsequent PTT Transmit Request the permission to transmit.

Moreover, as soon as the first SIP session for a confirmed or unconfirmed group call have been successfully established between the Home RFSS and a Serving RFSS, the Home RFSS SHALL start a Tgchhangtime timer that is reinitialized every time the Home RFSS detects voice activity on the RTP sessions for the group call.

As each RTP session is established, the Home RFSS SHALL bridge it to other RTP session(s) established earlier, and SHALL control the ensuing talk spurts

among the various RFSSs participating in the group call, following the procedures in Section 7 of this document.

6.6.1.3.2 Actions Required at Serving RFSS

[No change from existing text in this section of [102BACA].]

6.6.2 Subsequent Modification of ISSI SIP Session Characteristics

[No change from existing text in this section of [102BACA].]

6.6.3 Handling (re-)INVITE Request Collisions

[No change from existing text in this section of [102BACA].]

6.6.4 ISSI SIP Heartbeat

[No change from existing text in this section of [102BACA].]

6.6.5 ISSI SIP Session Termination by Home RFSS

[No change from existing text in this section of [102BACA].]

6.6.6 ISSI SIP Session Termination by Serving RFSS

[No change from existing text in this section of [102BACA].]

6.6.7 ISSI Group Voice Call Release

[No change from existing text in this section of [102BACA].]

7. Push-to-Talk (PTT) Management

[No change from existing text in this section of [102BACA], except for selected subsections thereof, as shown below.]

7.1 SAP Primitive Model

[No change from existing text in this section of [102BACA].]

7.2 State Model

[No change from existing text in this section of [102BACA].]

7.3 Management of Header Fields

[No change from existing text in this section of [102BACA].]

7.4 Management of Transmission Sequence Numbers

[No change from existing text in this section of [102BACA].]

7.5 Losing Audio Processing

[Replace existing text in this section of [102BACA] as shown below.]

At any particular time, a call MAY carry 0, 1 or more than one talk spurt. A talk spurt is uniquely identified on a call segment by the Transmission Sequence Number described in Section 7.4.

Each talk spurt also has a volatile characteristic known as the “losing audio flag”. When an RFSS forwards more than one talk spurt to a peer, it SHALL set the losing audio flag on packets associated with all streams except the preferred one. Upon receiving a talk spurt from an SMF, depending on the current state of the call, the MMF MAY re-mark the **losing audio flag** before distributing the talk spurts to other SMFs.

When an RFSS forwards only one talk spurt to a peer, the losing audio flag on packets associated with the stream SHOULD be set to 0.

At any particular time:

1. **When there are multiple streams in a particular direction in an RTP session, only one stream in that direction MAY be the preferred talk spurt (i.e., the one with the losing audio flag cleared). For transmission over an air interface, an RFSS SHOULD prefer talk spurts with the losing audio flag cleared.**

2. When there is only one stream in a particular direction in an RTP session, this stream SHALL be the preferred talk spurt in that direction regardless of whether the losing audio flag is cleared

When an MMF begins receiving a new talk spurt (due to issuing a PTT Transmit Grant as in Section 7.9.1 or due to receiving unsolicited PTT Transmit Progress messages as in Section 7.9.6), or when a talk spurt is completed (as in Section 7.9.7), it MAY select a new preferred talk spurt, and set the losing audio flag on existing talk spurts appropriately.

7.6 Transmission Vocoder Mode

[No change from existing text in this section of [102BACA].]

7.7 Handling Service Options

[Replace existing text in this section of [102BACA] with the text below.]

The Service Options in the ISSI Packet Type block SHALL be used by the SMF and the MMF to maintain the state of a talk spurt.

- The protected state of a talk spurt SHALL depend on the Service Options “P” bit.
- The emergency status of a talk spurt SHALL depend on the Service Options “E” bit.
- The priority of a talk spurt SHALL depend on the Service Options “Priority level” bits.

NOTE: The emergency status of a talk spurt (the “E” bit in the Service Options) could change during the talk spurt if the ISSI call was set up before the emergency status of the SU’s transmission was known. In this case the default value (non-emergency) is initially chosen and once the emergency status is known, and if it is determined to be in the emergency state, the “E” bit in the next PTT Progress is updated. The emergency state of an SU transmission might not be initially known if the air interface is Message Trunked and the SU transmitted on the voice channel without first requesting to talk on the voice/control channel. .

7.8 Sending RTP Packets

[No change from existing text in this section of [102BACA], except for Subsection 7.8.2 below.]

7.8.1 Sending Heartbeats

[No change from existing text in this section of [102BACA].]

7.8.2 Initiation of a Serving Transmission

[No change from existing text in this section of [102BACA], except for Subsection 7.8.2.1 below.]

7.8.2.1 Requesting Permission for a Transmission (PTT Transmit Request)

[Replace existing text in this section of [102BACA] with the text below.]

A SMF MAY, at any time, request permission to transmit RTP voice payload packets to an MMF. Upon sending the initial PTT Transmit Request for a talk spurt, **the SMF SHALL start the RequestTimeout.**

If the transmit priority of the talk spurt is lower than or equal to that of the most recently transmitting RFSS, then, in order to allow for loss or delayed receipt of PTT Transmit End packets, and to optimize bandwidth utilization over the ISSI, an RFSS SHOULD only request permission to transmit if:

- 1) **MinKeybackTime** seconds have elapsed since the receipt of the first (of up to **Render**) PTT Transmit End packets received from the MMF.

OR,

- 2) The RFSS has not received any non-heartbeat RTP packet for the session for more than **EndLossTimeout** seconds.

To ensure that the PTT Transmit Request reaches the MMF despite possible packet losses over the IP-based backbone, the RFSS SHALL continue to send PTT Transmit Request packets to the MMF at a nominal rate of once in **Trequest** seconds until one of the following events occurs:

1. The SMF receives PTT Transmit Grant from the MMF, in which case it SHALL behave as described in Section 7.9.3.
2. The SMF receives PTT Transmit Deny from the MMF, in which case it SHALL behave as described in Section 7.9.4.

3. The SMF receives PTT Transmit Wait from t.he MMF, in which case it SHALL behave as described in Section 7.9.5.
4. The **RequestTimeout** expires, in which case it SHALL behave as described in Section Section 7.10.1.
5. The SMF receives an end trigger, in which case it SHALL behave as described in Section 7.8.5.

An MMF SHALL NOT send PTT Transmit Request to a SMF. Talk spurts outbound from a MMF to an SMF are initiated via PTT Transmit Start.

PTT Transmit Request packets MAY include voice content, which SHALL be processed in accordance with Section 7.11.

7.8.2.2 Initiating a Talk Spurt without a-Priori Permission (Unstoppable Triggers)
[No change from existing text in this section of [102BACA].]

7.8.3 Initiation of a Home Talk Spurt (PTT Transmit Start)
[No change from existing text in this section of [102BACA].]

7.8.4 Sending PTT Transmit Progress Packets
[No change from existing text in this section of [102BACA].]

7.8.5 Terminating a Talk Spurt
[No change from existing text in this section of [102BACA].]

7.8.6 Muting a Talk Spurt
[No change from existing text in this section of [102BACA].]

7.8.7 Unmuting a Talk Spurt
[No change from existing text in this section of [102BACA].]

7.9 Processing of RTP Packets

[No change from existing text in this section of [102BACA], except for selected subsections thereof, as shown below.]

7.9.1 Processing PTT Transmit Request

[Replace existing text in this section of [102BACA] with the text below.]

An MMF that receives a PTT Transmit Request SHALL behave as follows:

- 1) It SHALL arbitrate the request in accordance with local policies and procedures governed by the guidelines below.
- 2) In the event that the MMF wishes to allow the talk spurt to proceed, it SHALL **grant** the talk spurt by sending a single instance of the PTT Transmit Grant message. It SHALL also initiate talk spurts to other SMFs that it determines SHOULD be joined to the transmission, by sending PTT Transmit Start messages as described in Section 7.8.3 for each. The MMF MAY also select a new preferred talk spurt and set the losing audio flag on existing talk spurt appropriately, as described in Section 7.5.
- 3) In the event that the MMF wishes to prevent the talk spurt from proceeding, it SHALL deny the talk spurt by sending a single transmission of the PTT Transmit Deny message.
- 4) In the event that the MMF wishes to defer making a decision about the talk spurt (e.g., Confirmed group call processing), it SHALL reply with a single transmission of the PTT Transmit Wait message. Subsequently, the MMF SHALL reply with a single instance of PTT Transmit Grant or PTT Transmit Deny, as described above.
- 5) It SHALL process any voice content in accordance with Section 7.11.

The decision by a MMF to Grant, Deny or Defer a decision on a talk spurt SHALL be made in accordance with the following guidelines:

- 1) If the talk spurt has already been denied, the MMF SHALL Deny the talk spurt.
- 2) If the talk spurt has already been granted, the MMF SHALL Grant the talk spurt.
- 3) If a Wait decision has been made, and no subsequent decision has been made, the MMF SHALL continue to defer the decision.
- 4) If no talker is active for the session, the MMF MAY grant the talk spurt immediately.

- 5) If no talker is active for the session, the MMF MAY reserve judgment for Telection seconds, and make a decision based on all requests received in that time.
- 6) A console-initiated talk spurt SHOULD be preferred over an SU initiated talk spurt.
- 7) An emergency talk spurt SHOULD be preferred over any other talk spurt, regardless of priority.
- 8) A strictly higher priority talk spurt SHOULD be preferred over a lower priority talk spurt.
- 9) A talk spurt with voice payload SHOULD be preferred over a talk spurt without voice payload.
- 10) An earlier arriving request SHOULD be preferred over a later arriving request.
- 11) A strictly higher priority console initiated transmission request SHOULD be preferred over a lower priority console transmission.
- 12) If the protected state of the talk spurt (indicated in the Service Options field of the ISSI Packet Type block) is not supported by the destination SG, the MMF SHALL Deny the talk spurt.

7.9.2 Processing PTT Transmit Start

[No change from existing text in this section of [102BACA].]

7.9.3 Processing PTT Transmit Grant

[Replace existing text in this section of [102BACA] with the text below.]

A SMF receiving a PTT Transmit Grant SHALL:

- Stop sending PTT Transmit Request packets and SHALL cancel the **RequestTimeout**,

OR

- Cancel the **WaitTimeout** if it has received a **PTT Transmit Wait** prior to receiving the **PTT Transmit Grant**

The SMF SHALL then begin to send PTT Transmit Progress packets as described in Section 7.8.4, taking such other actions (such as granting an RF channel to a requesting SU) as MAY be required by local policy.

7.9.4 Processing PTT Transmit Deny

[No change from existing text in this section of [102BACA].]

7.9.5 Processing PTT Transmit Wait

[Replace existing text in this section of [102BACA] with the text below.]

A SMF receiving a PTT Transmit Wait after requesting permission to transmit as described in Section 7.8.2.1 and not having received a response SHALL stop sending PTT Transmit Request packets, SHALL cancel the **RequestTimeout**, and SHALL start the **WaitTimeout**. After receiving a response, or while transmitting PTT Transmit Progress packets, the SMF SHALL silently discard PTT Transmit Wait.

It SHALL furthermore take other actions in accordance with local policy to notify the source of the Transmit Trigger that it is awaiting a final response (grant or denial) for its PTT Transmit Request, in accordance with the procedures for the particular source.

7.9.6 Processing PTT Transmit Progress

[No change from existing text in this section of [102BACA].]

7.9.7 Processing PTT Transmit End

[No change from existing text in this section of [102BACA].]

7.9.8 Processing PTT Transmit Mute

[No change from existing text in this section of [102BACA].]

7.9.9 Processing of PTT Transmit Unmute

[No change from existing text in this section of [102BACA].]

7.9.10 Processing Heartbeats

[No change from existing text in this section of [102BACA].]

7.9.11 Processing of the M-bit

[No change from existing text in this section of [102BACA].]

7.10 Processing of Timeouts

[No change from existing text in this section of [102BACA].]

7.11 Processing of Voice Content

[No change from existing text in this section of [102BACA].]

8. RFSS Service Capability Polling

[No change from existing text in this section of [102BACA].]

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Annex A (Normative) – Generic SIP Responses Over the ISSI

[No change from existing text in this section of [102BACA].]

Annex B (Normative) – Table of Timers and Constants

[No change from existing text in this section of [102BACA].]

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Annex C (Informative) – TIA-102 Vocoder Error Mitigation Data⁹

[No change from existing text in this section of [102BACA].]

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⁹ The text in this Annex was sourced, with minor editorial modifications, from a contribution, entitled “P25 Vocoder Error Mitigation Data”, that John C. Hardwick, of Digital Voice Systems, Inc., submitted to the APCO Interface Committee Project 25 Vocoder Task Group, 11 October 2005. Appearance of “P25” throughout this Annex indicates the item has been adopted or may be adopted by the APCO/NASTD/FED Project 25 Steering Committee for use in its Standard.

Annex D (Informative) – SU-to-SU Call Flows

*[No change from existing text in this section of **[102BACA].**]*

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Annex E (Informative) – Group Call Flows

[No change from existing text in this section of [102BACA].]

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Annex F (Informative) – SU-to-SU Call Flows Involving Multiple Vocoder Modes

*[No change from existing text in this section of **[102BACA]**.]*

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Annex G (Informative) – Group Call Flows Involving Multiple Vocoder Modes

[No change from existing text in this section of [102BACA].]

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Annex H (informative) – Future Considerations

[No change from existing text in this section of [102BACA], except for subsection H.3 below.]

H.1 ISSI SIP and RTP Security Services

[No change from existing text in this section of [102BACA].]

H.2 ISSI SIP and RTP Compression Services

[No change from existing text in this section of [102BACA].]

~~H.3 IP Based Backbone Bandwidth Reservation~~

~~RFC 2474 defines a replacement header field, called the DS (Differentiated Services) field, superseding the existing definitions of the IPv4 TOS (Type of Service) octet. This new field includes a 6-bit DSCP (Differentiated Services Codepoint) which could be flexibly mapped to particular forwarding treatment, or per-hop behavior (PHB), at each Differentiated Services compliant network node (e.g., routers, etc.) along its path, for efficient routing of SIP messages and RTP packets to their destinations, according to the urgency of the call request (e.g., preemptive priority call). Appropriate assignment and mapping of DSCPs for ISSI applications are for further study.~~

H.3 Mixed IANA and ISSI Payload Carrying Trunked, Conventional, and Additional Data

[No change from existing text in this section of [102BACA].]

H.4 Refreshing Service Profile Information

[No change from existing text in this section of [102BACA].]

Annex I (informative) – Console Functionality

[No change from existing text in this section of [102BACA].]

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